MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3130 FOUNDATION DESIGN OF CONVEYOR 3130-CNV-002

DESIGN CALCULATION

AS BUILT

MD-216-3130-EG-CV-CAL-1002

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RL	:	Reduced Level
C.L.	:	Centre Line
CJ	:	Construction Joint
FDN	:	Foundation
TOC	:	Top of concrete
FGL	:	Finished Ground Level
FFL	:	Finished Floor Level
BOF	:	Bottom of footing
TYP	:	Typical
TOC	:	Top of concrete
IP	:	Insert Plate
RCC	:	Reinforced Cement Concrete
MKD.	:	Marked
THK.	:	Thick

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1. INTENT

The document is prepared for Concrete design calculation of Conveyor Gallery 3130-CNV-002.
This conveyor is between TP-4 and Tertiary Screen house and belt width of the conveyor is 1200 mm.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for foundation design of Conveyor 3130-CNV-002. All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The column and beam reinforcement design have been performed in Staad-Pro Software. The footings of Ground Portal GR-1 to GR-5; TR-4 & VGTU have been performed in MS Excel.
Foundation of TR-1, TR-2 & TR-3 has been designed by FEM analysis in Staad-Pro Software.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 1m & 3m depth below FGL (ie. RL 911.500m & RL 909.500m respectively) refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

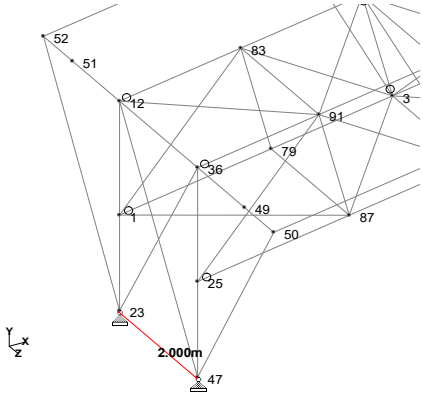
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 INPUT LOADS FROM SUPERSTRUCTURE

For load calculation of superstructure refer Document No: MD-216-3130-EG-ST-CAL-1002

5.1 Support Reaction from TR-1

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
23	1 DL	0.0	35.5	-4.6
	2 LL	0.0	24.1	0.9
	3 ML - CONVEYOR MATERIAL	0.0	17.1	1.7
	4 WL (+Z)	0.0	-33.7	-12.5
	5 WL (-Z)	0.0	33.7	12.6
	6 TL1	0.0	0.2	6.3
	7 TL2	0.0	-0.2	-6.3
	8 EARTHQUAKE LOAD (+X)	-0.9	-0.4	0.0
	9 EARTHQUAKE LOAD (+Z)	0.0	-1.0	-0.4
47	1 DL	0.0	75.2	0.6
	2 LL	0.0	23.9	-0.9
	3 ML - CONVEYOR MATERIAL	0.0	16.8	-1.7
	4 WL (+Z)	0.0	33.7	-12.0
	5 WL (-Z)	0.0	-33.7	12.0
	6 TL1	0.0	-0.2	-6.3
	7 TL2	0.0	0.2	6.3
	8 EARTHQUAKE LOAD (+X)	-0.9	-0.4	0.0
	9 EARTHQUAKE LOAD (+Z)	0.0	1.0	-0.4



Staad model of 12m Gallery End portal which is acting as TR1

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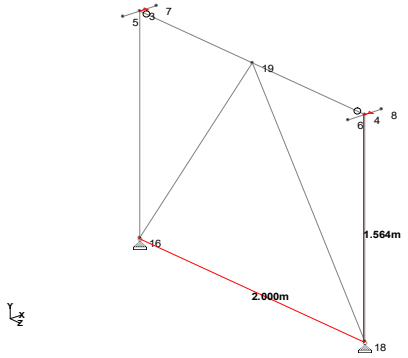
5.2 Support Reaction from TR-2

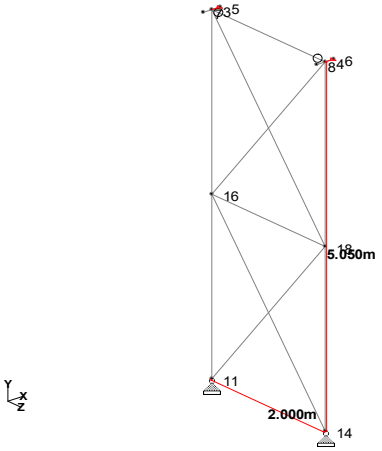
Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
16	1 DEAD LOAD	1.3	79.5	-4.7
	2 LIVE LOAD	1.0	60.0	0.0
	3 MATERIAL LOAD	0.6	41.0	0.0
	4 WL (+Z)	-1.2	-106.7	-32.0
	5 WL (-Z)	1.2	106.6	31.9
	6 WL (+X)	-0.5	0.0	0.0
	7 WL (-X)	0.5	0.0	0.0
	8 TL1	0.0	0.0	0.0
	9 TL2	0.0	0.0	0.0
	10 EL (+X)	0.0	0.1	0.0
	11 EL (+Z)	0.0	-7.4	-3.6
18	1 DEAD LOAD	2.1	183.3	-4.9
	2 LIVE LOAD	0.9	59.9	0.0
	3 MATERIAL LOAD	0.6	40.9	0.0
	4 WL (+Z)	1.2	106.7	-32.0
	5 WL (-Z)	-1.2	-106.6	31.9
	6 WL (+X)	-0.5	0.0	0.0
	7 WL (-X)	0.5	0.0	0.0
	8 TL1	0.0	0.0	0.0
	9 TL2	0.0	0.0	0.0
	10 EL (+X)	-0.1	0.0	0.0
	11 EL (+Z)	0.0	7.4	-3.6



Staad model of TR2 superstructure (along with dimension)

5.3 Support Reaction from TR-3

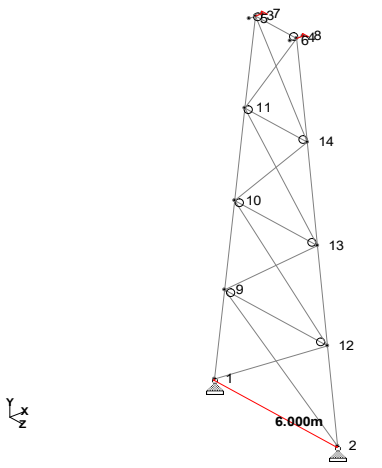
Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
11	1 DEAD LOAD	0.1	77.7	2.3
	2 LIVE LOAD	0.0	71.9	3.7
	3 MATERIAL LOAD	0.0	48.3	2.5
	4 WL (+Z)	0.0	-267.2	-40.7
	5 WL (-Z)	0.0	269.9	41.8
	6 WL (+X)	-1.5	0.0	0.0
	7 WL (-X)	1.5	0.0	0.0
	8 TL1	0.0	0.1	17.7
	9 TL2	0.0	-0.1	-17.7
	10 EL (+X)	0.0	0.1	0.0
	11 EL (+Z)	0.0	-13.7	-2.4
14	1 DEAD LOAD	0.0	234.1	-13.6
	2 LIVE LOAD	0.0	72.1	-3.7
	3 MATERIAL LOAD	0.0	47.9	-2.4
	4 WL (+Z)	0.0	267.2	-40.7
	5 WL (-Z)	0.0	-269.9	41.7
	6 WL (+X)	-1.5	0.0	0.0
	7 WL (-X)	1.5	0.0	0.0
	8 TL1	0.0	-0.1	-17.7
	9 TL2	0.0	0.1	17.7
	10 EL (+X)	0.0	-0.1	0.0
	11 EL (+Z)	0.0	13.7	-2.4



Staad model of TR3 superstructure (along with dimension)

5.4 Support Reaction from TR-4

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
1	1 DEAD LOAD	0.0	132.9	26.8
	2 LIVE LOAD	0.0	72.2	13.1
	3 MATERIAL LOAD	0.0	55.1	10.0
	4 WL (+Z)	0.0	-182.1	-24.4
	5 WL (-Z)	0.0	244.8	49.0
	6 WL (+X)	-4.6	0.0	0.0
	7 WL (-X)	4.6	0.0	0.0
	8 TL1	0.0	0.0	57.7
	9 TL2	0.0	0.0	-57.7
	10 EL (+X)	0.0	0.1	0.0
	11 EL (+Z)	0.0	-11.7	-2.2
2	1 DEAD LOAD	0.0	228.3	-38.1
	2 LIVE LOAD	0.0	71.8	-13.0
	3 MATERIAL LOAD	0.0	55.1	-10.0
	4 WL (+Z)	0.0	182.1	-24.3
	5 WL (-Z)	0.0	-244.8	49.0
	6 WL (+X)	-4.6	0.0	0.0
	7 WL (-X)	4.6	0.0	0.0
	8 TL1	0.0	0.0	-57.7
	9 TL2	0.0	0.0	57.7
	10 EL (+X)	0.0	-0.1	0.0
	11 EL (+Z)	0.0	11.7	-2.2



Staad model of TR4 superstructure (along with dimension)

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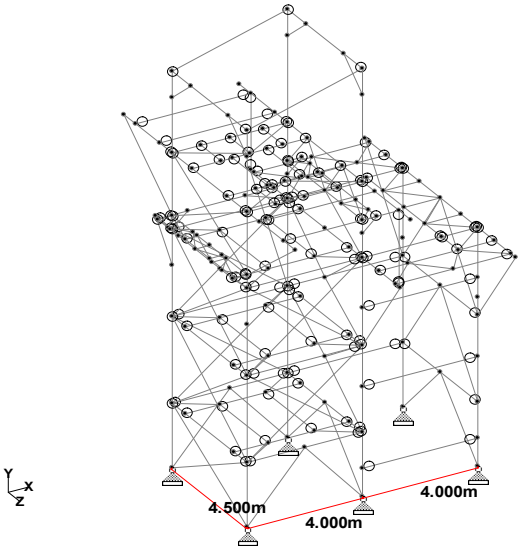
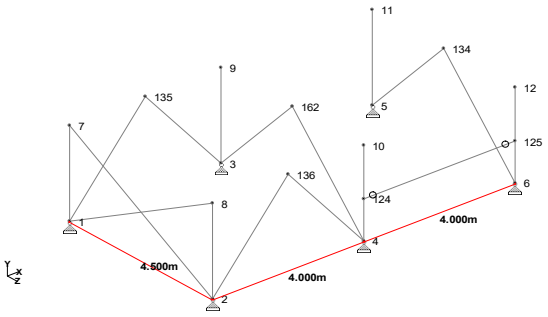
5.5 Support Reaction from VGTU

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
1	1 DEAD LOAD	23.1	256.5	11.3
	2 LIVE LOAD	-0.1	138.6	7.3
	3 MATERIAL LOAD	-0.1	23.9	1.4
	4 WL (+Z)	1.3	-122.8	-30.5
	5 WL (-Z)	-1.3	122.8	30.5
	6 WL (+X)	-49.1	-206.7	-7.9
	7 WL (-X)	49.1	206.7	7.9
	8 TL1	1.7	4.1	38.2
	9 TL2	-1.7	-4.1	-38.2
	10 EL (+X)	-1.5	-7.2	-0.3
	11 EL (+Z)	0.1	-6.7	-1.6
2	1 DEAD LOAD	22.9	306.2	-16.5
	2 LIVE LOAD	1.0	126.7	-7.3
	3 MATERIAL LOAD	0.1	23.4	-1.2
	4 WL (+Z)	-1.3	122.8	-30.7
	5 WL (-Z)	1.3	-122.8	30.7
	6 WL (+X)	-48.9	-208.0	8.3
	7 WL (-X)	48.9	208.0	-8.3
	8 TL1	1.3	-4.1	-37.2
	9 TL2	-1.3	4.1	37.2
	10 EL (+X)	-1.6	-7.4	0.3
	11 EL (+Z)	-0.1	6.7	-1.6
3	1 DEAD LOAD	21.1	60.3	-1.5
	2 LIVE LOAD	-1.0	138.5	0.6
	3 MATERIAL LOAD	-0.1	27.6	-0.1
	4 WL (+Z)	1.8	-122.9	-25.6
	5 WL (-Z)	-1.8	122.9	25.6
	6 WL (+X)	-49.1	205.4	-0.2
	7 WL (-X)	49.1	-205.4	0.2
	8 TL1	-1.2	-5.4	-0.2
	9 TL2	1.2	5.4	0.2
	10 EL (+X)	-1.5	7.1	0.0
	11 EL (+Z)	0.2	-6.7	-1.2
4	1 DEAD LOAD	20.9	117.7	-4.2
	2 LIVE LOAD	0.1	165.7	-1.4
	3 MATERIAL LOAD	0.0	28.3	-0.2
	4 WL (+Z)	-1.8	122.9	-25.6
	5 WL (-Z)	1.8	-122.9	25.6
	6 WL (+X)	-50.0	209.2	-0.8
	7 WL (-X)	50.0	-209.2	0.8
	8 TL1	-1.2	5.4	-1.5
	9 TL2	1.2	-5.4	1.5
	10 EL (+X)	-1.6	7.4	0.0
	11 EL (+Z)	-0.2	6.7	-1.2
5	1 DEAD LOAD	0.0	15.7	0.1
	2 LIVE LOAD	0.0	51.3	0.5
	3 MATERIAL LOAD	0.0	0.4	0.1
	4 WL (+Z)	0.0	-29.2	-11.5
	5 WL (-Z)	0.0	29.2	11.5
	6 WL (+X)	-1.0	1.2	0.3
	7 WL (-X)	1.0	-1.2	-0.3
	8 TL1	-0.1	1.2	0.8
	9 TL2	0.1	-1.2	-0.8
	10 EL (+X)	0.0	0.1	0.0
	11 EL (+Z)	0.0	-0.9	-0.2
6	1 DEAD LOAD	0.0	17.8	-0.6
	2 LIVE LOAD	0.0	48.5	0.5
	3 MATERIAL LOAD	0.0	-0.4	0.1
	4 WL (+Z)	0.0	29.2	-11.5
	5 WL (-Z)	0.0	-29.2	11.5
	6 WL (+X)	0.1	-1.2	0.3
	7 WL (-X)	-0.1	1.2	-0.3
	8 TL1	-0.5	-1.2	-0.2
	9 TL2	0.5	1.2	0.2
	10 EL (+X)	0.0	-0.1	0.0
	11 EL (+Z)	0.0	0.9	-0.2



Staad model of VGTU superstructure (along witgh dimension)



Node number and orientation of VGTU superstructure

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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

5.6 LOAD COMBINATIONS

5.6.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD COMB 301 1.0 DL	(Eq. 2.4.1-1 of SBC 301)
LOAD COMB 302 1.0 (DL + TL1) + 1.0 (LL+ML)	(Eq. 2.4.1-2 of SBC 301)
LOAD COMB 303 1.0 (DL + TL1) + 1.0 (LL+ML)	
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75 (LL+ML)	(Eq. 2.4.1-6 of SBC 301)
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75 (LL+ML)	
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75 (LL+ML)	
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75 (LL+ML)	
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75 (LL+ML)	
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75 (LL+ML)	
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75 (LL+ML)	
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75 (LL+ML)	
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)	(Eq. 2.4.1-7 of SBC 301)
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)	
LOAD COMB 314 0.6 DL + 1.0 WL(+X)	
LOAD COMB 315 0.6 DL + 1.0 WL(-X)	
LOAD COMB 316 0.6 DL + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD COMB 317 0.6 DL + 0.7 EL(-X)	
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)	
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)	

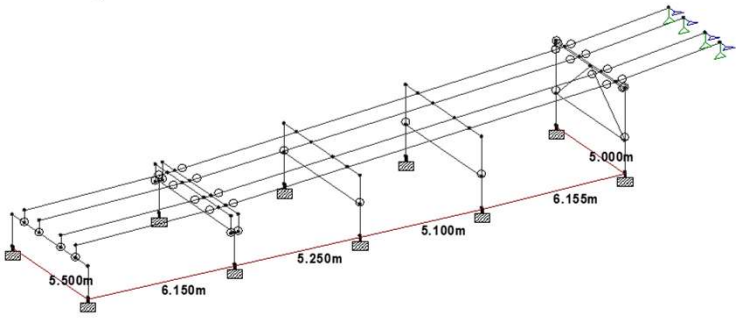
5.6.2 Load Combination for Concrete design

Load Case	Reference
LOAD COMB 401 1.4 DL	(Eq. 2.3.2-1 of SBC 301)
LOAD COMB 402 1.4 (DL + TL1) + 1.7 (LL+ML)	(Eq. 2.3.2-2 of SBC 301)
LOAD COMB 403 1.4 (DL + TL1) + 1.7 (LL+ML)	
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5 (LL+ML)	(Eq. 2.3.2-4 of SBC 301)
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5 (LL+ML)	
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5 (LL+ML)	
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5 (LL+ML)	
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5 (LL+ML)	(Eq. 2.3.2-5 of SBC 301)
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5 (LL+ML)	
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5 (LL+ML)	
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5 (LL+ML)	
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)	(Eq. 2.3.2-6 of SBC 301)
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)	
LOAD COMB 414 0.9 DL + 1.6 WL(+X)	
LOAD COMB 415 0.9 DL + 1.6 WL(-X)	
LOAD COMB 416 0.9 DL + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD COMB 417 0.9 DL + 1.0 EL(-X)	
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)	
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)	

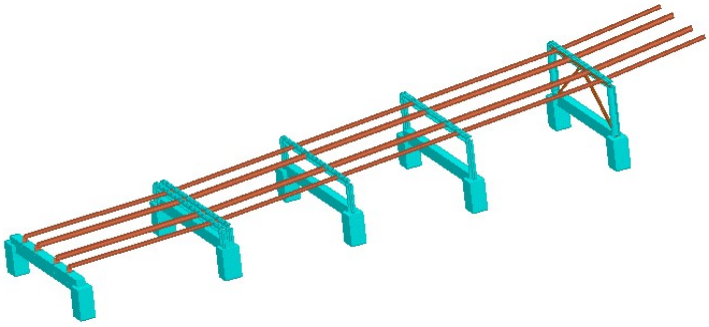
Eq. 2.3.2-3 of SBC 301 - 2007 has not been considered as it would not produce worst load combination in the absence of Roof Live Load & Rain Load

6 GEOMETRY

6.1 Geometry of Ground Portals



Staad model of Ground Portals (along witgh dimension)



Isometric view of Ground Portals (From Staad)

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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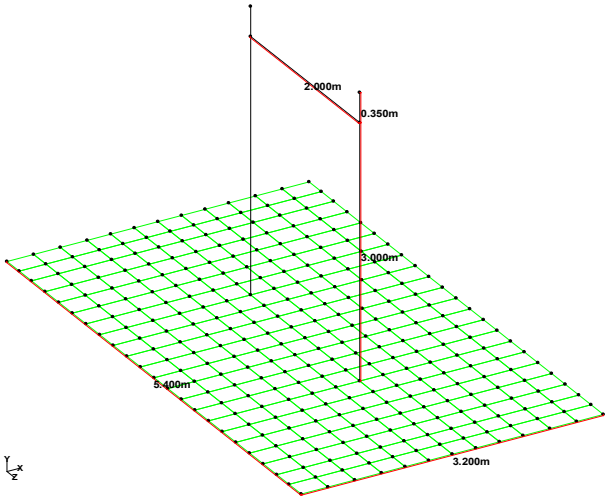
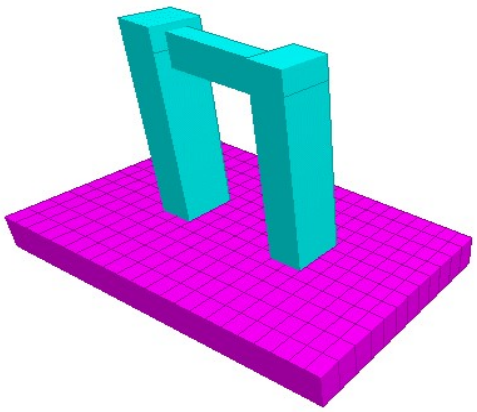
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

6.2 Geometry of TR-1, TR-2 & TR-3 Foundation

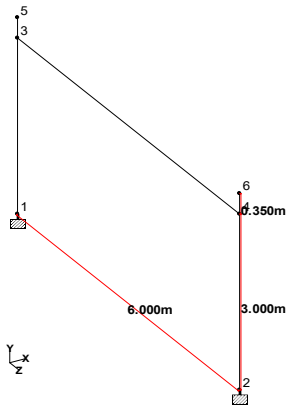


Staad Model of TR1,TR2,TR3 Foundation mkd F2 (Along with dimension)

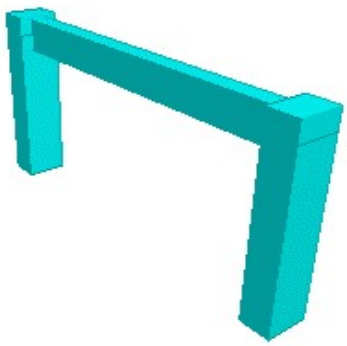


Isometric view of TR1,TR2,TR3 Foundation mkd F2

6.3 Geometry of TR-4 Foundation

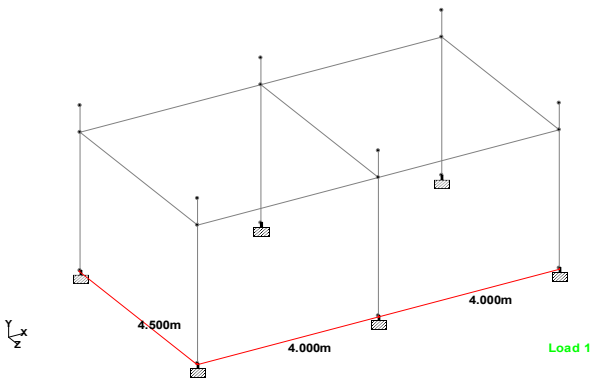


Staad Model of TR4 Foundation framework (Along with dimension)

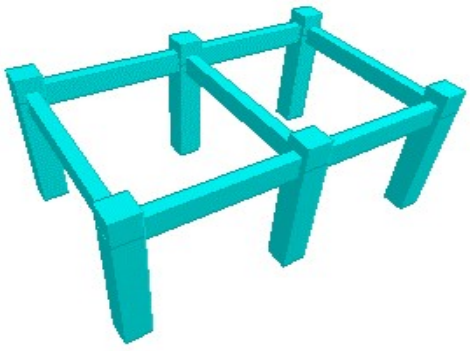


Isometric view of of TR4 Foundation mkd F5

6.4 Geometry of VGTU Foundation



Staad Model of VGTU Foundation (Along with dimension)



Isometric view of VGTU foundation (From Staad)

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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7. DESIGN OF CONCRETE COMPONENTS

All foundations have been designed using worksheet in accordance to SBC-304.

All beams & columns have been designed in Staad-Pro by ACI 318M.

For Design Results, Refer Attachemnt 3 to 8 - Staad Output file.

7.1 Design Summary of Column

7.1.1 Column Reinforcement Summary

Column mark	Width (Along conv)	Depth (Across conv)	Reinforcement	Stirrup	Remarks	Numbers
P1	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP1	2
P2	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP3 & GP4	4
P2A	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP5	2
P3	600	600	12 No.- D20	3 Nos D10 @ 200c/c	TR1 & TR2	4
P4	700	500	12 No.- D20	3 Nos D10 @ 200c/c	TR3 & TR4	4
P4A	700	500	12 No.- D20	3 Nos D10 @ 200c/c	GP2	2
P5	600	600	12 No.- D20	3 Nos D10 @ 200c/c	VGTU Grid C-1, C-2, B-1 & B-2	4
P6	600	600	12 No.- D20	3 Nos D10 @ 200c/c	VGTU Grid A-1 & A-2	2
P7	450	450	8 No.- D20	2 Nos D10 @ 200c/c	VGTU Post	2

7.2 Design Summary of Beam

7.2.1 Beam Reinforcement Summary

Beam mark	Width (mm)	Depth (mm)	Top Reinforcement	Bottom Reinforcement	Stirrup	Numbers
TB1	350	600	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	1
TB2	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	4
TB3	300	500	3 Nos D20	3 Nos D20	2L D10 STRPS @ 200c/c	3
TB4	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	3
TB5	300	600	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	4
TB6	300	600	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	1

7.3 Design of Foundation

7.3.1 Foundation marking, Max/min Base pressure and Reinforcement Summary

Structure Name	Foundation mark	Length (Along conv) (mm)	Width (Across conv) (mm)	Thickness (mm)	Max pressure (kN/m^2)	Top reinforcement	Bottom reinforcement	F.O.S.	Numbers
GP1	F1A	1500	2000	300	69	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.13	2
GP2	**F1**	**1500**	**2000**	**300**	**79**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.15**	**2**
GP3	F1	1500	2000	300	63	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.12	2
GP4	F1	1500	2000	300	63	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.12	2
GP5	F1	1500	2000	300	51	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.10	2
TR1	F2	3200	5400	550	80	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.14	1
TR2	F2	3200	5400	550	107	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.19	1
TR3	**F2**	**3200**	**5400**	**550**	**121**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.22**	**1**
VGTU Grid A-1, A-2	F3	2500	2500	450	106	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.19	2
VGTU Grid C-1, C-2	**F3**	**2500**	**2500**	**450**	**195**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.35**	**2**
TR4	F3	2500	2500	450	157	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.28	2
VGTU Grid B-1 & B-2	**F4**	**3000**	**2500**	**450**	**143**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.26**	**2**

The design of the most critical foundation of the group is attached in the following pages.

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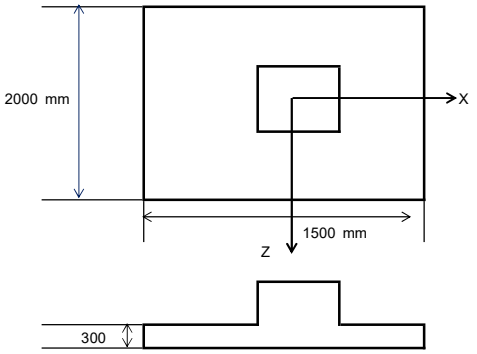
7.3.2 Design of foundation F1 for GP-2
(Refer staad file 3130-CNV-002 ground portal foundation.std)

Finished Ground level	F.G.L	=	0.000	m	Staad support node number	=	79	
Bottom of foundation	B.O.F	=	-1.000	m	Weight of soil	=	33.26 KN	*inputs
Top of footing (staad support)	T.O.F	=	-0.700	m	Weight of foundation	=	21.60 KN	
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN	
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0 KN	
Length of footing	Lz	=	2.00	m	Pressure on top for tension	=	18.3 KN/m^2	
Width of footing	Lx	=	1.50	m				
Thickness of footing	D	=	0.30	m				
Depth of pedestal	Bx	=	0.60	m				
Width of pedestal	Bz	=	0.60	m				
Height of Load point from bottom of footing	H	=	0.150	m				
Section Modulus of Pedestal	Zxx	=	1.00	m^3				
Section Modulus of Pedestal	Zzz	=	0.75	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	518	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	50	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	16	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

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CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	0.6	57.56	13.6	10.42	-0.1	-1.3	112.4	49.64	25.31	1	0.10
302	3.5	90.39	20.2	19.94	1.3	-7.2	145.3	77.90	18.93	1	0.15
303	2.9	90.53	19.6	18.67	0.29	-5.9	145.4	75.01	21.92	1	0.14
304	3.1	83.07	19.7	19.04	1.32	-6.3	137.9	73.40	18.55	1	0.14
305	2.8	81.21	17.8	16.73	1.22	-5.8	136.1	69.86	20.86	1	0.13
306	2.4	83.21	19.0	17.76	0.3	-5.0	138.1	70.50	21.55	1	0.14
307	2.2	81.35	17.2	15.46	0.21	-4.6	136.2	66.98	23.83	1	0.13

Max P_{max} = 77.90 KN/m^2

Max P_{min} = 18.55 KN/m^2

Resultant F.O.S = 0.15

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	0.8	80.6	19.0	14.60	-0.2	-1.83	43.90	9.83
402	5.5	136.5	30.1	30.38	1.86	-11.34	90.98	-0.01
403	4.7	136.7	29.2	28.59	0.45	-9.59	86.92	4.18
404	2.9	87.4	22.1	20.64	1.59	-6.01	57.77	0.47
405	2.4	83.4	18.1	15.73	1.36	-5.02	50.23	5.37
406	2.2	87.5	21.3	19.11	0.38	-4.51	54.29	4.06
407	1.7	83.6	17.3	14.20	0.15	-3.53	46.75	8.96

Max Net factored B.P = 90.98 KN/m^2

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.45 m
Moment about Z: Muz = 9.21 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 399.85 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 100.9 mm^2

Effective Depth Along X = 242 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 435.6 mm^2
A_{req} = 100.9 mm^2
Required Spacing = **461.6** mm

Length of cantilever along Z = 0.7 m
Moment about X: Mux = 22.29 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 348.73 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 262.8 mm^2

Effective Depth Along X = 226 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 406.8 mm^2
A_{req} = 262.8 mm^2
Required Spacing = **494.3** mm

Provide Bottom Reinforcement = **16 mm dia @ 200** c/c
Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 0.45 m
Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 406.49 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Effective Depth Along X = 244 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 219.6 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **515.0** mm

Length of cantilever along Z = 0.7 m
Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 367.49 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Effective Depth Along Z = 232 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 208.8 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **541.7** mm

Provide top Reinforcement = **12 mm dia @ 200 c/c**
Ast prov = 565.5 mm^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	26-03-2020	24-04-2020	01-11-2023	Date	26-03-2020	24-04-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
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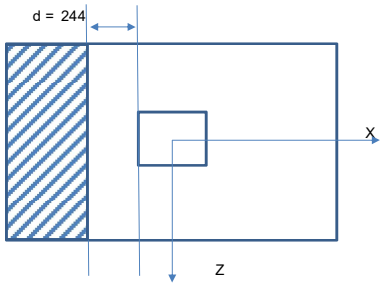
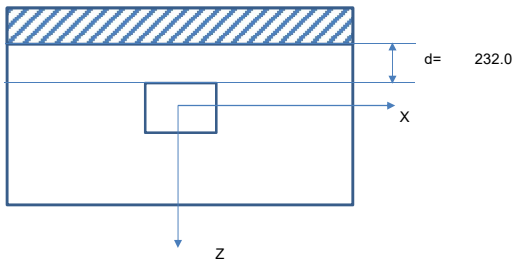
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

ONE WAY SHEAR CKECK



Along X Direction	Cantilever	L	=	0.206	m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0041							*Table 22.5.5.1 SBC 304*
Effective Depth Along X		d	=	244	mm				λ	=	1 Normal concrete
Shear Force at d distance		V_u	=	18.74	KN				Mu	=	1.93 KNm
Allowable shear strength		φVc	=	186.04	KN	O.K			φ	=	0.75 for shear
									φVc	=	203.6 KN
								ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc	=	186.0 KN *22.5.5.1*
								iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
										=	314.0 KN

ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.468	m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0043							*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	232	mm				λ	=	1 Normal concrete
Shear Force at d distance		V_u	=	42.58	KN				Mu	=	9.96 KNm
Allowable shear strength		φVc	=	177.41	KN	O.K			φ	=	0.75 for shear
									φVc	=	177.4 KN
								ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc	=	177.5 KN
								iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
									φVc	=	298.5 KN

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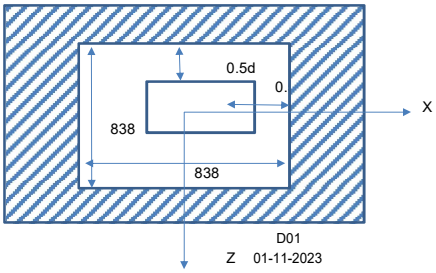
TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



D01
01-11-2023
AS BUILT
RPK
SBY
PRI

					Total area	=	3 m2
					Punching area	=	0.70 m2
					λ	=	1
					β	=	1
					αs	=	40
					b0	=	3352.0 mm
				Allowable shear stress	least of		*Table 22.6.5.2 SBC 304*
				v_c	= $0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm^2
				v_c	= $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm^2
				v_c	= $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.38 N/mm^2

2
01-11-2023
AS BUILT
RPK
SBY
PRI

					Design vc	=	1.95 N/mm^2	
Check at 0.5d from column face								
Perimeter at 0.5d from column face	u	=	3.35 m		Vc	=	$v_c b_0 d$	
Depth at section	d	=	238 mm			=	1557.50 KN	
Shear Force at section	Vu	=	209.0 KN		φ	=	0.75 for shear	
Allowable shear strength	φVc	=	1168.13 KN	O.K	φV_n	=	φVc	
					φVc	=	1168.1 KN	

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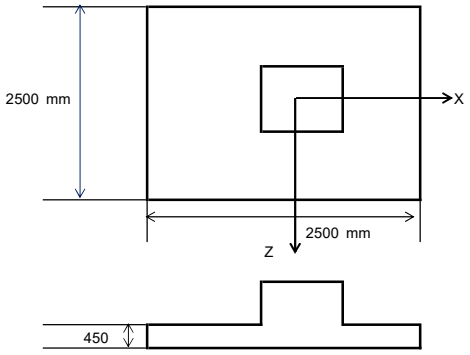
7.3.3 Design of foundation F3 for VGTU grid A
(Refer staad file 3130-CNV-002-VGTU_Fdn.std)

Finished Ground level	F.G.L	=	0.000	m	Staad support node number	=	2	
Bottom of foundation	B.O.F	=	-3.000	m	Weight of soil	=	270.35	KN
Top of footing (staad support)	T.O.F	=	-2.550	m	Weight of foundation	=	67.50	KN
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00	KN
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0	KN
Length of footing	Lz	=	2.50	m	Pressure on top for tension	=	54.1	KN/m^2
Width of footing	Lx	=	2.50	m				
Thickness of footing	D	=	0.45	m				
Depth of pedestal	Bx	=	0.60	m				
Width of pedestal	Bz	=	0.60	m				
Height of Load point from bottom of footing	H	=	0.225	m				
Section Modulus of Pedestal	Zxx	=	2.60	m^3				
Section Modulus of Pedestal	Zzz	=	2.60	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	554	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	50	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	16	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

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CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	-12.9	367.29	1.8	7.23	0.1	35.3	705.1	129.13	96.51	1	0.23
302	-13.2	513.92	-9.2	-5.95	-0.05	36.4	851.8	152.53	120.04	1	0.28
303	-13.9	522.45	8.7	15.67	-0.09	37.4	860.3	158.01	117.29	1	0.29
304	-12.7	584.64	22.9	72.41	0.67	34.6	922.5	188.69	106.50	1	0.34
305	-14.0	376.25	-22.3	-61.47	-0.75	38.3	714.1	152.55	75.96	1	0.28
306	5.0	308.48	1.6	6.76	-0.14	-18.0	646.3	112.93	93.90	1	0.20
307	-31.7	652.44	-1.0	4.13	0.06	90.9	990.3	194.95	121.95	1	0.35
308	-13.0	476.23	0.3	5.52	-0.04	35.3	814.1	145.92	114.59	1	0.26
309	-13.8	484.69	0.2	5.37	-0.04	37.6	822.5	148.13	115.09	1	0.27
310	-13.3	484.38	1.1	7.83	0	36.3	822.2	148.52	114.59	1	0.27
311	-13.4	476.55	-0.5	3.07	-0.08	36.6	814.4	145.52	115.08	1	0.26
312	-6.8	359.28	31.2	93.61	0.98	18.7	697.1	154.67	68.42	1	0.28
313	-8.6	81.42	-29.1	-84.89	-0.91	23.6	419.3	108.73	25.44	1	0.20
314	16.8	-8.94	2.8	6.09	-0.11	-51.5	328.9	74.73	30.52	1	0.13
315	-32.2	449.68	-0.7	2.58	0.17	93.8	787.5	163.00	89.01	1	0.29
316	-7.2	214.73	1.1	4.43	0.04	19.6	552.6	97.63	79.19	1	0.18
317	-8.2	226.01	1.0	4.24	0.02	22.7	563.9	100.57	79.86	1	0.18
318	-7.7	225.59	2.1	7.51	0.1	21.0	563.4	101.10	79.20	1	0.18
319	-7.8	215.15	0.0	1.16	-0.02	21.3	553.0	97.10	79.86	1	0.18

Max P_{max} = 194.95 KN/m^2

Max P_{min} = 25.44 KN/m^2

Resultant F.O.S = 0.35

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	-18.0	514.2	2.5	10.12	0.1	49.36	105.11	59.43
402	-18.7	764.8	-13.4	-9.05	-0.11	51.39	145.57	99.15
403	-19.6	776.7	11.6	21.22	-0.16	52.78	152.69	95.86
404	-14.3	738.5	49.3	150.33	1.52	39.17	190.92	45.38
405	-17.2	293.9	-47.1	-135.27	-1.51	46.98	117.00	-22.96
406	23.4	149.3	3.9	10.30	-0.22	-73.09	55.91	-8.13
407	-54.9	883.1	-1.6	4.68	0.22	159.30	204.26	78.32
408	-15.0	508.1	1.2	7.63	0.01	40.85	99.92	62.68
409	-16.5	524.3	1.0	7.34	-0.01	45.35	104.12	63.64
410	-15.7	523.7	2.7	12.03	0.07	42.90	104.88	62.69
411	-15.8	508.7	-0.4	2.95	-0.07	43.31	99.16	63.63
412	-10.1	552.8	49.8	149.35	1.57	27.80	156.47	20.43
413	-13.0	108.2	-46.6	-136.24	-1.46	35.59	83.30	-48.67
414	27.6	-36.3	4.4	9.31	-0.18	-84.47	30.20	-41.83
415	-50.7	697.5	-1.2	3.71	0.27	147.92	169.82	53.37
416	-10.8	322.5	1.7	6.65	0.05	29.47	65.47	37.73
417	-12.3	338.6	1.5	6.37	0.04	33.98	69.67	38.68
418	-11.5	338.0	3.1	11.04	0.12	31.51	70.43	37.74
419	-11.6	323.1	0.1	1.97	-0.03	31.94	64.72	38.68

Max Net factored B.P = 204.26 KN/m^2

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.95 m
Moment about Z: Muz = 92.17 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 1049.16 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 628.4 mm^2

Effective Depth Along X = 392 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 705.6 mm^2
A_{req} = 628.4 mm^2
Required Spacing = **285.0** mm

Length of cantilever along Z = 0.95 m
Moment about X: Mux = 92.17 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 965.26 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 655.8 mm^2

Effective Depth Along X = 376 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 676.8 mm^2
A_{req} = 655.8 mm^2
Required Spacing = **297.1** mm

Provide Bottom Reinforcement = **16 mm dia @ 200** c/c
Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 0.95 m
Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 1059.89 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Effective Depth Along X = 394 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 354.6 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **318.9** mm

Length of cantilever along Z = 0.95 m
Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 996.31 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Effective Depth Along Z = 382 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 343.8 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **329.0** mm

Provide top Reinforcement = **12 mm dia @ 200 c/c**
Ast prov = 565.5 mm^2

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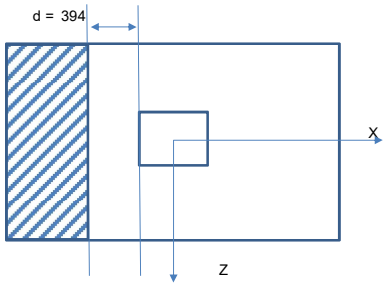
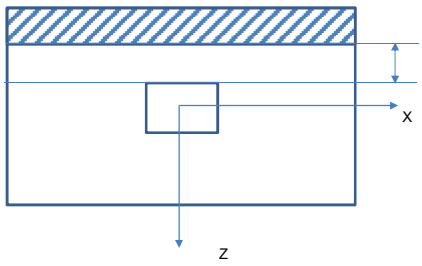


Along X Direction	Cantilever	L	=	0.556	m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0026							*Table 22.5.5.1 SBC 304*
Effective Depth Along X		d	=	394	mm				λ	=	1 Normal concrete
Shear Force at d distance		V_u	=	113.57	KN				Mu	=	31.57 KNm
Allowable shear strength		φVc	=	292.53	KN	O.K			φ	=	0.75 for shear
									φVc	=	297.9 KN
								ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc	=	292.5 KN *Table 22.5.5.1 SBC 304*
								iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
										=	507.0 KN

ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.568	m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0026							*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	382	mm				λ	=	1 Normal concrete
Shear Force at d distance		V_u	=	116.02	KN				Mu	=	32.95 KNm
Allowable shear strength		φVc	=	284.01	KN	O.K			φ	=	0.75 for shear
									φVc	=	288.4 KN
								ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc	=	284.0 KN
								iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
									φVc	=	491.5 KN

Global Project ID	**900707**
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
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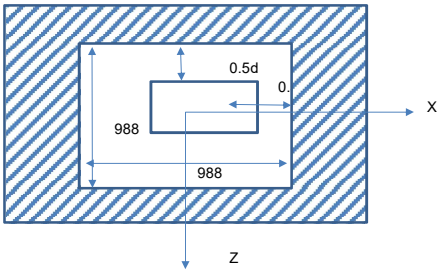
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



	Total area	=	6.25	m2
	Punching area	=	0.98	m2
	λ	=	1	
	β	=	1	
	αs	=	40	
	b0	=	3952.0	mm
Allowable shear stress	least of		*Table 22.6.5.2 SBC 304*	
v_c	= $0.33\lambda\sqrt{f_c'}$	=	1.95	N/mm^2
v_c	= $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02	N/mm^2
v_c	= $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.91	N/mm^2
Design vc	=	1.95	N/mm^2	

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.95	m		Vc	=	$v_c b_0 d$
Depth at section	d	=	388	mm			=	2993.62 KN
Shear Force at section	Vu	=	1077.3	KN		φ	=	0.75 for shear
Allowable shear strength	φVc	=	2245.21	KN	O.K	φV_n	=	φVc
						φVc	=	2245.2 KN

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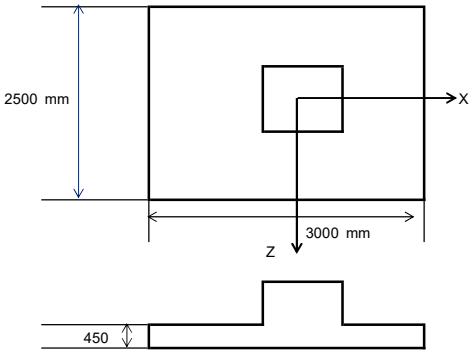
7.3.4 Design of foundation F4 VGTU grid B
(Refer staad file 3130-CNV-002-VGTU_Fdn.std)

Finished Ground level	F.G.L	=	0.000	m	Staad support node number	=	4
Bottom of foundation	B.O.F	=	-3.000	m	Weight of soil	=	327.73 KN
Top of footing (staad support)	T.O.F	=	-2.550	m	Weight of foundation	=	81.00 KN
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0 KN
Length of footing	Lz	=	2.50	m	Pressure on top for tension	=	54.5 KN/m^2
Width of footing	Lx	=	3.00	m			
Thickness of footing	D	=	0.45	m			
Depth of pedestal	Bx	=	0.60	m			
Width of pedestal	Bz	=	0.60	m			
Height of Load point from bottom of footing	H	=	0.225	m			
Section Modulus of Pedestal	Zxx	=	3.13	m^3			
Section Modulus of Pedestal	Zzz	=	3.75	m^3			
Allowable Net Bearing Capacity		=	500	KN/m^2			
Gross Bearing Capacity		=	554	KN/m^2			
Grade of concrete		=	35	N/mm^2			
Grade of steel		=	420	N/mm^2			
Clear cover		=	50	mm			
Top Reinfocement		=	12	mm			
Bottom Reinforcement		=	16	mm			
Co-efficient of friction between soil & concrete		=	0.3				
Unit wt of soil		=	18	KN/m^3			
Unit wt of concrete		=	24	KN/m^3			

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

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CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	-17.3	177.67	4.4	10.02	0.4	40.5	586.4	92.20	64.17	1	0.17
302	-17.8	375.83	4.6	11.85	0.51	41.7	784.6	119.52	89.69	1	0.22
303	-17.7	365.04	4.3	9.62	0.28	41.9	773.8	117.43	88.91	1	0.21
304	-16.4	424.21	22.8	65.24	2.25	39.0	832.9	142.33	79.79	1	0.26
305	-18.8	220.3	-14.0	-44.14	-1.48	44.0	629.0	109.72	58.02	1	0.20
306	11.5	479.25	4.7	11.29	0.37	-26.0	888.0	128.95	107.84	1	0.23
307	-46.7	165.24	4.2	9.83	0.4	109.0	574.0	108.74	44.32	1	0.20
308	-17.0	326.13	4.4	10.55	0.39	40.0	734.9	112.03	83.93	1	0.20
309	-18.2	318.36	4.4	10.57	0.38	43.0	727.1	111.79	82.10	1	0.20
310	-17.5	326.05	5.0	12.32	0.47	41.3	734.8	112.94	83.00	1	0.20
311	-17.7	318.44	3.9	8.79	0.29	41.6	727.2	110.87	83.04	1	0.20
312	-8.8	242.55	27.2	78.93	2.7	21.0	651.3	117.68	55.99	1	0.21
313	-12.0	-29.32	-21.9	-66.93	-2.28	27.6	379.4	79.37	21.80	1	0.14
314	28.4	315.94	3.0	6.99	0.19	-65.7	724.7	116.38	76.87	1	0.21
315	-49.1	-102.74	2.3	5.03	0.23	114.3	306.0	72.90	8.70	1	0.13
316	-9.5	111.78	2.6	6.00	0.21	22.3	520.5	77.28	61.52	1	0.14
317	-11.2	101.42	2.7	6.03	0.2	26.3	510.1	76.96	59.08	1	0.14
318	-10.3	111.68	3.4	8.36	0.3	24.1	520.4	78.50	60.28	1	0.14
319	-10.5	101.52	1.9	3.66	0.09	24.5	510.2	75.74	60.33	1	0.14

Max P_{max} = 142.33 KN/m^2

Max P_{min} = 8.70 KN/m^2

Resultant F.O.S = 0.26

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	-24.2	248.7	6.2	14.03	0.5	56.75	52.79	13.54
402	-25.0	584.0	6.4	16.80	0.73	58.77	98.91	56.82
403	-24.9	568.9	6.1	13.69	0.4	59.07	95.98	55.72
404	-18.4	527.1	44.6	129.04	4.43	43.92	123.29	17.28
405	-23.5	92.1	-34.0	-104.32	-3.54	54.56	60.21	-35.65
406	41.0	644.5	5.8	13.95	0.41	-94.75	115.67	56.21
407	-82.9	-25.4	4.8	10.82	0.47	193.31	51.63	-58.39
408	-19.7	317.0	5.3	12.36	0.45	46.47	58.61	25.91
409	-22.2	302.2	5.3	12.40	0.43	52.09	58.15	22.43
410	-20.8	316.8	6.4	15.75	0.62	49.00	60.35	24.14
411	-21.1	302.3	4.2	9.02	0.27	49.56	56.41	24.21
412	-13.0	377.4	43.2	125.67	4.3	31.12	98.84	1.81
413	-18.1	-57.6	-35.3	-107.68	-3.66	41.77	37.92	-53.27
414	46.4	494.9	4.5	10.59	0.28	-107.55	98.05	33.91
415	-77.5	-175.0	3.5	7.45	0.35	180.51	27.18	-73.86
416	-14.3	167.3	4.0	9.00	0.32	33.67	34.17	10.45
417	-16.8	152.5	4.0	9.04	0.31	39.29	33.70	6.96
418	-15.4	167.2	5.1	12.39	0.49	36.21	35.91	8.67
419	-15.7	152.7	2.8	5.65	0.14	36.76	31.96	8.74

Max Net factored B.P = 123.29 KN/m^2

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.2 m

Moment about Z: Muz = 88.77 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1049.16 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 604.9 mm^2

Effective Depth Along X = 392 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

greater of

or $A_{stmin} = 0.0014bd$

A_{stmin} = 705.6 mm^2

A_{req} = 604.9 mm^2

Required Spacing = **285.0** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 55.63 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 965.26 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 393.8 mm^2

Effective Depth Along X = 376 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement $A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *24.4.3.2 SBC 304*

greater of

or $A_{stmin} = 0.0014bd$

A_{stmin} = 676.8 mm^2

A_{req} = 393.8 mm^2

Required Spacing = **297.1** mm

Provide Bottom Reinforcement = **16 mm dia @ 200** c/c

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 1.2 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 1059.89 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 394 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o

greater of

or A_{stmin} = 50% of 0.0014bd more prov ir bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 354.6 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **318.9** mm

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 996.31 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 382 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o

greater of

or A_{stmin} = 50% of 0.0014bd more prov ir bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 343.8 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **329.0** mm

Provide top Reinforcement = **12 mm dia @ 200 c/c**

Ast prov = 565.5 mm^2

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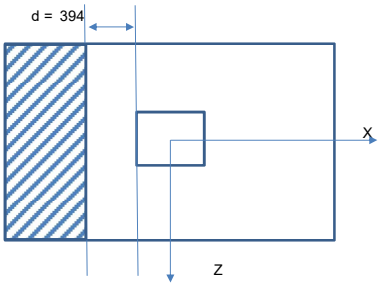
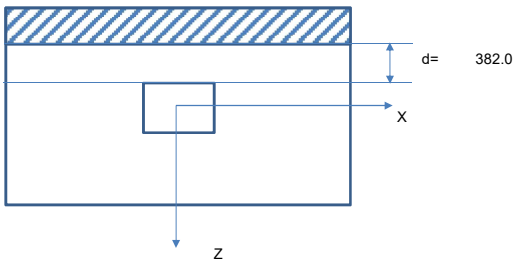
ONE WAY SHEAR CKECK



Along X Direction	Cantilever	L	=	0.806	m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0026							*Table 22.5.5.1 SBC 304*
Effective Depth Along X		d	=	394	mm				λ	=	1 Normal concrete
Shear Force at d distance		V_u	=	99.37	KN				Mu	=	40.05 KNm
Allowable shear strength		φVc	=	292.24	KN	O.K			φ	=	0.75 for shear
									φVc	=	292.2 KN
								ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc	=	292.5 KN *Table 22.5.5.1 SBC 304*
								iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
										=	507.0 KN

ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.568	m		Least of	i)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0026							*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	382	mm				λ	=	1 Normal concrete
Shear Force at d distance		V_u	=	70.03	KN				Mu	=	19.89 KNm
Allowable shear strength		φVc	=	284.01	KN	O.K			φ	=	0.75 for shear
									φVc	=	288.4 KN
								ii)	V_c	=	$\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc	=	284.0 KN
								iii)	V_c	=	$0.29\lambda\sqrt{f_c'}b_w d$
									φVc	=	491.5 KN

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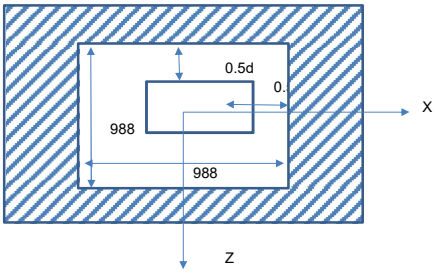
TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



	Total area	=	7.5	m2	
	Punching area	=	0.98	m2	
	λ	=	1		
	β	=	1		
	αs	=	40		
	b0	=	3952.0	mm	
Allowable shear stress	least of				*Table 22.6.5.2 SBC 304*
v_c	= $0.33\lambda\sqrt{f_c'}$	=	1.95	N/mm^2	
v_c	= $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02	N/mm^2	
v_c	= $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.91	N/mm^2	
Design vc	=	1.95	N/mm^2		

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.95	m		Vc	=	$v_c b_0 d$	
Depth at section	d	=	388	mm			=	2993.62	KN
Shear Force at section	Vu	=	804.3	KN		φ	=	0.75	for shear
Allowable shear strength	φVc	=	2245.21	KN	O.K	φV_n	=	φVc	
						φVc	=	2245.2	KN

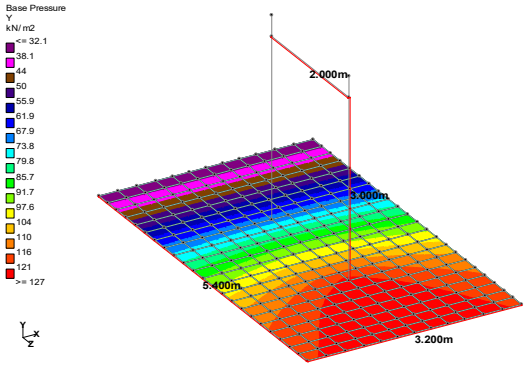
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	26-03-2020	24-04-2020	01-11-2023	Date	26-03-2020	24-04-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK	Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

7.3.5 Design of combined footing for Trestle TR3 mkd. as F2

For Calculation of Base pressure and design forces for column & foundation Raft, Refer STAAD File **3130-CNV-002-TR2_Fdn.STD**

Staad Supoort Nodes are 5 & 6



Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +913.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	-0.15	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.200	m	
Width of footing	Lx	=	5.400	m	
Thickness of footing	D	=	0.55	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	26.1	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	127.4	KN/m^2	Load Case 304
Minimum Base Pressur		=	-11.7	KN/m^2	Load Case 312

2) Design of foundation Raft :

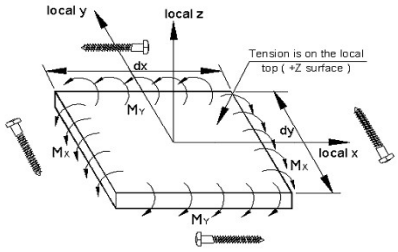
Slab Thickness =	550	mm	Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Width =	1000	mm	Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Clear Cover =	75	mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Compressive Strength of Concrete , fc' =	35	N/mm^2	trength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	38.430	kN-m	Corresponding Mxy	0.490	kN-m	*(Plate No. - 115 , L/C No. - 412)*
Min. Mx	-84.120	kN-m	Corresponding Mxy	13.830	kN-m	*(Plate No. - 24 , L/C No. - 404)*
Max. My	92.800	kN-m	Corresponding Mxy	1.890	kN-m	*(Plate No. - 114 , L/C No. - 412)*
Min. My	-144.140	kN-m	Corresponding Mxy	4.890	kN-m	*(Plate No. - 25 , L/C No. - 404)*

FOR TOP FACE REINFORCEMRNT :



Mx1 =	Mx + abs(Mxy) =	38.43 + 0.49 =	38.920 kN-m
My1 =	My + abs(Mxy) =	92.8 + 1.89 =	94.690 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	38.43 + 0.003 =	38.433 kN-m
My2 =	My+abs(Mxy^2/Mx) =	92.8 + 0.062 =	92.862 kN-m

DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	38.92 kN-m
Myd =	94.69 kN-m

ALONG LOCAL X - DIRECTION :

Mu =	38.92	kN-m						
Φ*Kn =	Mu/bd^2					deff. =	469	mm
Mu/bd^2 =	38.92X10^6/(1000X469^2) =		0.177	N/mm^2		<	6.9	N/mm^2
Pt =	0.0005		Reqd. Ast =		0.0005X1000X469/100 =		2.35	cm^2
Provided	12	mm @	200	C/C	(Provide Ast =		5.65	)cm^2
					Total Provide Ast =		5.65	cm^2
					Total Provide Pt =		0.0012	

O.K.

ALONG LOCAL Y - DIRECTION :

Mu =	94.69	kN-m						
Φ*Kn =	Mu/bd^2					deff. =	457	mm
Mu/bd^2 =	94.69X10^6/(1000X457^2) =		0.43	N/mm^2		<	6.9	N/mm^2
Pt =	0.0011		Reqd. Ast =		0.0011X1000X469/100 =		5.03	cm^2
Provided	12	mm @	200	C/C	(Provide Ast =		5.65	)cm^2
					Total Provide Ast =		5.65	cm^2
					Total Provide Pt =		0.0012	

O.K.

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -84.12 - 13.83 = -97.95 kN-m
My1 = My - abs(Mxy) = -144.14 - 4.89 = -149.03 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -84.12 - 1.327 = -85.447 kN-m
My2 = My-abs(Mxy^2/Mx) = -144.14 - 0.284 = -143.856 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -97.95 kN-m
Myd = -149.03 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 97.95 kN-m
Φ*Kn = Mu/bd^2 deff. = 467 mm
Mu/bd^2 = 97.95X10^6/(1000X467^2) = 0.449 N/mm^2 < 6.9 N/mm^2
Pt = 0.0012 Reqd. Ast = 0.0012X1000X467/100 = 5.6 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0022

ALONG LOCAL Y - DIRECTION :

Mu = 149.03 kN-m
Φ*Kn = Mu/bd^2 deff. = 451 mm
Mu/bd^2 = 149.03X10^6/(1000X451^2) = 0.733 N/mm^2 < 6.9 N/mm^2
Pt = 0.0020 Reqd. Ast = 0.002X1000X451/100 = 9.02 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0022

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.2 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 288 , L/C No. - 404)*
Vu = 93.40 kN

Provided Pt = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X93.4X10^3X467/97.95/10^6)X1000X467/10^3
= 449.827 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X467/10^3
= 459.515 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X467/10^3
= 801.215 kN
HENCE, Vc = 449.827 kN
Φt*Vc = 0.75X449.827 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 337.37 kN > 93.4T , Hence O.K.

ALONG LOCAL Y - DIRECTION :

SQy = -0.29 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 169 , L/C No. - 404)*
Vu = 130.79 kN

Pt provided = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X130.79X10^3X451/149.03/10^3)X1000X451/10^3
= 433.58 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X451/10^3
= 443.772 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X451/10^3
= 773.764 kN
HENCE, Vc = 433.58 kN
Φt*Vc = 0.75X433.58 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 325.185 kN > 130.79T , Hence O.K.

Global Project ID	**900707**
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	26-03-2020	24-04-2020	01-11-2023				Date	26-03-2020	24-04-2020	01-11-2023		
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Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 1 **MECHANICAL GENERAL ARRANGEMENT DRAWINGS**

DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1005
REV: A01

ABBREVIATION

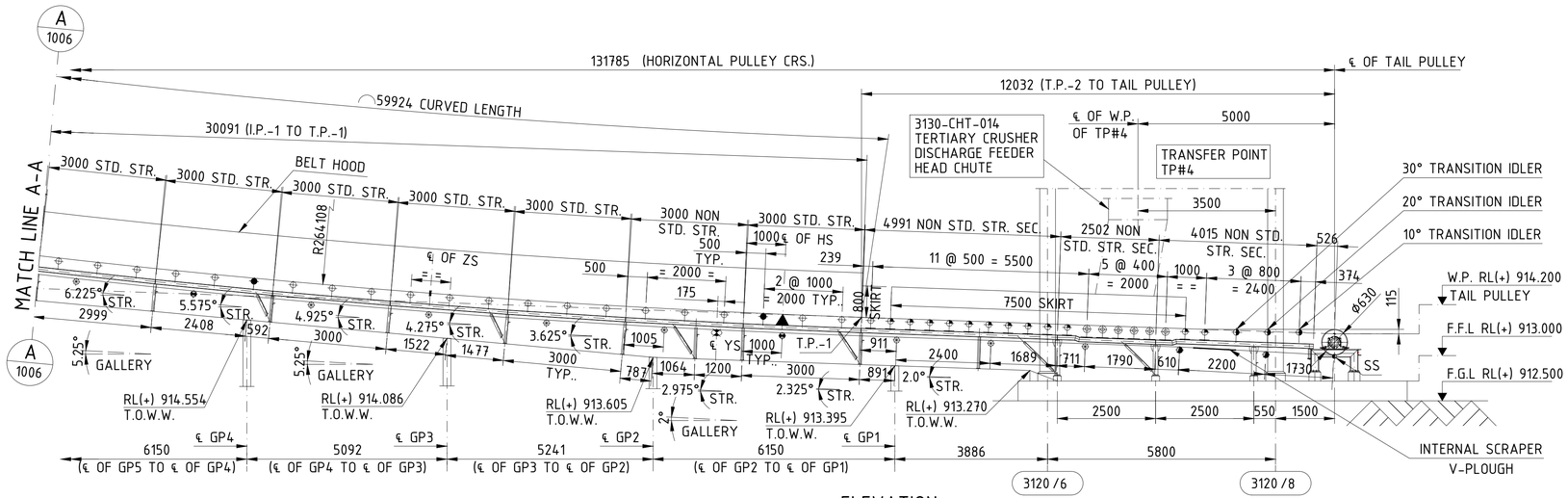
CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
PROJ.:	PROJECTION	BRG.:	BEARING
OPNG.:	OPENING		

CONVEYOR CHARACTERISTICS

	PARTICULARS	SPECIFICATION	QUANTITY
MISCELLANEOUS EQUIPMENT	TAKE-UP / TRAVEL	VGTU. / 4000 mm	
	TAKE-UP LOAD	~4890 kg.(APPROXIMATELY) RCC BLOCKS, 40 kg EACH	
	SCRAPER AT DISCHARGE PULLEY	EXTERNAL / MAIN & PRE CLEANER	1 NO. + 1 NO.
	SCRAPER AT TAKE-UP	INTERNAL V-PLOUGH	1 NO.
	SCRAPER AT TAIL PULLEY	INTERNAL V-PLOUGH	1 NO.
	BELT SWAY SWITCH (ZS)	@ 50 m INTERVAL	3 PAIRS
	PULL CORD SWITCH (HS)	@ 30 m INTERVAL	5 PAIRS
	ZERO SPEED SWITCH (SS)	AT TAIL PULLEY	1 NO.
	BELT RIP SWITCH (YS)	AFTER FEEDING ZONE	1 NO.
	DECK PLATE	3.15 mm THK. CONTINUOUS	1 LOT.
	BELT HOOD	GALVANIZED CORRUGATED SHEET	1 LOT.

CONVEYOR CHARACTERISTICS

	PARTICULARS		SPECIFICATION	QUANTITY
CONVEYOR	EQUIPMENT TAG NO.		3130-CNV-002	1 NO.
	EQUIPMENT DESCRIPTION		TERTIARY SCREEN FEED CONVEYOR	
	LOCATION		TP#4 TO SCREEN HOUSE	
	CAPACITY RATED / DESIGN	(t/h)	782 / 870	
	BELT WIDTH	(mm)	1200	
	BELT SPEED	(m/sec)	1.673	
	TRIPPER FIXED / MOVABLE		N.A.	
	CONVEYOR LENGTH / LIFT	(mm)	131785 / 25500	
MATERIAL	MATERIAL		GOLD ORE : FRESH ORE / OXIDE ORE	
	BULK DENSITY	(t/m³)	1.6	
	MAX. LUMP SIZE	(mm)	(-)21.5	
	MOISTURE CONTENT	(%)	2	
	SURCHARGE ANGLE	(deg.)	20	
BELTING	MAX. RUNNING TENSION	(T1 kN)	82.4	
	SLACK SIDE TENSION	(T2 kN)	31.358	
	TAIL TENSION	(T0 kN)	31.805	
	BELT DESIGNATION		EP-1000 / 4 (HEAVY DUTY)	
	TOP / BOTTOM COVER THICKNESS	(mm)	6 / 3	
	COVER GRADE / FABRIC		DIN-X / EP	
PULLEYS	DISCHARGE WITH DRIVE	(MKD. x DIA.)	D-5 x 800 mm	1 NO.
	DISCHARGE WITHOUT DRIVE	(MKD. x DIA.)	N.A.	
	SNUB	(MKD. x DIA.)	ND-7 x 500 mm	1 NO.
	TAIL	(MKD. x DIA.)	ND-6 x 630 mm	1 NO.
	TAKE-UP	(MKD. x DIA.)	ND-6 x 630 mm	1 NO.
	DEFLECTOR	(MKD. x DIA.)	DR-3 x 273 mm	1 NO.
	TRIPPER DISCHARGE	(MKD. x DIA.)	N.A.	
	TRIPPER BEND	(MKD. x DIA.)	N.A.	
	BEND	(MKD. x DIA.)	ND-6 x 630 mm	2 NOS.
	LAGGING	(mm)	DRIVE: 16thk CERAMIC	
			NON-DRIVE: 16thk PLAIN RUBBER	
BEARING/SLEEVE PLUMMER BLOCK	DISCHARGE WITH DRIVE		TBA	2 SETS
	DISCHARGE WITHOUT DRIVE		N.A.	
	SNUB		TBA	2 SETS
	TAIL		TBA	2 SETS
	TAKE-UP		TBA	2 SETS
	DEFLECTOR		TBA	2 SETS
	BEND		TBA	4 SETS
	TRIPPER DISCHARGE		N.A.	
	TRIPPER BEND		N.A.	
IDLER	TROUGHING	(DIA. x deg.)	152.4 x 35°	113 NOS.
	TROUGHING RETRACTABLE	(DIA. x deg.)	152.4 x 45°	12 NOS.
	TROUGHING TRAINER	(DIA. x deg.)	152.4 x 35°	9 NOS.
	RETURN	(DIA. x deg.)	152.4 x 10°	32 NOS.
	SELF ALIGNING RETURN TRAINER INVERTED IDLER	(DIA. x deg.)	152.4 x 10°	6 NOS.
	IMPACT RETRACTABLE	(DIA. x deg.)	190 x 45°	6 NOS.
	TRANSITION (ADJUSTABLE)	(DIA. x deg.)	152.4 x 10°/20°/30°	(1+2+2) NOS.
	RETURN SELF CLEANING	(DIA. x deg.)	190 x 0°	2 NOS.
	NON MAGNETIC TROUGHING	(DIA. x deg.)	N.A.	
	WEIGHING IDLER	(DIA. x deg.)	N.A.	
	SPECIAL MACHINED IDLER	(DIA. x deg.)	N.A.	
	RETURN FLAT IDLER	(DIA. x deg.)	152.4 x 0°	4 NOS.
DRIVE	MOTOR: RATING	kW/rpm (SYN)	132 / 1800	1 NO.
	MOTOR: FRAME SIZE		TBA	
	MOTOR: TYPE		SQUIRREL CAGE	
	MOTOR: SUPPLY		4.16kV, 60Hz	
	REDUCER: TYPE		BEVEL HELICAL	1 NO.
	REDUCER: MODEL NO./COOLING/RATIO		TBA	
	REDUCER: RATIO		45	
	REDUCER: HOLD BACK		YES	
	BRAKE: TYPE		ELECTRO HYDRAULIC THRUSTER	1 NO.
	BRAKE: MODEL NO.		TBA	
	BRAKE: DRAM DIAMETER		TBA	
	COUPLING: HIGH SPEED: TYPE		FLUID COUPLING	1 NO.
	COUPLING: HIGH SPEED: MODEL NO.		TBA	
	COUPLING: LOW SPEED: TYPE		GEAR	1 NO.
	COUPLING: LOW SPEED: MODEL NO.		TBA	



ELEVATION
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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-ELV-1107	CONV.3130-CNV-002-ELEVATION								
MD-216-3130-EG-ME-ELV-1006	CONV.3130-CNV-002-ELEVATION								
MD-216-3130-EG-ME-LPX-1003	CONV.3130-CNV-002-PLAN	A01	06/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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Saudi Arabian Mining Company شركة معادن العربية السعودية
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LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_100_H_G_16114_002			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707__GAU01__ZC01__MTL01__31338		OU100669079	0
DRAWN	HKD	03/02/2020	
CHECKED	BRD	04/02/2020	
ENGINEER	BD	04/02/2020	
APPROVED	BGR	06/02/2020	
EMPLOYER			
SCALE	1:50	A1	

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
CONV. 3130-CNV-002 - ELEVATION

DRAWING NUMBER	REV
MD-216-3130-EG-ME-ELV-1005	A01

DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1006 | REV: A01

TABLE-2 : LOAD FOR TAKE UP BEND FRAME

FOR CONV. 3130-CNV-002

POINTS	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD	
		ALONG (IN "t")	ACROSS (IN "t")
F2	(+) 4.00	(+) 1.50	(±) 0.40
F3	(+) 4.00	(-) 1.50	(±) 0.40

SIGN CONVENTION

UPWARD (-)
DOWNWARD (+)
TOWARDS DRIVE PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

Note: Wind load to be considered during civil & structural design

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W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
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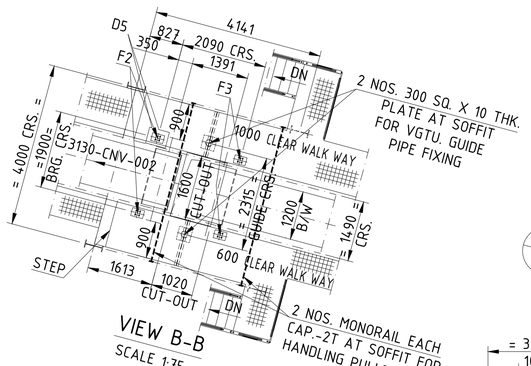

VIEW B-B
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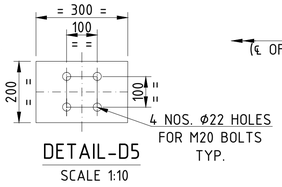

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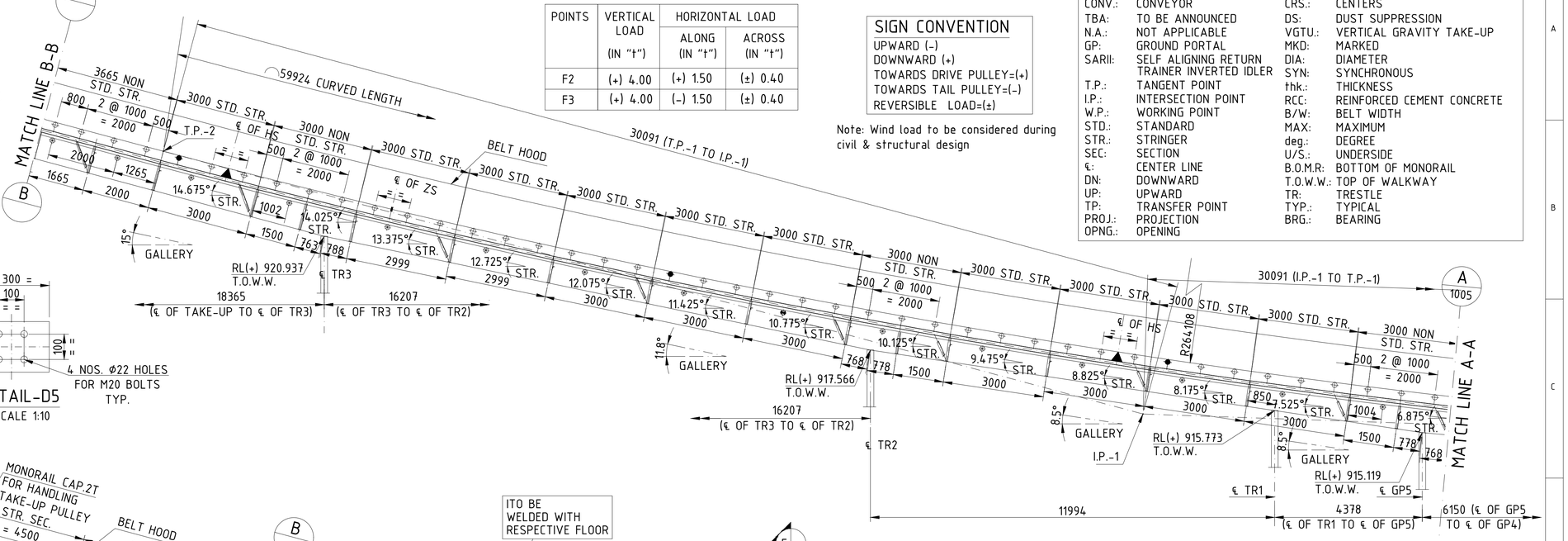
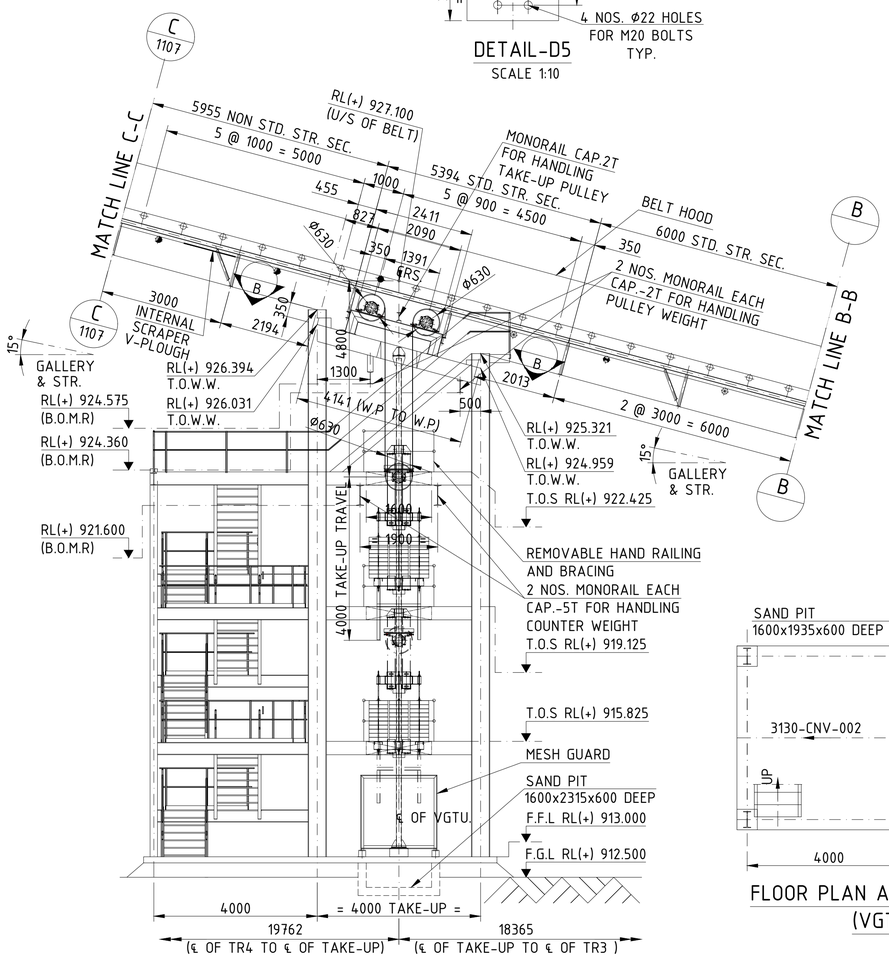
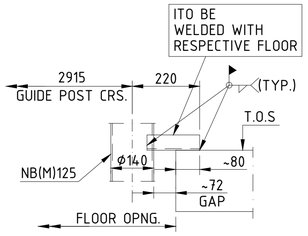
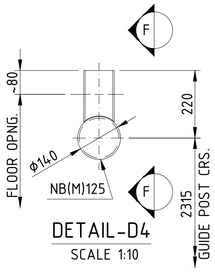


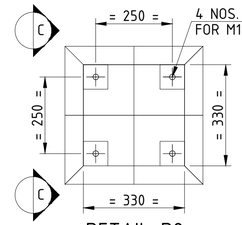




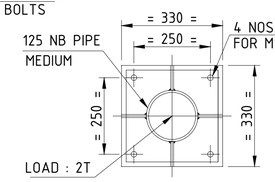
VIEW F-F
SCALE 1:10



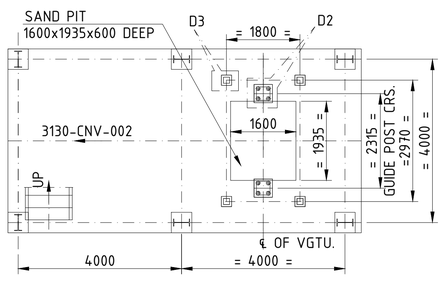
DETAIL-D4
SCALE 1:10



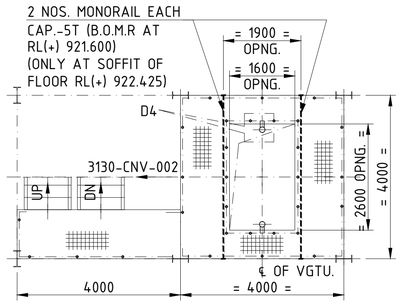
DETAIL-D2
SCALE 1:10



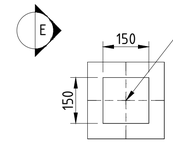
VIEW D-D
SCALE 1:10



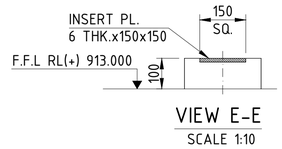
FLOOR PLAN AT RL(+) 913.000 (F.F.L)
(VGTU. TOWER)
SCALE 1:75



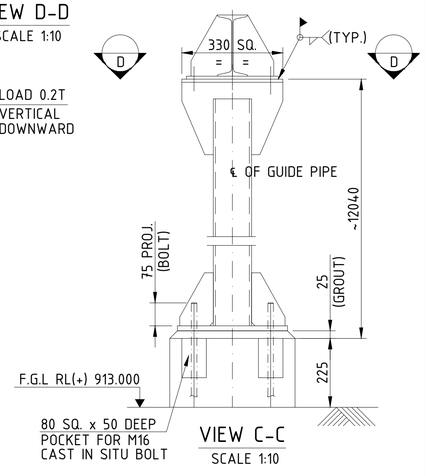
FLOOR PLAN AT RL(+) 922.425 T.O.S
FLOOR PLAN AT RL(+) 919.125 T.O.S
FLOOR PLAN AT RL(+) 915.825 T.O.S
(VGTU. TOWER)
SCALE 1:75



DETAIL-D3
SCALE 1:10



VIEW E-E
SCALE 1:10



VIEW C-C
SCALE 1:10

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-ELV-1107	CONV.3130-CNV-002-ELEVATION								
MD-216-3130-EG-ME-ELV-1005	CONV.3130-CNV-002-ELEVATION								
MD-216-3130-EG-ME-LPX-1003	CONV.3130-CNV-002-PLAN	A01	06/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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LARSEN & TOUBRO DRAWING NUMBER: 019045_100_H_G_16114_003 | REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31339 | DOCUMENT ID: OU100669080 | REVISION: 0

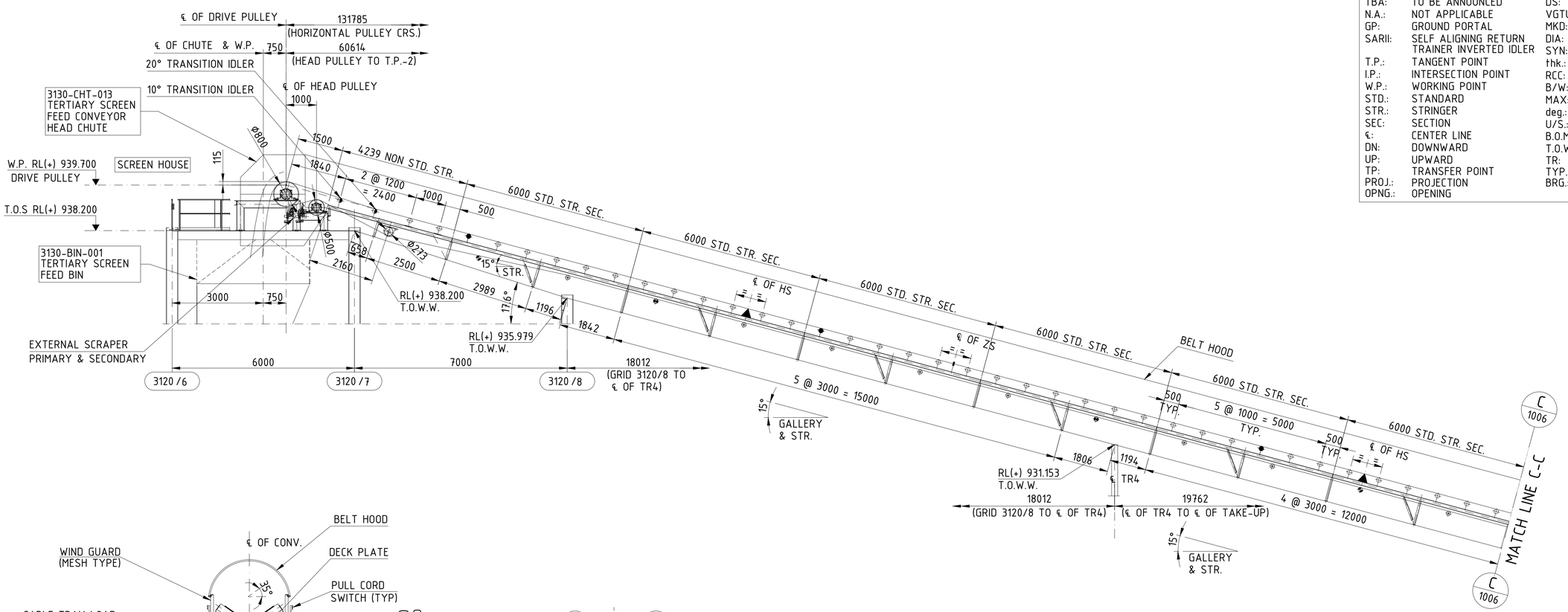
DRAWN	HKD	03/02/2020
CHECKED	BRD	04/02/2020
ENGINEER	BD	04/02/2020
APPROVED	BGR	06/02/2020
EMPLOYER		
SCALE	1:50	A1

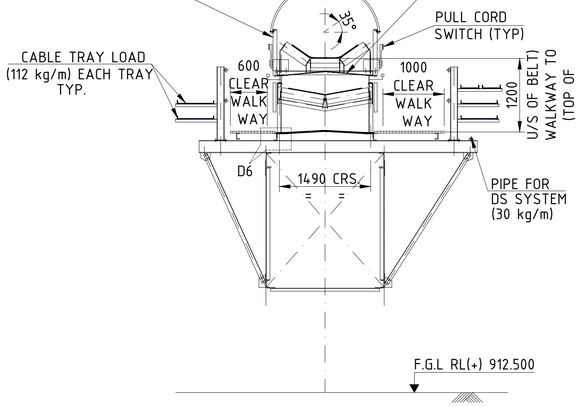
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
CONV. 3130-CNV-002 - ELEVATION

DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1006 | REV: A01

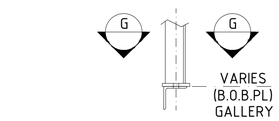
DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1107 | REV: A01

ABBREVIATION			
CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
PROJ.:	PROJECTION	BRG.:	BEARING
OPNG.:	OPENING		

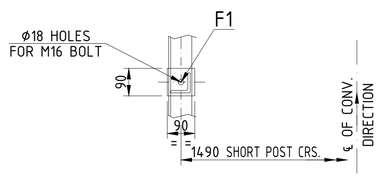




TYPICAL CONVEYOR SECTION
ON GALLERY
(1200 B/W)
SCALE 1:50



DETAIL-D6
SCALE 1:10



VIEW G-G
SCALE 1:10

TABLE-1 : LOAD FOR CONVEYOR SHORT POST

FOR CONV. 3130-CNV-002

POINT	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
	DEAD LOAD	LIVE LOAD	ALONG	ACROSS
F1	(+)0.65	(+)0.80	(±)0.15	(±)0.08

SIGN CONVENTION
UPWARD (-)
DOWNWARD (+)
TOWARDS DRIVE PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

Note: Wind load to be considered during civil & structural design

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-ELV-1006	CONV.3130-CNV-002-ELEVATION								
MD-216-3130-EG-ME-ELV-1005	CONV.3130-CNV-002-ELEVATION								
MD-216-3130-EG-ME-LPX-1003	CONV.3130-CNV-002-PLAN	A01	06/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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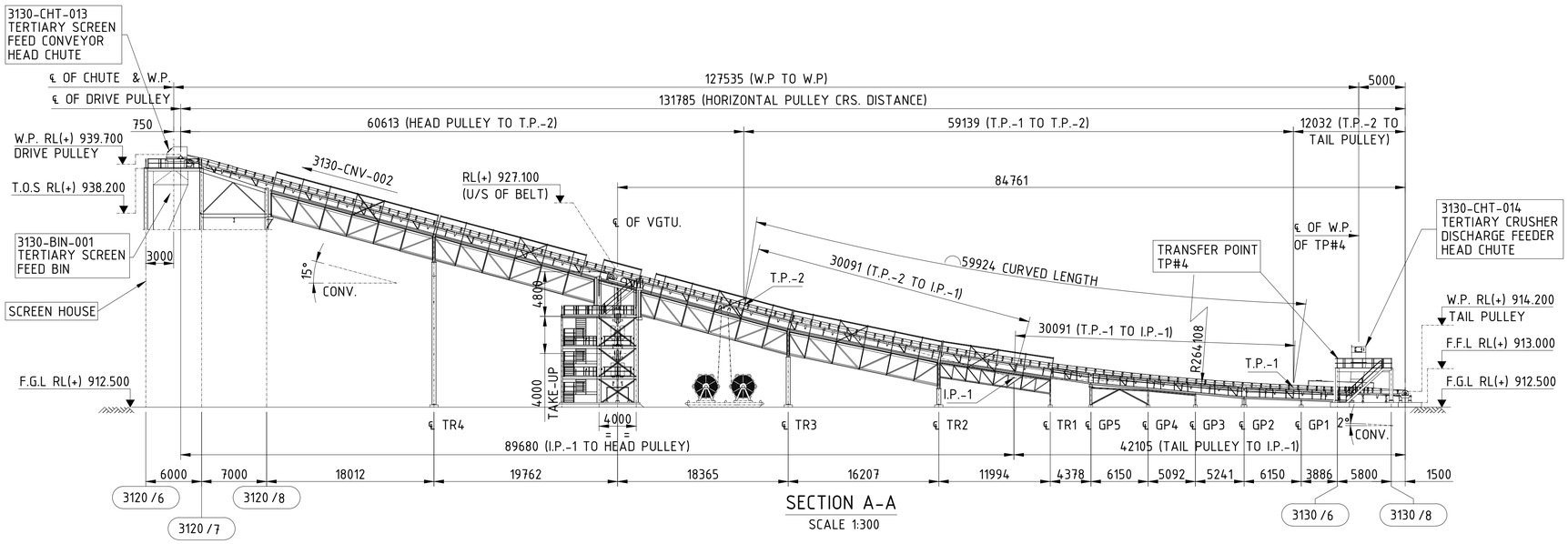
LARSEN & TOUBRO DRAWING NUMBER	019045_100_H_G_16114_004		REVISION 0
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31343	DOCUMENT ID OU100669106	REVISION 0

DRAWN	HKD	03/02/2020
CHECKED	BRD	04/02/2020
ENGINEER	BD	04/02/2020
APPROVED	BGR	06/02/2020
EMPLOYER		
SCALE	1:50	A1

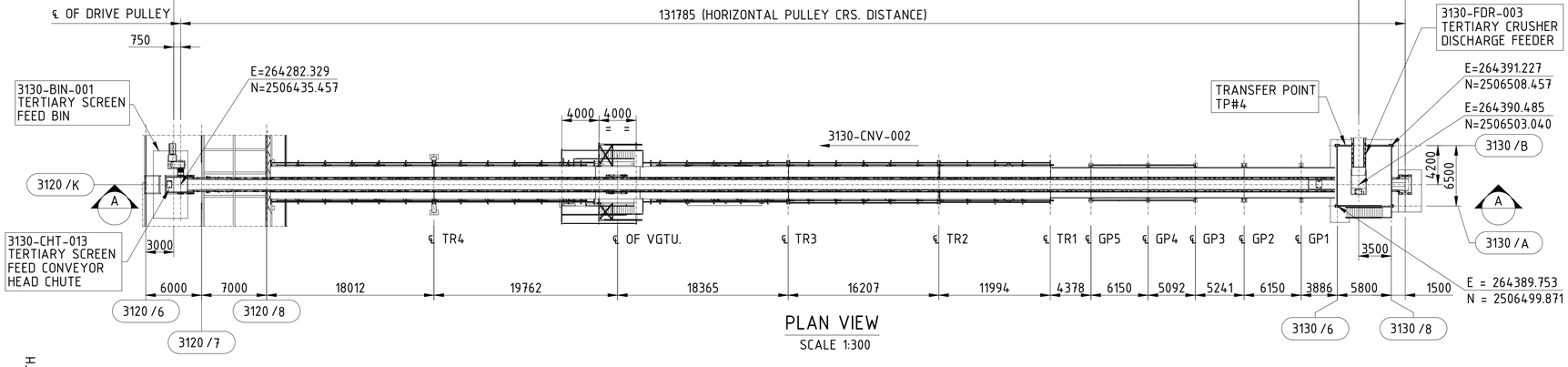
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
CONV. 3130-CNV-002 - ELEVATION

DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1107 | REV: A01

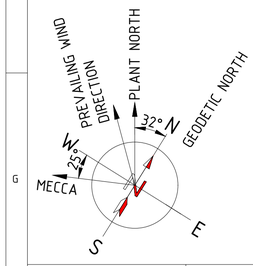
DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1003
REV: A01



SECTION A-A
SCALE 1:300



PLAN VIEW
SCALE 1:300



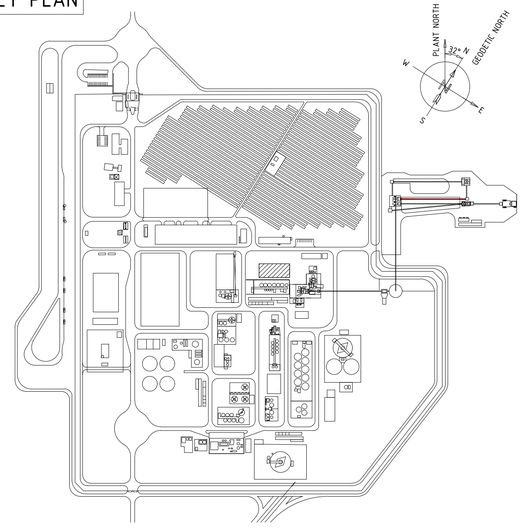
ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
PROJ.:	PROJECTION	BRG.:	BEARING
OPNG.:	OPENING		

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+) 913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+) 912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

- CONCRETE FLOOR
- CHECKERED PLATE
- GRATING
- BRICK WALL
- CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-ELV-1107	CONV.3130-CNV-002-ELEVATION								
MD-216-3130-EG-ME-ELV-1006	CONV.3130-CNV-002-ELEVATION								
MD-216-3130-EG-ME-ELV-1005	CONV.3130-CNV-002-ELEVATION	A01	06/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_100_H_G_16114_001		0
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707__GAU01__ZC01__MTL01__31327	OU100669068	0

DRAWN	HKD	03/02/2020
CHECKED	BRD	04/02/2020
ENGINEER	BD	04/02/2020
APPROVED	BGR	06/02/2020
EMPLOYER		
SCALE	1:50	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
CONV. 3130-CNV-002 - PLAN

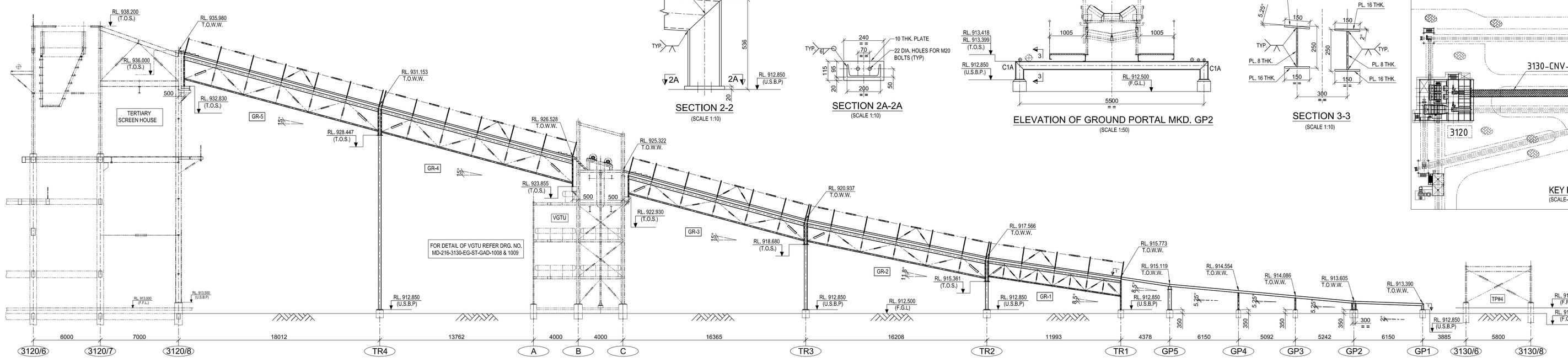
DRAWING NUMBER: MD-216-3130-EG-ME-LPX-1003
REV: A01

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

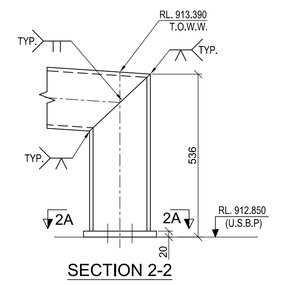
Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	26-03-2020	24-04-2020	01-11-2023				Date	26-03-2020	24-04-2020	01-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 2 **STRUCTURAL GENERAL ARRANGEMENT DRAWING CONTAINING BASE PLATE DETAIL**

DRAWING NUMBER: MD-216-3130-EG-ST-GAD-1006 | REV: A01

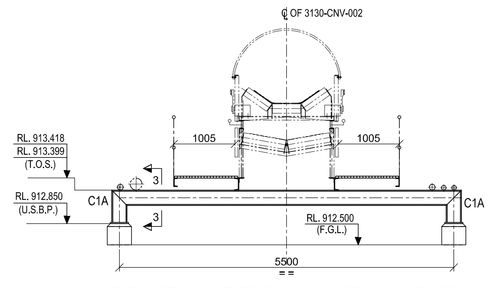


ELEVATION
(SCALE - 1:150)

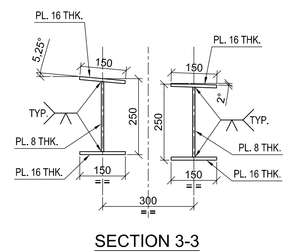


SECTION 2-2
(SCALE 1:10)

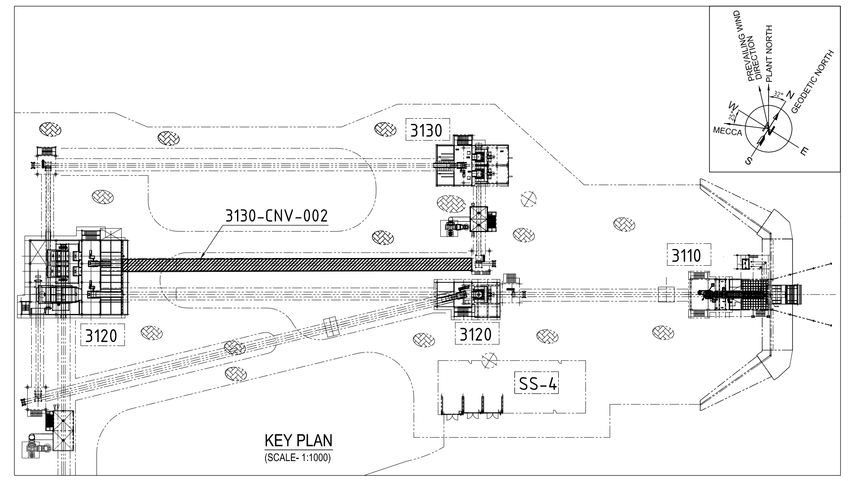
SECTION 2A-2A
(SCALE 1:10)



ELEVATION OF GROUND PORTAL MKD. GP2
(SCALE 1:50)



SECTION 3-3
(SCALE 1:10)

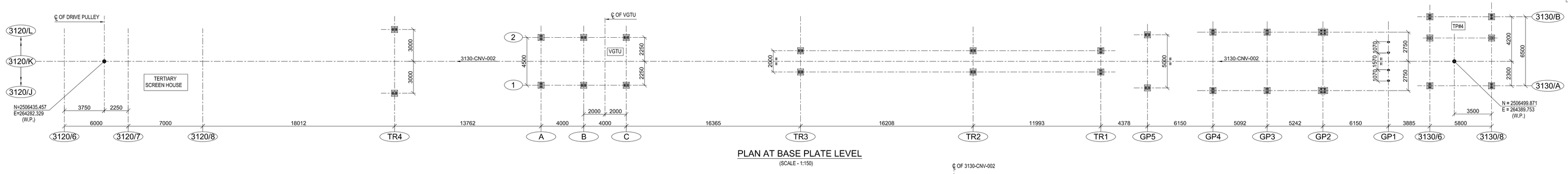


KEY PLAN
(SCALE- 1:1000)

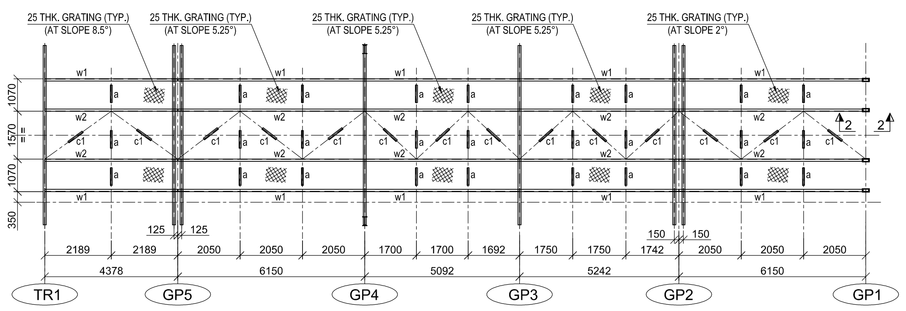
SCHEDULE OF MEMBERS FOR GP-2 TO GP4

MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C1A	1. PL. 150X16 2. PL. 218X8	1 2 1	-	-	GUSSET PLATE 8 THK.
a	L 50X50X6	L	70	-	
c1	2L 75X75X8	L	70	-	
h1	2L 75X75X8		50	-	
w1	UPN 140	[	-	-	
w2	UPN 200	[	-	-	

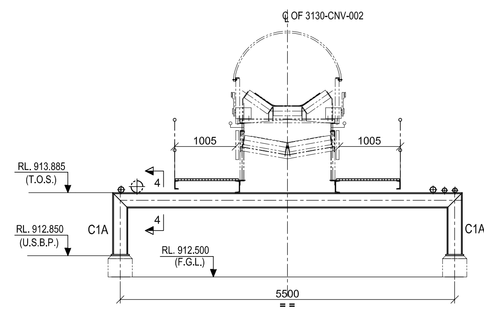
GRADE OF MATERIAL IS S275 JR
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.



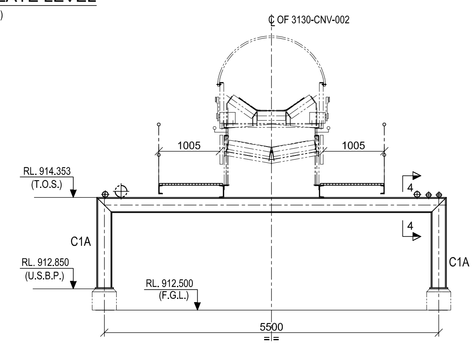
PLAN AT BASE PLATE LEVEL
(SCALE - 1:150)



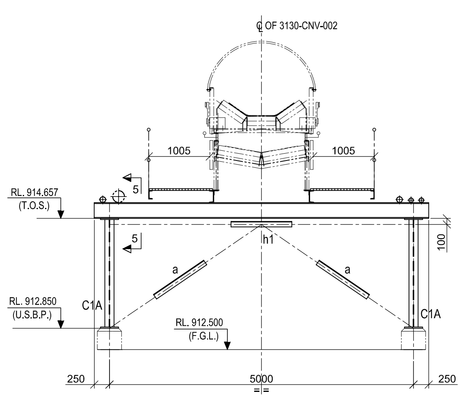
VIEW 1-1
SCALE - 1:100



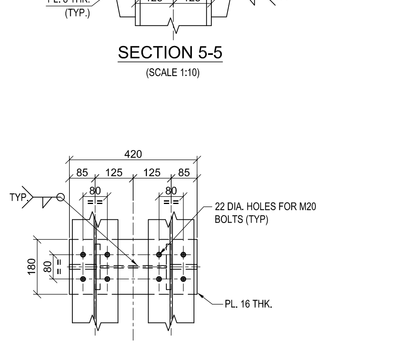
ELEVATION OF GROUND PORTAL MKD. GP3
(SCALE 1:50)



ELEVATION OF GROUND PORTAL MKD. GP4
(SCALE 1:50)



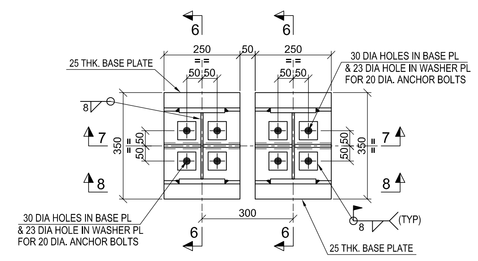
ELEVATION OF GROUND PORTAL MKD. GP5
(SCALE 1:50)



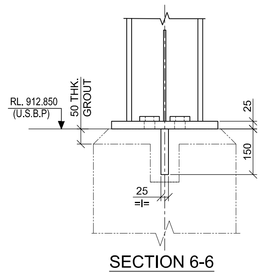
SECTION 4-4
(SCALE 1:10)

SECTION 5A-5A
(SCALE 1:10)

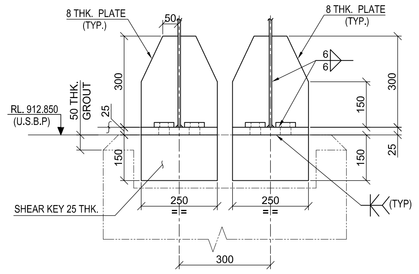
SECTION 5-5
(SCALE 1:10)



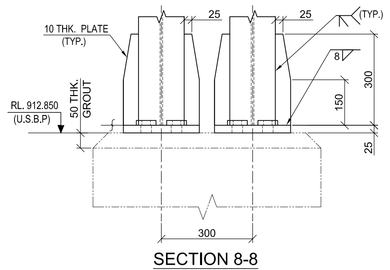
BASE PLATE DETAIL FOR GP2
(SCALE - 1:10)



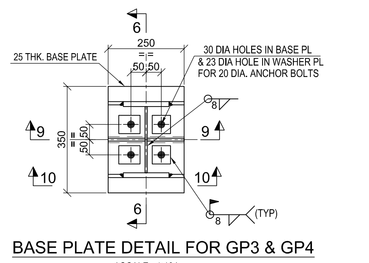
SECTION 6-6
(SCALE - 1:10)



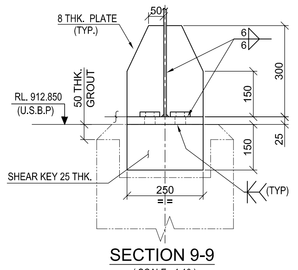
SECTION 7-7
(SCALE - 1:10)



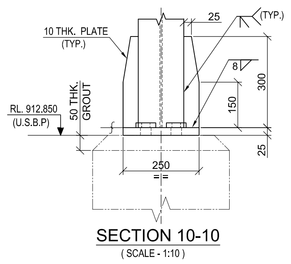
SECTION 8-8
(SCALE - 1:10)



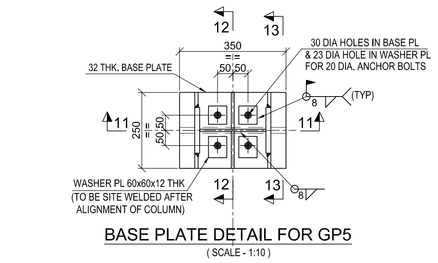
BASE PLATE DETAIL FOR GP3 & GP4
(SCALE - 1:10)



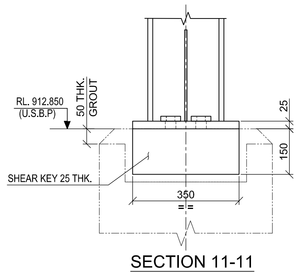
SECTION 9-9
(SCALE - 1:10)



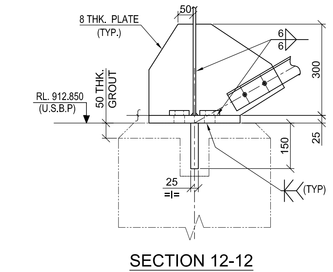
SECTION 10-10
(SCALE - 1:10)



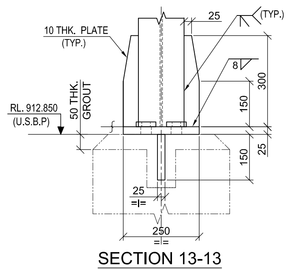
BASE PLATE DETAIL FOR GP5
(SCALE - 1:10)



SECTION 11-11
(SCALE - 1:10)



SECTION 12-12
(SCALE - 1:10)



SECTION 13-13
(SCALE - 1:10)

LEGENDS :-

T.O.S. - TOP OF STEEL
R.L. - REDUCED LEVEL
F/S-FAR SIDE
N/S-NEAR SIDE
TYP.-TYPICAL
W.P.-WORKING POINT
MM-MILLIMETER
PL.-PLATE
MKD.-MARKED
℄ - CENTRE LINE
T.O.C. - TOP OF CONCRETE.
MR. - MONORAIL
B.O.M.R. - BOTTOM OF MONORAIL
B/S - BOTH SIDE
CHQD. PL. - CHEQUERED PLATE
U.S.B.P. - UNDER SIDE OF BASE PLATE
THK. - THICK
DN - DOWN
MC - MOMENT CONNECTION
T.O.W/W-TOP OF WALKWAY
O/P - OVER PLATE
VGTU -VERTICAL GRAVITY TAKE UP UNIT
TP -TRANSFER POINT

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES, UNLESS OTHERWISE STATED.
2. GRADE OF STEEL USED S275JR AS PER EN10025
3. THIS DRG. SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3130-EG-ST-GAD-1007, 1008 ,1009 & 1010
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-LPX-1003-RA01	CONV.3130-CNV-002-PLAN								
MD-216-3130-EG-ME-ELV-1005-RA01	CONV.3130-CNV-002-ELEVATION								
MD-216-3130-EG-ME-ELV-1006-RA01	CONV.3130-CNV-002-ELEVATION								
MD-216-3130-EG-ME-ELV-1107-RA01	CONV.3130-CNV-002-ELEVATION	A01	27/03/2020	ISSUED FOR REFERENCE	HSK	SBY	RPK	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER: 019045_100_S_G_26013_018 | REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31366 | DOCUMENT ID: OU100670434 | REVISION: 0

DRAWN	HSK	24/03/2020
CHECKED	SBY	25/03/2020
ENGINEER	RPK	20/03/2020
APPROVED	PRI	27/03/2020
EMPLOYER		

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
STRUCTURAL GA FOR 3130-CNV-002, SH1
PLAN, ELEVATIONS & DETAILS

SCALE: 1:150,1:100,1:50,1:10 | A0
DRAWING NUMBER: MD - 216 - 3130 - EG - ST - GAD - 1006 | REV: A01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	26-03-2020	24-04-2020	01-11-2023				Date	26-03-2020	24-04-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 3 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR GROUND PORTAL**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-Mar-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 6 0.35 0; 2 11.4 0.35 0; 3 16.5 0.35 0; 4 6 1.099 0; 5 11.4 1.586 0;
6 16.5 2.054 0; 7 6 0.35 5.5; 8 11.4 0.35 5.5; 9 16.5 0.35 5.5; 10 6 1.099 5.5;
11 11.4 1.586 5.5; 12 16.5 2.054 5.5; 13 6 1.099 0.895; 14 6 1.099 1.965;
15 6 1.099 3.535; 16 6 1.099 4.605; 17 11.4 1.586 0.895; 18 16.5 2.054 0.895;
23 27 3.274 0.895; 24 11.4 1.586 1.965; 25 16.5 2.054 1.965; 26 27 3.274 1.965;
27 11.4 1.586 3.535; 28 16.5 2.054 3.535; 29 27 3.274 3.535;
30 11.4 1.586 4.605; 31 16.5 2.054 4.605; 32 27 3.274 4.605; 33 0 0.89 0.895;
34 0 0.89 1.965; 35 0 0.89 3.535; 36 0 0.89 4.605; 41 6.3 0.35 0;
42 6.3 1.118 0; 43 6.3 0.35 5.5; 44 6.3 1.118 5.5; 45 6.3 1.118 0.895;
46 6.3 1.118 1.965; 47 6.3 1.118 3.535; 48 6.3 1.118 4.605; 49 22.65 0.35 0.25;
50 22.65 2.539 0.25; 51 22.65 0.35 5.25; 52 22.65 2.539 5.25;
53 22.65 2.619 0.25; 54 22.65 2.619 0.895; 55 22.65 2.619 1.965;
56 22.65 2.619 3.535; 57 22.65 2.619 4.605; 58 22.65 2.619 5.25;
61 0 0.35 0.895; 62 0 0.35 1.965; 63 0 0.35 3.535; 64 0 0.35 4.605;
65 22.65 2.539 2.75; 66 0 0.35 0; 67 0 0.35 5.5; 68 6.15 0.35 0;
69 6.15 0.35 5.5; 70 11.4 -1 0; 71 16.5 -1 0; 72 11.4 -1 5.5; 73 16.5 -1 5.5;
74 22.65 -1 0.25; 75 22.65 -1 5.25; 76 0 -1 0; 77 0 -1 5.5; 78 6.15 -1 0;
79 6.15 -1 5.5;
MEMBER INCIDENCES
1 1 4; 2 2 5; 3 3 6; 4 7 10; 5 8 11; 6 9 12; 7 4 13; 8 5 17; 9 6 18; 10 13 14;
11 14 15; 12 15 16; 13 16 10; 14 17 24; 15 13 45; 16 18 25; 17 17 18; 22 18 54;
23 24 27; 24 14 46; 25 25 28; 26 24 25; 28 25 55; 29 27 30; 30 15 47; 31 28 31;
32 27 28; 34 28 56; 35 30 11; 36 16 48; 37 31 12; 38 30 31; 40 31 57; 41 33 13;
42 34 14; 43 35 15; 44 36 16; 52 41 42; 53 43 44; 54 42 45; 55 45 46; 56 46 47;
57 47 48; 58 48 44; 59 49 50; 60 51 52; 61 53 54; 62 54 55; 63 55 56; 64 56 57;
65 57 58; 66 49 65; 67 51 65; 68 50 53; 69 52 58; 70 50 65; 77 54 23; 78 57 32;
87 45 17; 88 46 24; 89 47 27; 90 48 30; 91 55 26; 92 56 29; 93 64 36; 94 63 35;
95 62 34; 96 61 33; 97 65 52; 98 66 61; 99 61 62; 100 62 63; 101 63 64;
102 64 67; 105 2 8; 106 3 9; 107 49 51; 108 1 68; 109 7 69; 110 68 41;
111 69 43; 112 76 66; 113 77 67; 114 78 68; 115 79 69; 116 70 2; 117 72 8;
118 71 3; 119 73 9; 120 74 49; 121 75 51; 122 68 69;
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
```

```
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
1 TO 8 10 TO 14 23 29 35 52 TO 60 68 -
69 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
9 16 25 31 37 61 TO 65 TAPERED 0.25 0.008 0.25 0.2 0.016 0.2 0.016
15 17 22 36 38 40 41 44 77 78 87 90 93 96 TABLE ST UPN140
24 26 28 30 32 34 42 43 88 89 91 92 94 95 TABLE ST UPN200
66 67 TABLE LD L50X50X6 SP 0.01
70 97 TABLE LD L75X75X8 SP 0.01
MEMBER PROPERTY EUROPEAN
98 TO 102 PRIS YD 0.6 ZD 0.35
105 TO 111 122 PRIS YD 0.5 ZD 0.3
112 113 116 TO 121 PRIS YD 0.6 ZD 0.6
114 115 PRIS YD 0.7 ZD 0.5
CONSTANTS
BETA 90 MEMB 1 TO 6 52 53 68 69
MATERIAL STEEL MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 44 52 TO 70 77 -
78 87 TO 97
MATERIAL CONCRETE MEMB 98 TO 102 105 TO 122
SUPPORTS
23 26 29 32 FIXED BUT FX MX MY MZ
70 TO 79 FIXED
MEMBER TRUSS
66 67
MEMBER RELEASE
1 TO 6 52 53 59 TO 61 70 77 78 87 TO 96 START MP 0.99
22 28 34 40 TO 44 65 97 END MP 0.99
LOAD 1 DL
SELFWEIGHT Y -1.1
*WALKWAY LOAD @50KG/SQ. M Including Seal Plate & H/R
MEMBER LOAD
15 17 22 36 38 40 41 44 77 78 87 90 UNI GY -0.32
24 26 28 30 32 34 42 43 88 89 91 92 UNI GY -0.47
*PIPE DEAD LOAD
MEMBER LOAD
7 TO 9 54 61 CON GY -4.08
13 35 37 58 65 CON GY -7.14
*CONVEYOR DEAD LOAD
MEMBER LOAD
42 43 CON GY -5 0.84
42 43 CON GY -5 3.9
88 89 CON GY -5 0.59
88 89 CON GY -5 3.59
26 32 CON GY -5 1.48
26 32 CON GY -5 4.6
```

```
28 34 CON GY -5 2.36
28 34 CON GY -5 5.4
91 92 CON GY -5 2.3
*
42 43 CON GZ 0.8 0.84
42 43 CON GZ 0.8 3.9
88 89 CON GZ 0.8 0.59
88 89 CON GZ 0.8 3.59
26 32 CON GZ 0.8 1.48
26 32 CON GZ 0.8 2.1
26 32 CON GZ 0.8 4.6
28 34 CON GZ 0.8 2.36
28 34 CON GZ 0.8 5.4
91 92 CON GZ 0.8 2.3
*
MEMBER LOAD
42 43 CON GX 1.5 0.84
42 43 CON GX -0.8 3.9
88 89 CON GX -0.8 0.59
88 89 CON GX -0.8 3.59
26 32 CON GX -0.8 1.48
26 32 CON GX -0.8 2.1
26 32 CON GX -0.8 4.6
28 34 CON GX -0.8 2.36
28 34 CON GX -0.8 5.4
91 92 CON GX -0.8 2.3
LOAD 2 LL
*WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
MEMBER LOAD
15 17 22 24 26 28 30 32 34 36 38 40 TO 44 77 78 87 TO 92 UNI GY -2
LOAD 3 ML
*CONVEYOR MATERIAL LOAD
MEMBER LOAD
42 43 CON GY -5.5 0.84
42 43 CON GY -5.5 3.9
88 89 CON GY -5.5 0.59
88 89 CON GY -5.5 3.59
26 32 CON GY -5.5 1.48
26 32 CON GY -5.5 4.6
28 34 CON GY -5.5 2.36
28 34 CON GY -5.5 5.4
91 92 CON GY -5.5 2.3
** Spillage Load @75kg/ Sq. m on seal plate
24 26 28 30 32 34 42 43 88 89 91 92 UNI GY -0.375
LOAD 4 WIND LOAD +Z
**WIND LOAD ON CONVEYOR HOOD
MEMBER LOAD
42 43 CON GZ 0.7 0.84
42 43 CON GZ 0.7 3.9
88 89 CON GZ 0.7 0.59
```

```
88 89 CON GZ 0.7 3.59
26 32 CON GZ 0.7 1.48
26 32 CON GZ 0.7 4.6
28 34 CON GZ 0.7 2.36
28 34 CON GZ 0.7 5.4
91 92 CON GZ 0.7 2.3
**
42 CON GY 1.6 0.84
43 CON GY -1.6 0.84
42 CON GY 1.6 3.9
43 CON GY -1.6 3.9
88 CON GY 1.6 0.59
89 CON GY -1.6 0.59
88 CON GY 1.6 3.59
89 CON GY -1.6 3.59
26 CON GY 1.6 1.48
32 CON GY -1.6 1.48
26 CON GY 1.6 4.6
32 CON GY -1.6 4.6
28 CON GY 1.6 2.36
34 CON GY -1.6 2.36
28 CON GY 1.6 5.4
34 CON GY -1.6 5.4
91 CON GY 1.6 2.3
92 CON GY -1.6 2.3
LOAD 5 WIND LOAD -Z
**WIND LOAD ON CONVEYOR HOOD
MEMBER LOAD
42 43 CON GZ -0.7 0.84
42 43 CON GZ -0.7 3.9
88 89 CON GZ -0.7 0.59
88 89 CON GZ -0.7 3.59
26 32 CON GZ -0.7 1.48
26 32 CON GZ -0.7 4.6
28 34 CON GZ -0.7 2.36
28 34 CON GZ -0.7 5.4
91 92 CON GZ -0.7 2.3
**
42 CON GY -1.6 0.84
43 CON GY 1.6 0.84
42 CON GY -1.6 3.9
43 CON GY 1.6 3.9
88 CON GY -1.6 0.59
89 CON GY 1.6 0.59
88 CON GY -1.6 3.59
89 CON GY 1.6 3.59
26 CON GY -1.6 1.48
32 CON GY 1.6 1.48
26 CON GY -1.6 4.6
32 CON GY 1.6 4.6
```

```
28 CON GY -1.6 2.36
34 CON GY 1.6 2.36
28 CON GY -1.6 5.4
34 CON GY 1.6 5.4
91 CON GY -1.6 2.3
92 CON GY 1.6 2.3
LOAD 6 TL1
TEMPERATURE LOAD
1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 44 TEMP 25
LOAD 7 TL2
TEMPERATURE LOAD
41 42 TEMP -25
****LOAD COMBINATION FOR BEARING PRESSURE CHECK
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 6 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
1 1.0 7 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 (DL + TL1) + 0.75  (LL+ML) + 0.75 WL(+Z)
1 1.0 6 1.0 2 0.75 3 0.75 4 0.75
LOAD COMB 305 1.0 (DL + TL1) + 0.75  (LL+ML)  + 0.75 WL(-Z)
1 1.0 6 1.0 2 0.75 3 0.75 5 0.75
LOAD COMB 306 1.0 (DL + TL1) + 0.75  (LL+ML) + 0.75 WL(+Z)
1 1.0 7 1.0 2 0.75 3 0.75 4 0.75
LOAD COMB 307 1.0 (DL + TL1) + 0.75  (LL+ML)  + 0.75 WL(-Z)
1 1.0 7 1.0 2 0.75 3 0.75 5 0.75
****LOAD COMBINATION FOR FOUNDATION DESIGN
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7 (LL+ML)
1 1.4 6 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL2) + 1.7 (LL+ML)
1 1.4 7 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 (DL + TL1) + 0.5 (LL+ML) + 1.6 WL(+Z)
1 1.2 6 1.2 2 0.5 3 0.5 4 1.6
LOAD COMB 405 1.2 (DL + TL1) + 0.5 (LL+ML) + 1.6 WL(-Z)
1 1.2 6 1.2 2 0.5 3 0.5 5 1.6
LOAD COMB 406 1.2 (DL + TL2) + 0.5 (LL+ML) + 1.6 WL(+Z)
1 1.2 7 1.2 2 0.5 3 0.5 4 1.6
LOAD COMB 407 1.2 (DL + TL2) + 0.5 (LL+ML) + 1.6 WL(-Z)
1 1.2 7 1.2 2 0.5 3 0.5 5 1.6
PERFORM ANALYSIS
LOAD LIST 401 TO 407
START CONCRETE DESIGN
CODE ACI 2008
FYMAIN 420000 MEMB 98 TO 102 105 TO 122
CLT 0.05 MEMB 98 TO 102 105 TO 122
CLB 0.05 MEMB 98 TO 102 105 TO 122
DESIGN BEAM 98 TO 102 105 TO 107 122
```

```
DESIGN COLUMN 112 TO 121
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 25, 2020                *
*           Time=    19:35:12                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Conveyor CNV-3130-002\Civil\Staad\3130-CNV-002-Ground Portal Founda.. .S
     2. START JOB INFORMATION
     3. ENGINEER DATE 23-DEC-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 6 0.35 0; 2 11.4 0.35 0; 3 16.5 0.35 0; 4 6 1.099 0; 5 11.4 1.586 0
     9. 6 16.5 2.054 0; 7 6 0.35 5.5; 8 11.4 0.35 5.5; 9 16.5 0.35 5.5; 10 6 1.099 5.5
    10. 11 11.4 1.586 5.5; 12 16.5 2.054 5.5; 13 6 1.099 0.895; 14 6 1.099 1.965
    11. 15 6 1.099 3.535; 16 6 1.099 4.605; 17 11.4 1.586 0.895; 18 16.5 2.054 0.895
    12. 23 27 3.274 0.895; 24 11.4 1.586 1.965; 25 16.5 2.054 1.965; 26 27 3.274 1.965
    13. 27 11.4 1.586 3.535; 28 16.5 2.054 3.535; 29 27 3.274 3.535
    14. 30 11.4 1.586 4.605; 31 16.5 2.054 4.605; 32 27 3.274 4.605; 33 0 0.89 0.895
    15. 34 0 0.89 1.965; 35 0 0.89 3.535; 36 0 0.89 4.605; 41 6.3 0.35 0
    16. 42 6.3 1.118 0; 43 6.3 0.35 5.5; 44 6.3 1.118 5.5; 45 6.3 1.118 0.895
    17. 46 6.3 1.118 1.965; 47 6.3 1.118 3.535; 48 6.3 1.118 4.605; 49 22.65 0.35 0.25
    18. 50 22.65 2.539 0.25; 51 22.65 0.35 5.25; 52 22.65 2.539 5.25
    19. 53 22.65 2.619 0.25; 54 22.65 2.619 0.895; 55 22.65 2.619 1.965
    20. 56 22.65 2.619 3.535; 57 22.65 2.619 4.605; 58 22.65 2.619 5.25
    21. 61 0 0.35 0.895; 62 0 0.35 1.965; 63 0 0.35 3.535; 64 0 0.35 4.605
    22. 65 22.65 2.539 2.75; 66 0 0.35 0; 67 0 0.35 5.5; 68 6.15 0.35 0
    23. 69 6.15 0.35 5.5; 70 11.4 -1 0; 71 16.5 -1 0; 72 11.4 -1 5.5; 73 16.5 -1 5.5
    24. 74 22.65 -1 0.25; 75 22.65 -1 5.25; 76 0 -1 0; 77 0 -1 5.5; 78 6.15 -1 0
    25. 79 6.15 -1 5.5
    26. MEMBER INCIDENCES
    27. 1 1 4; 2 2 5; 3 3 6; 4 7 10; 5 8 11; 6 9 12; 7 4 13; 8 5 17; 9 6 18; 10 13 14
    28. 11 14 15; 12 15 16; 13 16 10; 14 17 24; 15 13 45; 16 18 25; 17 17 18; 22 18 54
    29. 23 24 27; 24 14 46; 25 25 28; 26 24 25; 28 25 55; 29 27 30; 30 15 47; 31 28 31
    30. 32 27 28; 34 28 56; 35 30 11; 36 16 48; 37 31 12; 38 30 31; 40 31 57; 41 33 13
    31. 42 34 14; 43 35 15; 44 36 16; 52 41 42; 53 43 44; 54 42 45; 55 45 46; 56 46 47
    32. 57 47 48; 58 48 44; 59 49 50; 60 51 52; 61 53 54; 62 54 55; 63 55 56; 64 56 57
    33. 65 57 58; 66 49 65; 67 51 65; 68 50 53; 69 52 58; 70 50 65; 77 54 23; 78 57 32
    34. 87 45 17; 88 46 24; 89 47 27; 90 48 30; 91 55 26; 92 56 29; 93 64 36; 94 63 35
    35. 95 62 34; 96 61 33; 97 65 52; 98 66 61; 99 61 62; 100 62 63; 101 63 64
    36. 102 64 67; 105 2 8; 106 3 9; 107 49 51; 108 1 68; 109 7 69; 110 68 41
    37. 111 69 43; 112 76 66; 113 77 67; 114 78 68; 115 79 69; 116 70 2; 117 72 8
    38. 118 71 3; 119 73 9; 120 74 49; 121 75 51; 122 68 69
```

```
39. DEFINE MATERIAL START
40. ISOTROPIC STEEL
41. E 2.05E+008
42. POISSON 0.3
43. DENSITY 76.8195
44. ALPHA 1.2E-005
45. DAMP 0.03
46. TYPE STEEL
47. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
48. ISOTROPIC CONCRETE
49. E 2.17185E+007
50. POISSON 0.17
51. DENSITY 23.5616
52. ALPHA 1E-005
53. DAMP 0.05
54. TYPE CONCRETE
55. STRENGTH FCU 27579
56. END DEFINE MATERIAL
57. MEMBER PROPERTY EUROPEAN
58. 1 TO 8 10 TO 14 23 29 35 52 TO 60 68 -
59. 69 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
60. 9 16 25 31 37 61 TO 65 TAPERED 0.25 0.008 0.25 0.2 0.016 0.2 0.016
61. 15 17 22 36 38 40 41 44 77 78 87 90 93 96 TABLE ST UPN140
62. 24 26 28 30 32 34 42 43 88 89 91 92 94 95 TABLE ST UPN200
63. 66 67 TABLE LD L50X50X6 SP 0.01
64. 70 97 TABLE LD L75X75X8 SP 0.01
65. MEMBER PROPERTY EUROPEAN
66. 98 TO 102 PRIS YD 0.6 ZD 0.35
67. 105 TO 111 122 PRIS YD 0.5 ZD 0.3
68. 112 113 116 TO 121 PRIS YD 0.6 ZD 0.6
69. 114 115 PRIS YD 0.7 ZD 0.5
70. CONSTANTS
71. BETA 90 MEMB 1 TO 6 52 53 68 69
72. MATERIAL STEEL MEMB 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 44 52 TO 70 77 -
73. 78 87 TO 97
74. MATERIAL CONCRETE MEMB 98 TO 102 105 TO 122
75. SUPPORTS
76. 23 26 29 32 FIXED BUT FX MX MY MZ
77. 70 TO 79 FIXED
78. MEMBER TRUSS
79. 66 67
80. MEMBER RELEASE
81. 1 TO 6 52 53 59 TO 61 70 77 78 87 TO 96 START MP 0.99
82. 22 28 34 40 TO 44 65 97 END MP 0.99
83. LOAD 1 DL
84. SELFWEIGHT Y -1.1
85. *WALKWAY LOAD @50KG/SQ. M INCLUDING SEAL PLATE _H/R
86. MEMBER LOAD
87. 15 17 22 36 38 40 41 44 77 78 87 90 UNI GY -0.32
88. 24 26 28 30 32 34 42 43 88 89 91 92 UNI GY -0.47
89. *PIPE DEAD LOAD
90. MEMBER LOAD
91. 7 TO 9 54 61 CON GY -4.08
92. 13 35 37 58 65 CON GY -7.14
93. *CONVEYOR DEAD LOAD
94. MEMBER LOAD
```

```
 95. 42 43 CON GY -5 0.84
 96. 42 43 CON GY -5 3.9
 97. 88 89 CON GY -5 0.59
 98. 88 89 CON GY -5 3.59
 99. 26 32 CON GY -5 1.48
100. 26 32 CON GY -5 4.6
101. 28 34 CON GY -5 2.36
102. 28 34 CON GY -5 5.4
103. 91 92 CON GY -5 2.3
104. *
105. 42 43 CON GZ 0.8 0.84
106. 42 43 CON GZ 0.8 3.9
107. 88 89 CON GZ 0.8 0.59
108. 88 89 CON GZ 0.8 3.59
109. 26 32 CON GZ 0.8 1.48
110. 26 32 CON GZ 0.8 2.1
111. 26 32 CON GZ 0.8 4.6
112. 28 34 CON GZ 0.8 2.36
113. 28 34 CON GZ 0.8 5.4
114. 91 92 CON GZ 0.8 2.3
115. *
116. MEMBER LOAD
117. 42 43 CON GX 1.5 0.84
118. 42 43 CON GX -0.8 3.9
119. 88 89 CON GX -0.8 0.59
120. 88 89 CON GX -0.8 3.59
121. 26 32 CON GX -0.8 1.48
122. 26 32 CON GX -0.8 2.1
123. 26 32 CON GX -0.8 4.6
124. 28 34 CON GX -0.8 2.36
125. 28 34 CON GX -0.8 5.4
126. 91 92 CON GX -0.8 2.3
127. LOAD 2 LL
128. *WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
129. MEMBER LOAD
130. 15 17 22 24 26 28 30 32 34 36 38 40 TO 44 77 78 87 TO 92 UNI GY -2
131. LOAD 3 ML
132. *CONVEYOR MATERIAL LOAD
133. MEMBER LOAD
134. 42 43 CON GY -5.5 0.84
135. 42 43 CON GY -5.5 3.9
136. 88 89 CON GY -5.5 0.59
137. 88 89 CON GY -5.5 3.59
138. 26 32 CON GY -5.5 1.48
139. 26 32 CON GY -5.5 4.6
140. 28 34 CON GY -5.5 2.36
141. 28 34 CON GY -5.5 5.4
142. 91 92 CON GY -5.5 2.3
143. ** SPILLAGE LOAD @75KG/ SQ. M ON SEAL PLATE
144. 24 26 28 30 32 34 42 43 88 89 91 92 UNI GY -0.375
145. LOAD 4 WIND LOAD +Z
146. **WIND LOAD ON CONVEYOR HOOD
147. MEMBER LOAD
148. 42 43 CON GZ 0.7 0.84
149. 42 43 CON GZ 0.7 3.9
150. 88 89 CON GZ 0.7 0.59
```

```
151. 88 89 CON GZ 0.7 3.59
152. 26 32 CON GZ 0.7 1.48
153. 26 32 CON GZ 0.7 4.6
154. 28 34 CON GZ 0.7 2.36
155. 28 34 CON GZ 0.7 5.4
156. 91 92 CON GZ 0.7 2.3
157. **
158. 42 CON GY 1.6 0.84
159. 43 CON GY -1.6 0.84
160. 42 CON GY 1.6 3.9
161. 43 CON GY -1.6 3.9
162. 88 CON GY 1.6 0.59
163. 89 CON GY -1.6 0.59
164. 88 CON GY 1.6 3.59
165. 89 CON GY -1.6 3.59
166. 26 CON GY 1.6 1.48
167. 32 CON GY -1.6 1.48
168. 26 CON GY 1.6 4.6
169. 32 CON GY -1.6 4.6
170. 28 CON GY 1.6 2.36
171. 34 CON GY -1.6 2.36
172. 28 CON GY 1.6 5.4
173. 34 CON GY -1.6 5.4
174. 91 CON GY 1.6 2.3
175. 92 CON GY -1.6 2.3
176. LOAD 5 WIND LOAD -Z
177. **WIND LOAD ON CONVEYOR HOOD
178. MEMBER LOAD
179. 42 43 CON GZ -0.7 0.84
180. 42 43 CON GZ -0.7 3.9
181. 88 89 CON GZ -0.7 0.59
182. 88 89 CON GZ -0.7 3.59
183. 26 32 CON GZ -0.7 1.48
184. 26 32 CON GZ -0.7 4.6
185. 28 34 CON GZ -0.7 2.36
186. 28 34 CON GZ -0.7 5.4
187. 91 92 CON GZ -0.7 2.3
188. **
189. 42 CON GY -1.6 0.84
190. 43 CON GY 1.6 0.84
191. 42 CON GY -1.6 3.9
192. 43 CON GY 1.6 3.9
193. 88 CON GY -1.6 0.59
194. 89 CON GY 1.6 0.59
195. 88 CON GY -1.6 3.59
196. 89 CON GY 1.6 3.59
197. 26 CON GY -1.6 1.48
198. 32 CON GY 1.6 1.48
199. 26 CON GY -1.6 4.6
200. 32 CON GY 1.6 4.6
201. 28 CON GY -1.6 2.36
202. 34 CON GY 1.6 2.36
203. 28 CON GY -1.6 5.4
204. 34 CON GY 1.6 5.4
205. 91 CON GY -1.6 2.3
206. 92 CON GY 1.6 2.3
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```
207. LOAD 6 TL1
208. TEMPERATURE LOAD
209. 1 TO 17 22 TO 26 28 TO 32 34 TO 38 40 TO 44 TEMP 25
210. LOAD 7 TL2
211. TEMPERATURE LOAD
212. 41 42 TEMP -25
213. ****LOAD COMBINATION FOR BEARING PRESSURE CHECK
214. LOAD COMB 301 1.0 DL
215. 1 1.0
216. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
217. 1 1.0 6 1.0 2 1.0 3 1.0
218. LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
219. 1 1.0 7 1.0 2 1.0 3 1.0
220. LOAD COMB 304 1.0 (DL + TL1) + 0.75  (LL+ML) + 0.75 WL(+Z)
221. 1 1.0 6 1.0 2 0.75 3 0.75 4 0.75
222. LOAD COMB 305 1.0 (DL + TL1) + 0.75  (LL+ML)  + 0.75 WL(-Z)
223. 1 1.0 6 1.0 2 0.75 3 0.75 5 0.75
224. LOAD COMB 306 1.0 (DL + TL1) + 0.75  (LL+ML) + 0.75 WL(+Z)
225. 1 1.0 7 1.0 2 0.75 3 0.75 4 0.75
226. LOAD COMB 307 1.0 (DL + TL1) + 0.75  (LL+ML)  + 0.75 WL(-Z)
227. 1 1.0 7 1.0 2 0.75 3 0.75 5 0.75
228. ****LOAD COMBINATION FOR FOUNDATION DESIGN
229. LOAD COMB 401 1.4 DL
230. 1 1.4
231. LOAD COMB 402 1.4 (DL + TL1) + 1.7 (LL+ML)
232. 1 1.4 6 1.4 2 1.7 3 1.7
233. LOAD COMB 403 1.4 (DL + TL2) + 1.7 (LL+ML)
234. 1 1.4 7 1.4 2 1.7 3 1.7
235. LOAD COMB 404 1.2 (DL + TL1) + 0.5 (LL+ML) + 1.6 WL(+Z)
236. 1 1.2 6 1.2 2 0.5 3 0.5 4 1.6
237. LOAD COMB 405 1.2 (DL + TL1) + 0.5 (LL+ML) + 1.6 WL(-Z)
238. 1 1.2 6 1.2 2 0.5 3 0.5 5 1.6
239. LOAD COMB 406 1.2 (DL + TL2) + 0.5 (LL+ML) + 1.6 WL(+Z)
240. 1 1.2 7 1.2 2 0.5 3 0.5 4 1.6
241. LOAD COMB 407 1.2 (DL + TL2) + 0.5 (LL+ML) + 1.6 WL(-Z)
242. 1 1.2 7 1.2 2 0.5 3 0.5 5 1.6
243. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**69**	**NUMBER OF MEMBERS**	**92**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**14**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 7, TOTAL DEGREES OF FREEDOM = 346
TOTAL LOAD COMBINATION CASES = 14 SO FAR.

```
244. LOAD LIST 401 TO 407
245. START CONCRETE DESIGN
246. CODE ACI 2008
247. FYMAIN 420000 MEMB 98 TO 102 105 TO 122
248. CLT 0.05 MEMB 98 TO 102 105 TO 122
249. CLB 0.05 MEMB 98 TO 102 105 TO 122
250. DESIGN BEAM 98 TO 102 105 TO 107 122
```

BEAM NO. 98 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

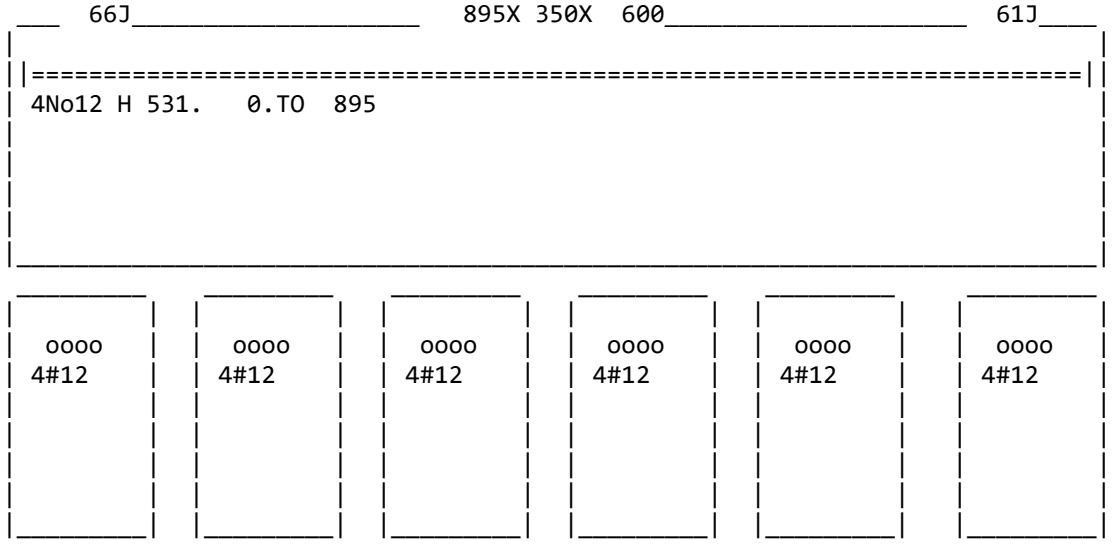
LEN - 895. MM FY - 420. FC - 28. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	531.	4 - 12MM	0.	895.	YES	YES

B E A M N O. 98 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 98 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 98 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



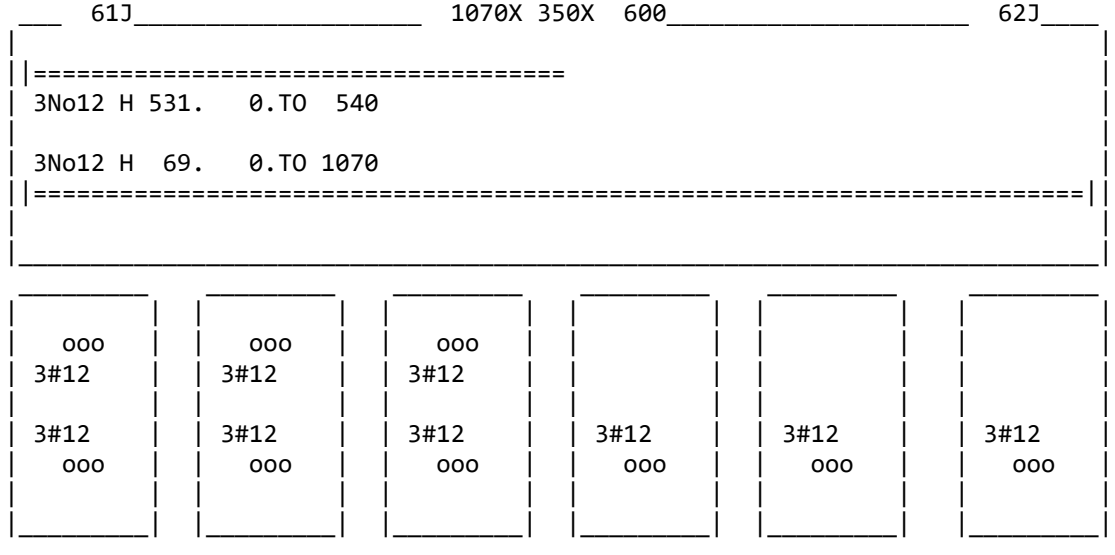
BEAM NO. 99 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1070. MM FY - 420. FC - 28. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	1070.	YES	YES
2	531.	3 - 12MM	0.	540.	YES	NO

B E A M N O. 99 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 22.73 KNS Vc= 175.03 KNS Vs= 0.00 KNS
Tu= 0.84 KN-MET Tc= 7.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 22.57 KNS Vc= 171.99 KNS Vs= 0.00 KNS
Tu= 0.84 KN-MET Tc= 7.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



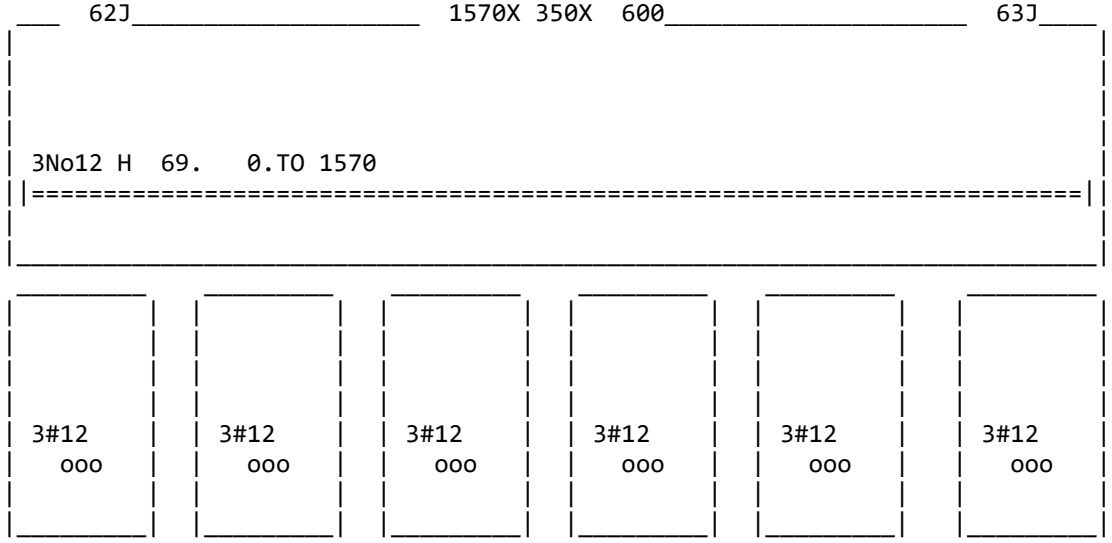
BEAM NO. 100 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1570. MM FY - 420. FC - 28. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	1570.	YES	YES

B E A M N O. 100 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 1.54 KNS Vc= 154.53 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 7.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 2.43 KNS Vc= 154.69 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 7.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



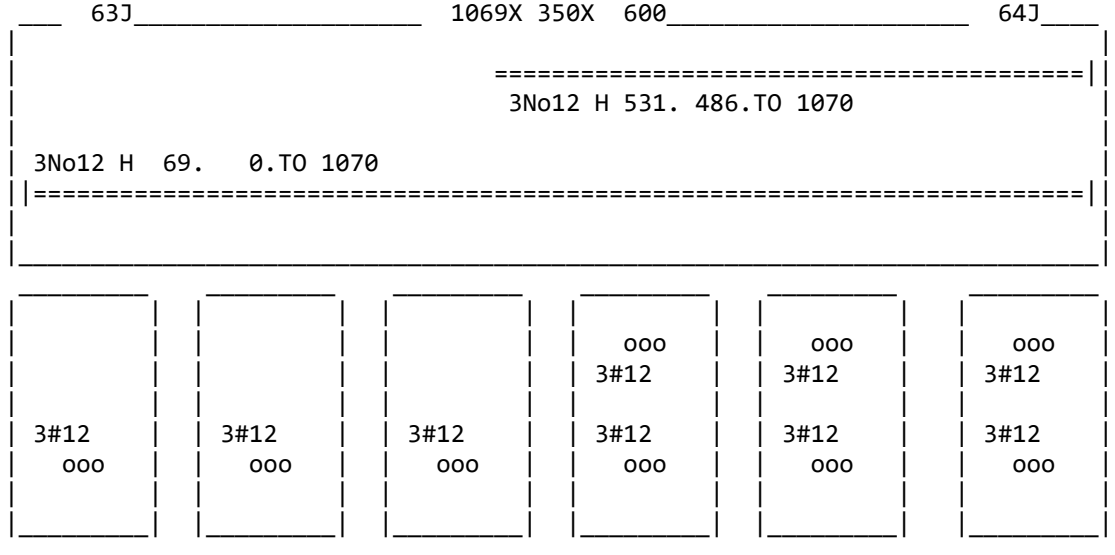
BEAM NO. 101 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1070. MM FY - 420. FC - 28. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	3 - 12MM	0.	1070.	YES	YES
2	531.	3 - 12MM	486.	1070.	NO	YES

B E A M N O. 101 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 23.51 KNS Vc= 175.32 KNS Vs= 0.00 KNS
Tu= 0.87 KN-MET Tc= 7.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 23.67 KNS Vc= 179.61 KNS Vs= 0.00 KNS
Tu= 0.87 KN-MET Tc= 7.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 102 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

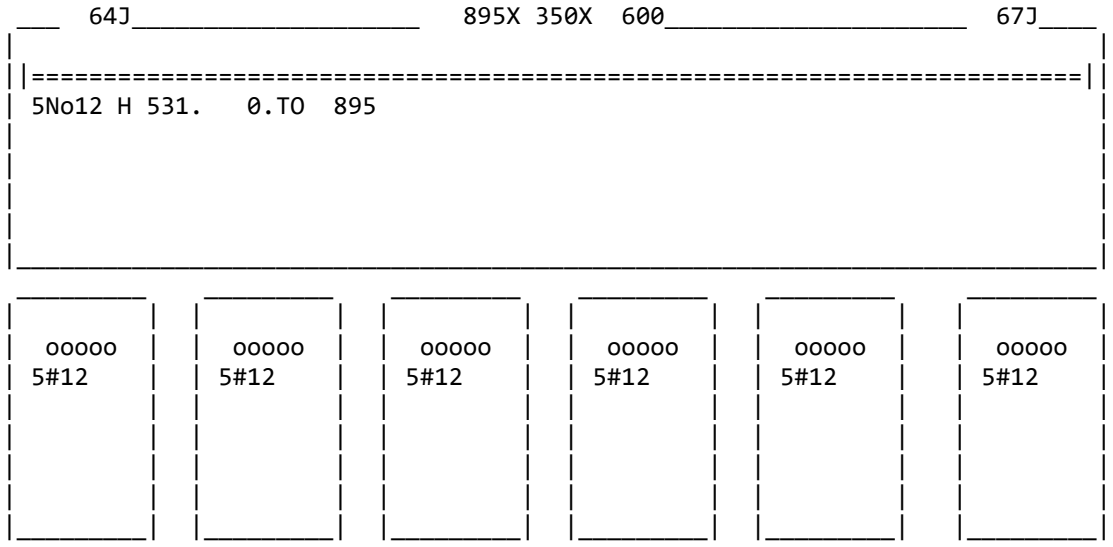
LEN - 895. MM FY - 420. FC - 28. MPA, SIZE - 350. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	531.	5 - 12MM	0.	895.	YES	YES

B E A M N O. 102 D E S I G N R E S U L T S - SHEAR

** LOCATION FOR DESIGN FOR SHEAR AT START OF MEMBER 102 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.

** LOCATION FOR DESIGN FOR SHEAR AT END OF MEMBER 102 IS BEYOND
THE MIDPOINT OF MEMBER. DESIGN FOR SHEAR AND TORSION NOT PERFORMED.



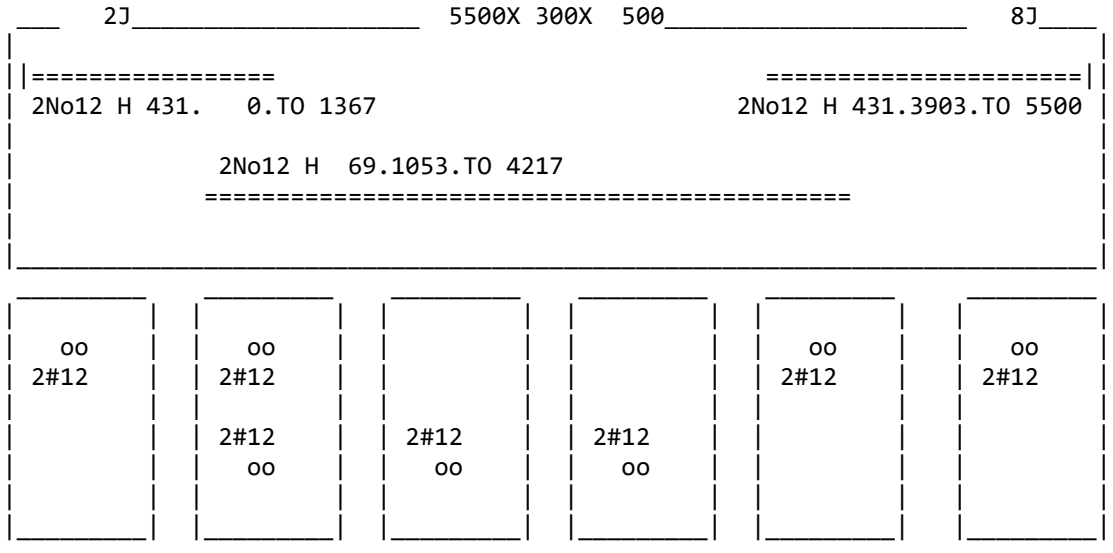
BEAM NO. 105 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	1053.	4217.	NO	NO
2	431.	2 - 12MM	0.	1367.	YES	NO
3	431.	2 - 12MM	3903.	5500.	NO	YES

B E A M N O. 105 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.45 KNS Vc= 111.65 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.87 KNS Vc= 111.65 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



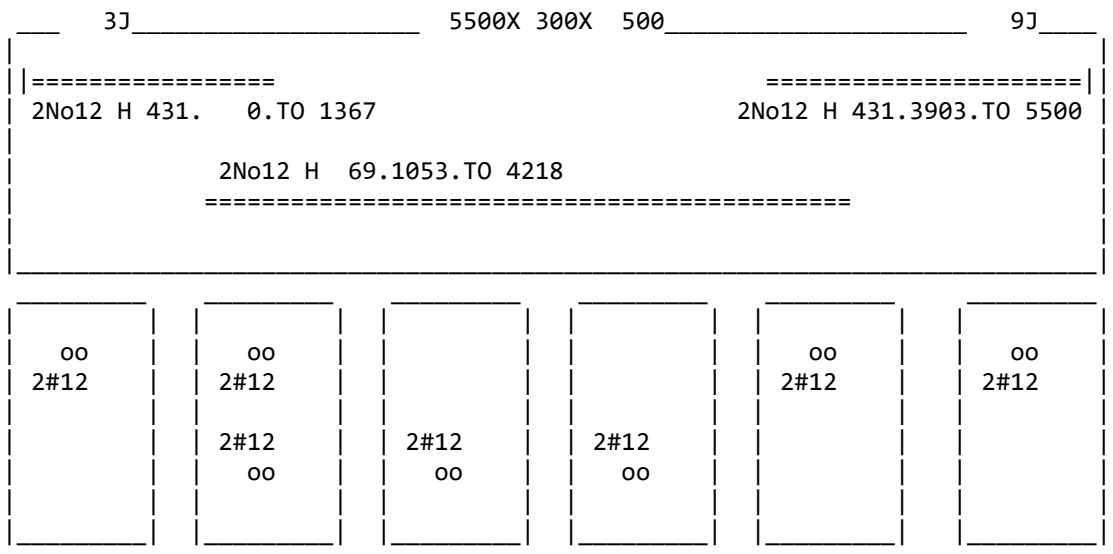
BEAM NO. 106 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	1053.	4218.	NO	NO
2	431.	2 - 12MM	0.	1367.	YES	NO
3	431.	2 - 12MM	3903.	5500.	NO	YES

B E A M N O. 106 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.44 KNS Vc= 112.35 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.87 KNS Vc= 112.35 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.6 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



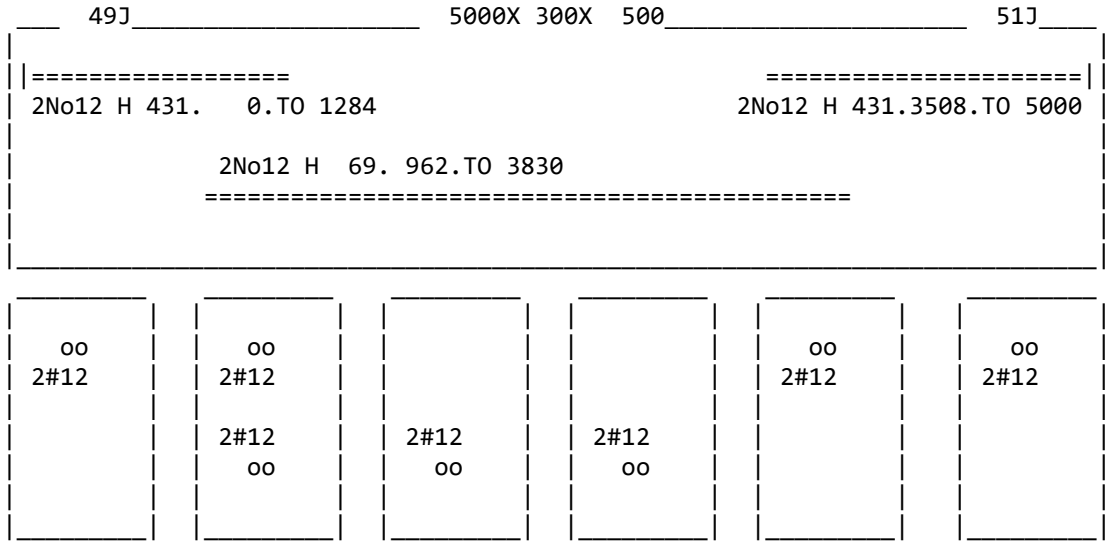
BEAM NO. 107 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	962.	3830.	NO	NO
2	431.	2 - 12MM	0.	1284.	YES	NO
3	431.	2 - 12MM	3508.	5000.	NO	YES

B E A M N O. 107 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.16 KNS Vc= 108.60 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 11.43 KNS Vc= 108.46 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



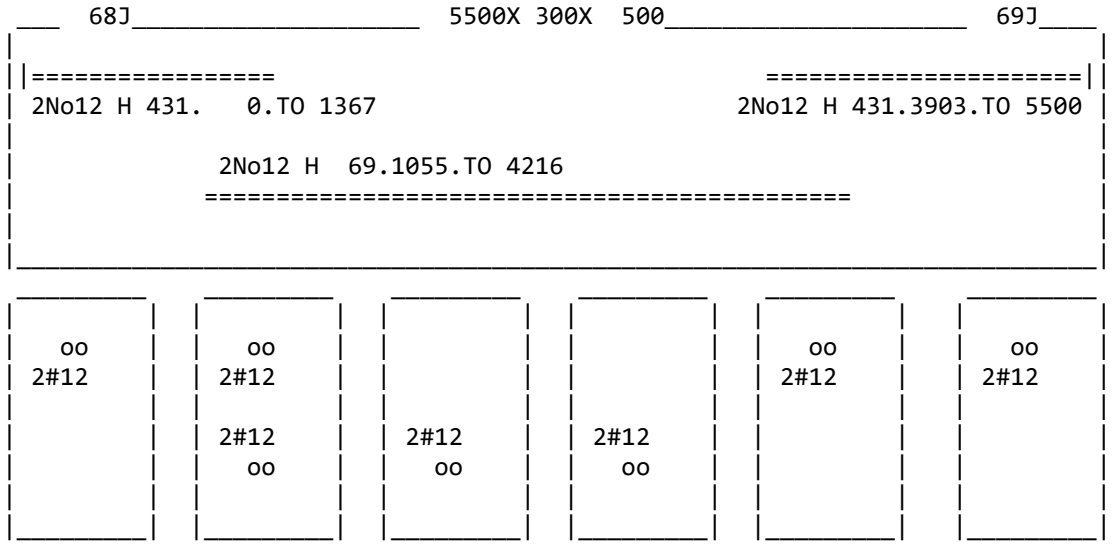
BEAM NO. 122 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 28. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	69.	2 - 12MM	1055.	4216.	NO	NO
2	431.	2 - 12MM	0.	1367.	YES	NO
3	431.	2 - 12MM	3903.	5500.	NO	YES

B E A M N O. 122 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.48 KNS Vc= 107.99 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.84 KNS Vc= 107.99 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 4.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

251. DESIGN COLUMN 112 TO 121

==========

COLUMN NO. 112 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 113 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 114 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 27.6 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.034	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 115 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 27.6 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.034	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 116 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 117 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 118 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 119 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 120 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 121 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 27.6 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
32 - 12 MM	1.005	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

252. END CONCRETE DESIGN
253. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 25,2020 TIME= 19:35:25 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01			
Date	26-03-2020	24-04-2020	01-11-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK	RPK	RPK			
Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI			

Outotec / L&T Revision No.:	0	1	2		
Date	26-03-2020	24-04-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI		

ATTACHMENT 4 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-1**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-Mar-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 2; 3 0 1.49012e-008 0; 4 0 1.49012e-008 2; 5 0 0.35 0;
6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7;
11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75;
15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7;
19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7;
24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75;
28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5;
33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25;
38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25;
42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25;
46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2;
51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75;
56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75;
60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5;
64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5;
69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25;
73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25;
77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25;
81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1;
86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75;
91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75;
95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75;
99 -0.75 -3 0.5; 100 -1.6 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5;
103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 107 1.6 -3 0.5;
108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25;
112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25;
116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0;
121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25;
126 -1.6 -3 -0.25; 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25;
130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25;
134 -0.75 -3 -0.5; 135 -1.6 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5;
138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5;
142 1.6 -3 -0.5; 143 -0.75 -3 -0.75; 144 -1.6 -3 -0.75; 145 -0.5 -3 -0.75;
146 -0.25 -3 -0.75; 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75;
150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7;
154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7;
158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25;
162 -0.5 -3 -1; 163 -0.5 -3 -1.25; 164 -0.25 -3 -1; 165 -0.25 -3 -1.25;
166 0 -3 -1; 167 0 -3 -1.25; 168 0.25 -3 -1; 169 0.25 -3 -1.25; 170 0.5 -3 -1;
171 0.5 -3 -1.25; 172 0.75 -3 -1; 173 0.75 -3 -1.25; 174 1.6 -3 -1;
175 1.6 -3 -1.25; 176 -0.75 -3 3.25; 177 -1.6 -3 3.25; 178 -0.75 -3 3;
179 -1.6 -3 3; 180 -0.5 -3 3.25; 181 -0.5 -3 3; 182 -0.25 -3 3.25;
183 -0.25 -3 3; 184 0 -3 3.25; 185 0 -3 3; 186 0.25 -3 3.25; 187 0.25 -3 3;
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188 0.5 -3 3.25; 189 0.5 -3 3; 190 0.75 -3 3.25; 191 0.75 -3 3;
192 1.6 -3 3.25; 193 1.6 -3 3; 194 -0.75 -3 -1.475; 195 -1.6 -3 -1.475;
196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475;
200 0.5 -3 -1.475; 201 0.75 -3 -1.475; 202 1.6 -3 -1.475; 203 -0.75 -3 3.475;
204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475;
208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475;
212 1.03333 -3 2.75; 213 1.03333 -3 2.5; 214 1.31667 -3 2.75;
215 1.31667 -3 2.5; 216 1.03333 -3 2.25; 217 1.31667 -3 2.25; 218 1.03333 -3 2;
219 1.31667 -3 2; 220 1.03333 -3 1.75; 221 1.31667 -3 1.75; 222 1.03333 -3 1.5;
223 1.31667 -3 1.5; 224 1.03333 -3 1.25; 225 1.31667 -3 1.25; 226 1.03333 -3 1;
227 1.31667 -3 1; 228 1.03333 -3 0.75; 229 1.31667 -3 0.75; 230 1.03333 -3 0.5;
231 1.31667 -3 0.5; 232 1.03333 -3 0.25; 233 1.31667 -3 0.25; 234 1.03333 -3 0;
235 1.31667 -3 0; 236 1.03333 -3 -0.25; 237 1.31667 -3 -0.25;
238 1.03333 -3 -0.5; 239 1.31667 -3 -0.5; 240 1.03333 -3 -0.75;
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244 1.03333 -3 -1.25; 245 1.31667 -3 -1.25; 246 1.03333 -3 3.25;
247 1.03333 -3 3; 248 1.31667 -3 3.25; 249 1.31667 -3 3; 250 1.03333 -3 -1.475;
251 1.31667 -3 -1.475; 252 1.03333 -3 -1.7; 253 1.31667 -3 -1.7;
254 1.03333 -3 3.7; 255 1.03333 -3 3.475; 256 1.31667 -3 3.7;
257 1.31667 -3 3.475; 258 -1.31667 -3 2.75; 259 -1.31667 -3 2.5;
260 -1.03333 -3 2.75; 261 -1.03333 -3 2.5; 262 -1.31667 -3 2.25;
263 -1.03333 -3 2.25; 264 -1.31667 -3 2; 265 -1.03333 -3 2;
266 -1.31667 -3 1.75; 267 -1.03333 -3 1.75; 268 -1.31667 -3 1.5;
269 -1.03333 -3 1.5; 270 -1.31667 -3 1.25; 271 -1.03333 -3 1.25;
272 -1.31667 -3 1; 273 -1.03333 -3 1; 274 -1.31667 -3 0.75;
275 -1.03333 -3 0.75; 276 -1.31667 -3 0.5; 277 -1.03333 -3 0.5;
278 -1.31667 -3 0.25; 279 -1.03333 -3 0.25; 280 -1.31667 -3 0;
281 -1.03333 -3 0; 282 -1.31667 -3 -0.25; 283 -1.03333 -3 -0.25;
284 -1.31667 -3 -0.5; 285 -1.03333 -3 -0.5; 286 -1.31667 -3 -0.75;
287 -1.03333 -3 -0.75; 288 -1.31667 -3 -1; 289 -1.03333 -3 -1;
290 -1.31667 -3 -1.25; 291 -1.03333 -3 -1.25; 292 -1.31667 -3 3.25;
293 -1.31667 -3 3; 294 -1.03333 -3 3.25; 295 -1.03333 -3 3;
296 -1.31667 -3 -1.475; 297 -1.03333 -3 -1.475; 298 -1.31667 -3 -1.7;
299 -1.03333 -3 -1.7; 300 -1.31667 -3 3.7; 301 -1.31667 -3 3.475;
302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
ELEMENT INCIDENCES SHELL
15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33;
20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41;
27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49;
34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56;
41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63;
48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69;
55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77;
60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85;
67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93;
74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99;
80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104;
84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111;

90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117;
96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121;
100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127;
105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131;
111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137;
114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140;
119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146;
122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149;
137 143 145 162 158; 138 158 162 163 160; 140 145 146 164 162;
141 162 164 165 163; 143 146 147 166 164; 144 164 166 167 165;
146 147 148 168 166; 147 166 168 169 167; 149 148 149 170 168;
150 168 170 171 169; 152 149 150 172 170; 153 170 172 173 171;
162 176 180 181 178; 163 178 181 17 14; 165 180 182 183 181; 166 181 183 19 17;
168 182 184 185 183; 169 183 185 20 19; 171 184 186 187 185; 172 185 187 22 20;
174 186 188 189 187; 175 187 189 24 22; 177 188 190 191 189; 178 189 191 26 24;
184 160 163 196 194; 185 194 196 153 152; 186 163 165 197 196;
187 196 197 154 153; 188 165 167 198 197; 189 197 198 7 154;
190 167 169 199 198; 191 198 199 155 7; 192 169 171 200 199;
193 199 200 156 155; 194 171 173 201 200; 195 200 201 157 156;
200 13 16 205 203; 201 203 205 180 176; 202 16 18 206 205; 203 205 206 182 180;
204 18 8 207 206; 205 206 207 184 182; 206 8 21 208 207; 207 207 208 186 184;
208 21 23 209 208; 209 208 209 188 186; 210 23 25 210 209; 211 209 210 190 188;
214 26 212 213 35; 215 212 214 215 213; 216 214 27 36 215; 217 35 213 216 44;
218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218;
222 217 45 53 219; 223 52 218 220 61; 224 218 219 221 220; 225 219 53 62 221;
226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79;
230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226;
234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229;
238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231;
241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233;
244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235;
247 123 234 236 132; 248 234 235 237 236; 249 235 124 133 237;
250 132 236 238 141; 251 236 237 239 238; 252 237 133 142 239;
253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241;
256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243;
259 172 242 244 173; 260 242 243 245 244; 261 243 174 175 245;
262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249;
265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214;
268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251;
271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253;
274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257;
277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248;
280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38;
284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264;
288 263 37 46 265; 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267;
292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73;
296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272;
300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275;
304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277;
307 100 276 278 109; 308 276 277 279 278; 309 277 99 108 279;

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310 109 278 280 118; 311 278 279 281 280; 312 279 108 117 281;
313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283;
316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285;
319 135 284 286 144; 320 284 285 287 286; 321 285 134 143 287;
322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289;
325 159 288 290 161; 326 288 289 291 290; 327 289 158 160 291;
328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295;
331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260;
334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297;
337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299;
340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303;
343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
214 TO 345
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
_SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 345
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
```

```
****************************************
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
JOINT LOAD
5 FX 0 FY -35.5 FZ 4.6
6 FX 0 FY -75.2 FZ -0.6
LOAD 2 LIVE LOAD
JOINT LOAD
5 FX 0 FY -24.1 FZ -0.9
6 FX 0 FY -23.9 FZ 0.9
LOAD 3 MATERIAL LOAD
JOINT LOAD
5 FX 0 FY -17.1 FZ -1.7
6 FX 0 FY -16.8 FZ 1.7
LOAD 4 WIND LOAD +Z
JOINT LOAD
5 FX 0 FY 33.7 FZ 12.5
6 FX 0 FY -33.7 FZ 12
LOAD 5 WIND LOAD -Z
JOINT LOAD
5 FX -0.04 FY -33.65 FZ -12.55
6 FX 0.04 FY 33.65 FZ -11.95
LOAD 6 WIND LOAD (+X)
JOINT LOAD
5 FX 0 FY -0.1 FZ 0
6 FX 0 FY 0.1 FZ 0
LOAD 7 WIND LOAD (-X)
JOINT LOAD
5 FX 0 FY 0.1 FZ 0
6 FX 0 FY -0.1 FZ 0
LOAD 8 TEMP LOAD 1
JOINT LOAD
5 FX -0.01 FY -0.22 FZ -6.33
6 FX 0.01 FY 0.22 FZ 6.32
LOAD 9 TEMP LOAD 2
JOINT LOAD
5 FX 0.01 FY 0.22 FZ 6.33
6 FX -0.01 FY -0.22 FZ -6.32
LOAD 10 EARTHQUAKE LOAD +X
JOINT LOAD
5 FX 0.92 FY 0.36 FZ 0.02
6 FX 0.92 FY 0.35 FZ -0.01
LOAD 11 EARTHQUAKE LOAD +Z
JOINT LOAD
5 FX 0 FY 1.03 FZ 0.39
```

```
6 FX 0 FY -1.03 FZ 0.37
*****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
```

```
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
```

```
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
****************************************
PERFORM ANALYSIS
****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 25, 2020                *
*           Time=    20:36:19                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Conveyor CNV-3130-002\Civil\Staad\3130-CNV-002 -TR1-Fdn.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 25-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 -3 0; 2 0 -3 2; 3 0 1.49012E-008 0; 4 0 1.49012E-008 2; 5 0 0.35 0
     9. 6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7
    10. 11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75
    11. 15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7
    12. 19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7
    13. 24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75
    14. 28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5
    15. 33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25
    16. 38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25
    17. 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25
    18. 46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2
    19. 51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75
    20. 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75
    21. 60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5
    22. 64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5
    23. 69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25
    24. 73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25
    25. 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25
    26. 81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1
    27. 86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75
    28. 91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75
    29. 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75
    30. 99 -0.75 -3 0.5; 100 -1.6 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5
    31. 103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 107 1.6 -3 0.5
    32. 108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25
    33. 112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25
    34. 116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0
    35. 121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25
    36. 126 -1.6 -3 -0.25; 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25
    37. 130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25
    38. 134 -0.75 -3 -0.5; 135 -1.6 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5
```

```
39. 138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5
40. 142 1.6 -3 -0.5; 143 -0.75 -3 -0.75; 144 -1.6 -3 -0.75; 145 -0.5 -3 -0.75
41. 146 -0.25 -3 -0.75; 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75
42. 150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7
43. 154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7
44. 158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25
45. 162 -0.5 -3 -1; 163 -0.5 -3 -1.25; 164 -0.25 -3 -1; 165 -0.25 -3 -1.25
46. 166 0 -3 -1; 167 0 -3 -1.25; 168 0.25 -3 -1; 169 0.25 -3 -1.25; 170 0.5 -3 -1.
47. 171 0.5 -3 -1.25; 172 0.75 -3 -1; 173 0.75 -3 -1.25; 174 1.6 -3 -1.
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51. 188 0.5 -3 3.25; 189 0.5 -3 3; 190 0.75 -3 3.25; 191 0.75 -3 3
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53. 196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475
54. 200 0.5 -3 -1.475; 201 0.75 -3 -1.475; 202 1.6 -3 -1.475; 203 -0.75 -3 3.475
55. 204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475
56. 208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475
57. 212 1.03333 -3 2.75; 213 1.03333 -3 2.5; 214 1.31667 -3 2.75
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85. 302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475
86. MEMBER INCIDENCES
87. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
88. ELEMENT INCIDENCES SHELL
89. 15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33
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139. 268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251
140. 271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253
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142. 277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248
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144. 284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264
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146. 292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73
147. 296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272
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149. 304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277
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152. 313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283
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157. 328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295
158. 331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260
159. 334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297
160. 337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299
161. 340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303
162. 343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294
163. START GROUP DEFINITION
164. ELEMENT
165. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
166. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
167. 108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
168. 165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
169. 214 TO 345
170. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
171. 111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
172. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
173. 214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
174. _SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
175. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
176. 214 TO 345
177. END GROUP DEFINITION
178. ELEMENT PROPERTY
179. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
180. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
181. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
182. 178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
183. DEFINE MATERIAL START
184. ISOTROPIC CONCRETE
185. E 2.17185E+007
186. POISSON 0.17
187. DENSITY 23.5616
188. ALPHA 1E-005
189. DAMP 0.05
190. TYPE CONCRETE
191. STRENGTH FCU 27579
192. END DEFINE MATERIAL
193. MEMBER PROPERTY AMERICAN
194. 1 TO 4 PRIS YD 0.7 ZD 0.5
195. 5 PRIS YD 0.5 ZD 0.3
196. CONSTANTS
197. MATERIAL CONCRETE ALL
198. ****************************************
199. SUPPORTS
200. 1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
```

JOINT	INFLUENCE AREA
1	0.062500
2	0.062500

7	0.028125
8	0.028125
9	0.015937
10	0.015937
11	0.015937
12	0.015937
13	0.030000
14	0.066666
15	0.035416
16	0.028125
17	0.062500
18	0.028125
19	0.062500
20	0.062500
21	0.028125
22	0.062500
23	0.028125
24	0.062500
25	0.030000
26	0.066666
27	0.035416
28	0.066666
29	0.035416
30	0.062500
31	0.062500
32	0.062500
33	0.062500
34	0.062500
35	0.066666
36	0.035416
37	0.066666
38	0.035416
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.066666
45	0.035416
46	0.066666
47	0.035416
48	0.062500
49	0.062500
50	0.062500
51	0.062500
52	0.066666
53	0.035416
54	0.066666
55	0.035416
56	0.062500
57	0.062500
58	0.062500
59	0.062500
60	0.062500
61	0.066666
62	0.035416

63	0.066666
64	0.035416
65	0.062500
66	0.062500
67	0.062500
68	0.062500
69	0.062500
70	0.066666
71	0.035416
72	0.066666
73	0.035416
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.066666
80	0.035416
81	0.066666
82	0.035416
83	0.062500
84	0.062500
85	0.062500
86	0.062500
87	0.062500
88	0.066666
89	0.035416
90	0.066666
91	0.035416
92	0.062500
93	0.062500
94	0.062500
95	0.062500
96	0.062500
97	0.066666
98	0.035416
99	0.066666
100	0.035416
101	0.062500
102	0.062500
103	0.062500
104	0.062500
105	0.062500
106	0.066666
107	0.035416
108	0.066666
109	0.035416
110	0.062500
111	0.062500
112	0.062500
113	0.062500
114	0.062500
115	0.066666
116	0.035416
117	0.066666
118	0.035416

119	0.062500
120	0.062500
121	0.062500
122	0.062500
123	0.066666
124	0.035416
125	0.066666
126	0.035416
127	0.062500
128	0.062500
129	0.062500
130	0.062500
131	0.062500
132	0.066666
133	0.035416
134	0.066666
135	0.035416
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.066666
142	0.035416
143	0.066666
144	0.035416
145	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.066666
151	0.035416
152	0.030000
153	0.028125
154	0.028125
155	0.028125
156	0.028125
157	0.030000
158	0.066666
159	0.035416
160	0.063333
161	0.033645
162	0.062500
163	0.059375
164	0.062500
165	0.059375
166	0.062500
167	0.059375
168	0.062500
169	0.059375
170	0.062500
171	0.059375
172	0.066666
173	0.063333
174	0.035416

175	0.033645
176	0.063333
177	0.033645
178	0.066666
179	0.035416
180	0.059375
181	0.062500
182	0.059375
183	0.062500
184	0.059375
185	0.062500
186	0.059375
187	0.062500
188	0.059375
189	0.062500
190	0.063333
191	0.066666
192	0.033645
193	0.035416
194	0.060000
195	0.031875
196	0.056250
197	0.056250
198	0.056250
199	0.056250
200	0.056250
201	0.060000
202	0.031875
203	0.060000
204	0.031875
205	0.056250
206	0.056250
207	0.056250
208	0.056250
209	0.056250
210	0.060000
211	0.031875
212	0.070834
213	0.070834
214	0.070834
215	0.070834
216	0.070834
217	0.070834
218	0.070834
219	0.070834
220	0.070834
221	0.070834
222	0.070834
223	0.070834
224	0.070834
225	0.070834
226	0.070834
227	0.070834
228	0.070834
229	0.070834
230	0.070834

231	0.070834
232	0.070834
233	0.070834
234	0.070834
235	0.070834
236	0.070834
237	0.070834
238	0.070834
239	0.070834
240	0.070834
241	0.070834
242	0.070834
243	0.070834
244	0.067292
245	0.067292
246	0.067292
247	0.070834
248	0.067292
249	0.070834
250	0.063750
251	0.063750
252	0.031875
253	0.031875
254	0.031875
255	0.063750
256	0.031875
257	0.063750
258	0.070834
259	0.070834
260	0.070834
261	0.070834
262	0.070834
263	0.070834
264	0.070834
265	0.070834
266	0.070834
267	0.070834
268	0.070834
269	0.070834
270	0.070834
271	0.070834
272	0.070834
273	0.070834
274	0.070834
275	0.070834
276	0.070834
277	0.070834
278	0.070834
279	0.070834
280	0.070834
281	0.070834
282	0.070834
283	0.070834
284	0.070834
285	0.070834
286	0.070834

287	0.070834
288	0.070834
289	0.070834
290	0.067292
291	0.067292
292	0.067292
293	0.070834
294	0.067292
295	0.070834
296	0.063750
297	0.063750
298	0.031875
299	0.031875
300	0.031875
301	0.063750
302	0.031875
303	0.063750
TOTAL	17.280014

```
201. ****************************************
202. LOAD 1 DEAD LOAD DL
203. SELFWEIGHT Y -1
204. ELEMENT LOAD
205. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
206. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
207. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
208. 178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
209. JOINT LOAD
210. 5 FX 0 FY -35.5 FZ 4.6
211. 6 FX 0 FY -75.2 FZ -0.6
212. LOAD 2 LIVE LOAD
213. JOINT LOAD
214. 5 FX 0 FY -24.1 FZ -0.9
215. 6 FX 0 FY -23.9 FZ 0.9
216. LOAD 3 MATERIAL LOAD
217. JOINT LOAD
218. 5 FX 0 FY -17.1 FZ -1.7
219. 6 FX 0 FY -16.8 FZ 1.7
220. LOAD 4 WIND LOAD +Z
221. JOINT LOAD
222. 5 FX 0 FY 33.7 FZ 12.5
223. 6 FX 0 FY -33.7 FZ 12
224. LOAD 5 WIND LOAD -Z
225. JOINT LOAD
226. 5 FX -0.04 FY -33.65 FZ -12.55
227. 6 FX 0.04 FY 33.65 FZ -11.95
228. LOAD 6 WIND LOAD (+X)
229. JOINT LOAD
230. 5 FX 0 FY -0.1 FZ 0
231. 6 FX 0 FY 0.1 FZ 0
232. LOAD 7 WIND LOAD (-X)
233. JOINT LOAD
234. 5 FX 0 FY 0.1 FZ 0
235. 6 FX 0 FY -0.1 FZ 0
236. LOAD 8 TEMP LOAD 1
237. JOINT LOAD
238. 5 FX -0.01 FY -0.22 FZ -6.33
```

```
239. 6 FX 0.01 FY 0.22 FZ 6.32
240. LOAD 9 TEMP LOAD 2
241. JOINT LOAD
242. 5 FX 0.01 FY 0.22 FZ 6.33
243. 6 FX -0.01 FY -0.22 FZ -6.32
244. LOAD 10 EARTHQUAKE LOAD +X
245. JOINT LOAD
246. 5 FX 0.92 FY 0.36 FZ 0.02
247. 6 FX 0.92 FY 0.35 FZ -0.01
248. LOAD 11 EARTHQUAKE LOAD +Z
249. JOINT LOAD
250. 5 FX 0 FY 1.03 FZ 0.39
251. 6 FX 0 FY -1.03 FZ 0.37
252. ****************************************
253. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
254. LOAD 301 1.0 DL
255. REPEAT LOAD
256. 1 1.0
257. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
258. REPEAT LOAD
259. 1 1.0 8 1.0 2 1.0 3 1.0
260. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
261. REPEAT LOAD
262. 1 1.0 9 1.0 2 1.0 3 1.0
263. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
264. REPEAT LOAD
265. 1 1.0 4 0.75 2 0.75 3 0.75
266. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
267. REPEAT LOAD
268. 1 1.0 5 0.75 2 0.75 3 0.75
269. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
270. REPEAT LOAD
271. 1 1.0 6 0.75 2 0.75 3 0.75
272. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
273. REPEAT LOAD
274. 1 1.0 7 0.75 2 0.75 3 0.75
275. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
276. REPEAT LOAD
277. 1 1.0 10 0.525 2 0.75 3 0.75
278. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
279. REPEAT LOAD
280. 1 1.0 10 -0.525 2 0.75 3 0.75
281. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
282. REPEAT LOAD
283. 1 1.0 11 0.525 2 0.75 3 0.75
284. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
285. REPEAT LOAD
286. 1 1.0 11 -0.525 2 0.75 3 0.75
287. LOAD 312 0.6 DL + 1.0 WL(+Z)
288. REPEAT LOAD
289. 1 0.6 4 1.0
290. LOAD 313 0.6 DL + 1.0 WL(-Z)
291. REPEAT LOAD
292. 1 0.6 5 1.0
293. LOAD 314 0.6 DL + 1.0 WL(+X)
294. REPEAT LOAD
```

```
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
****************************************
*****LOAD COMBINATION FOR FOUNDATION DESIGN
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
```

```
351. 1 0.9 5 1.6
352. LOAD 414 0.9 DL + 1.6 WL(+X)
353. REPEAT LOAD
354. 1 0.9 6 1.6
355. LOAD 415 0.9 DL + 1.6 WL(-X)
356. REPEAT LOAD
357. 1 0.9 7 1.6
358. LOAD 416 0.9 DL + 1.0 EL(+X)
359. REPEAT LOAD
360. 1 0.9 10 1.0
361. LOAD 417 0.9 DL + 1.0 EL(-X)
362. REPEAT LOAD
363. 1 0.9 10 -1.0
364. LOAD 418 0.9 DL + 1.0 EL(+Z)
365. REPEAT LOAD
366. 1 0.9 11 1.0
367. LOAD 419 0.9 DL + 1.0 EL(-Z)
368. REPEAT LOAD
369. 1 0.9 11 -1.0
370. ****************************************
371. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**303**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**264**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**299**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 921
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

```
372. ****************************************
373. LOAD LIST 401 TO 419
374. START CONCRETE DESIGN
```

```
375. CODE ACI 2008
376. FC 35000 MEMB 1 TO 5
377. FYMAIN 420000 MEMB 1 TO 5
378. TRACK 1 ALL
379. CLB 0.05 MEMB 1 TO 5
380. CLT 0.05 MEMB 1 TO 5
381. CLS 0.05 ALL
382. MAXMAIN 25 ALL
383. MINMAIN 16 ALL
384. MINSEC 8 ALL
385. REINF 0 MEMB 1 TO 4
386. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	84.92	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	84.92	0.27	0.00105

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37     76.46      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37     76.46      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 700.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3500.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 12 - 20 MM               1.077      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    181.17      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    181.17      7.99    0.01470
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	161.97	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	161.97	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

387. DESIGN BEAM 5

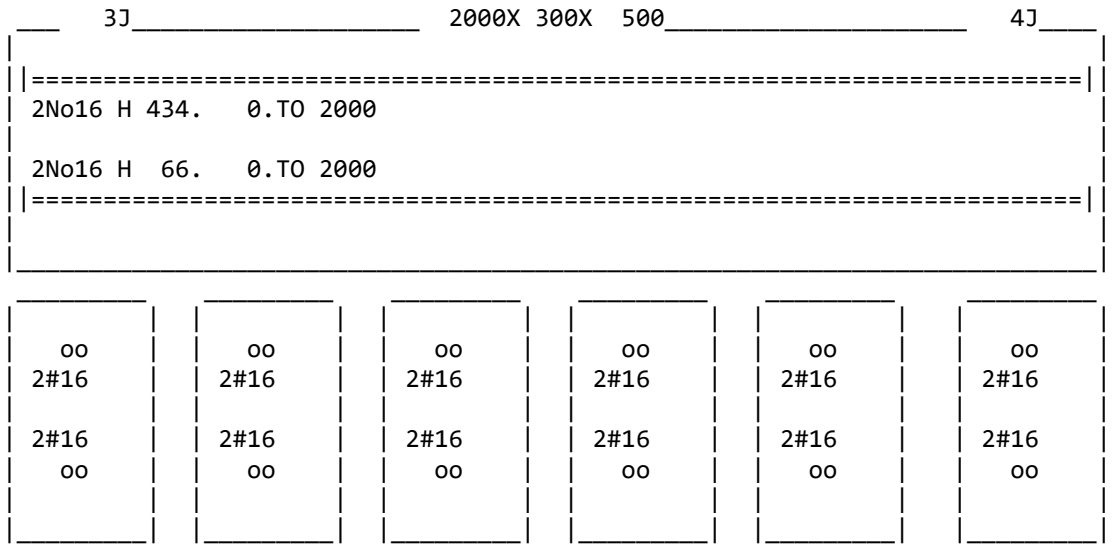
BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	2000.	YES	YES
	CRITICAL POS MOMENT= 29.76 KN-MET AT 0.MM, LOAD 404					
	REQD STEEL= 244.MM2, RHO=0.0019, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
2	434.	2 - 16MM	0.	2000.	YES	YES
	CRITICAL NEG MOMENT= 32.55 KN-MET AT 2000.MM, LOAD 404					
	REQD STEEL= 267.MM2, RHO=0.0020, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 0.69 KNS Vc= 122.74 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 6.29 KNS Vc= 143.40 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

388. END CONCRETE DESIGN
389. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 25,2020 TIME= 20:36:24 ****

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     ************************************************************
     *   For technical assistance on STAAD.Pro, please visit    *
     *   http://selectservices.bentley.com/en-US/               *
     *                                                          *
     *   Details about additional assistance from               *
     *   Bentley and Partners can be found at program menu      *
     *   Help->Technical Support                                *
     *                                                          *
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     *                http://www.bentley.com                    *
     ************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	26-03-2020	24-04-2020	01-11-2023				Date	26-03-2020	24-04-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 5 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-2**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-Mar-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 2; 3 0 1.49012e-008 0; 4 0 1.49012e-008 2; 5 0 0.35 0;
6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7;
11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75;
15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7;
19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7;
24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75;
28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5;
33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25;
38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25;
42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25;
46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2;
51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75;
56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75;
60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5;
64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5;
69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25;
73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25;
77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25;
81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1;
86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75;
91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75;
95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75;
99 -0.75 -3 0.5; 100 -1.6 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5;
103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 107 1.6 -3 0.5;
108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25;
112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25;
116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0;
121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25;
126 -1.6 -3 -0.25; 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25;
130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25;
134 -0.75 -3 -0.5; 135 -1.6 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5;
138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5;
142 1.6 -3 -0.5; 143 -0.75 -3 -0.75; 144 -1.6 -3 -0.75; 145 -0.5 -3 -0.75;
146 -0.25 -3 -0.75; 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75;
150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7;
154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7;
158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25;
162 -0.5 -3 -1; 163 -0.5 -3 -1.25; 164 -0.25 -3 -1; 165 -0.25 -3 -1.25;
166 0 -3 -1; 167 0 -3 -1.25; 168 0.25 -3 -1; 169 0.25 -3 -1.25; 170 0.5 -3 -1;
171 0.5 -3 -1.25; 172 0.75 -3 -1; 173 0.75 -3 -1.25; 174 1.6 -3 -1;
175 1.6 -3 -1.25; 176 -0.75 -3 3.25; 177 -1.6 -3 3.25; 178 -0.75 -3 3;
179 -1.6 -3 3; 180 -0.5 -3 3.25; 181 -0.5 -3 3; 182 -0.25 -3 3.25;
183 -0.25 -3 3; 184 0 -3 3.25; 185 0 -3 3; 186 0.25 -3 3.25; 187 0.25 -3 3;
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188 0.5 -3 3.25; 189 0.5 -3 3; 190 0.75 -3 3.25; 191 0.75 -3 3;
192 1.6 -3 3.25; 193 1.6 -3 3; 194 -0.75 -3 -1.475; 195 -1.6 -3 -1.475;
196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475;
200 0.5 -3 -1.475; 201 0.75 -3 -1.475; 202 1.6 -3 -1.475; 203 -0.75 -3 3.475;
204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475;
208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475;
212 1.03333 -3 2.75; 213 1.03333 -3 2.5; 214 1.31667 -3 2.75;
215 1.31667 -3 2.5; 216 1.03333 -3 2.25; 217 1.31667 -3 2.25; 218 1.03333 -3 2;
219 1.31667 -3 2; 220 1.03333 -3 1.75; 221 1.31667 -3 1.75; 222 1.03333 -3 1.5;
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227 1.31667 -3 1; 228 1.03333 -3 0.75; 229 1.31667 -3 0.75; 230 1.03333 -3 0.5;
231 1.31667 -3 0.5; 232 1.03333 -3 0.25; 233 1.31667 -3 0.25; 234 1.03333 -3 0;
235 1.31667 -3 0; 236 1.03333 -3 -0.25; 237 1.31667 -3 -0.25;
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241 1.31667 -3 -0.75; 242 1.03333 -3 -1; 243 1.31667 -3 -1;
244 1.03333 -3 -1.25; 245 1.31667 -3 -1.25; 246 1.03333 -3 3.25;
247 1.03333 -3 3; 248 1.31667 -3 3.25; 249 1.31667 -3 3; 250 1.03333 -3 -1.475;
251 1.31667 -3 -1.475; 252 1.03333 -3 -1.7; 253 1.31667 -3 -1.7;
254 1.03333 -3 3.7; 255 1.03333 -3 3.475; 256 1.31667 -3 3.7;
257 1.31667 -3 3.475; 258 -1.31667 -3 2.75; 259 -1.31667 -3 2.5;
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263 -1.03333 -3 2.25; 264 -1.31667 -3 2; 265 -1.03333 -3 2;
266 -1.31667 -3 1.75; 267 -1.03333 -3 1.75; 268 -1.31667 -3 1.5;
269 -1.03333 -3 1.5; 270 -1.31667 -3 1.25; 271 -1.03333 -3 1.25;
272 -1.31667 -3 1; 273 -1.03333 -3 1; 274 -1.31667 -3 0.75;
275 -1.03333 -3 0.75; 276 -1.31667 -3 0.5; 277 -1.03333 -3 0.5;
278 -1.31667 -3 0.25; 279 -1.03333 -3 0.25; 280 -1.31667 -3 0;
281 -1.03333 -3 0; 282 -1.31667 -3 -0.25; 283 -1.03333 -3 -0.25;
284 -1.31667 -3 -0.5; 285 -1.03333 -3 -0.5; 286 -1.31667 -3 -0.75;
287 -1.03333 -3 -0.75; 288 -1.31667 -3 -1; 289 -1.03333 -3 -1;
290 -1.31667 -3 -1.25; 291 -1.03333 -3 -1.25; 292 -1.31667 -3 3.25;
293 -1.31667 -3 3; 294 -1.03333 -3 3.25; 295 -1.03333 -3 3;
296 -1.31667 -3 -1.475; 297 -1.03333 -3 -1.475; 298 -1.31667 -3 -1.7;
299 -1.03333 -3 -1.7; 300 -1.31667 -3 3.7; 301 -1.31667 -3 3.475;
302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
ELEMENT INCIDENCES SHELL
15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33;
20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41;
27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49;
34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56;
41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63;
48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69;
55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77;
60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85;
67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93;
74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99;
80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104;
84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111;

90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117;
96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121;
100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127;
105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131;
111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137;
114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140;
119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146;
122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149;
137 143 145 162 158; 138 158 162 163 160; 140 145 146 164 162;
141 162 164 165 163; 143 146 147 166 164; 144 164 166 167 165;
146 147 148 168 166; 147 166 168 169 167; 149 148 149 170 168;
150 168 170 171 169; 152 149 150 172 170; 153 170 172 173 171;
162 176 180 181 178; 163 178 181 17 14; 165 180 182 183 181; 166 181 183 19 17;
168 182 184 185 183; 169 183 185 20 19; 171 184 186 187 185; 172 185 187 22 20;
174 186 188 189 187; 175 187 189 24 22; 177 188 190 191 189; 178 189 191 26 24;
184 160 163 196 194; 185 194 196 153 152; 186 163 165 197 196;
187 196 197 154 153; 188 165 167 198 197; 189 197 198 7 154;
190 167 169 199 198; 191 198 199 155 7; 192 169 171 200 199;
193 199 200 156 155; 194 171 173 201 200; 195 200 201 157 156;
200 13 16 205 203; 201 203 205 180 176; 202 16 18 206 205; 203 205 206 182 180;
204 18 8 207 206; 205 206 207 184 182; 206 8 21 208 207; 207 207 208 186 184;
208 21 23 209 208; 209 208 209 188 186; 210 23 25 210 209; 211 209 210 190 188;
214 26 212 213 35; 215 212 214 215 213; 216 214 27 36 215; 217 35 213 216 44;
218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218;
222 217 45 53 219; 223 52 218 220 61; 224 218 219 221 220; 225 219 53 62 221;
226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79;
230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226;
234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229;
238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231;
241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233;
244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235;
247 123 234 236 132; 248 234 235 237 236; 249 235 124 133 237;
250 132 236 238 141; 251 236 237 239 238; 252 237 133 142 239;
253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241;
256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243;
259 172 242 244 173; 260 242 243 245 244; 261 243 174 175 245;
262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249;
265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214;
268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251;
271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253;
274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257;
277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248;
280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38;
284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264;
288 263 37 46 265; 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267;
292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73;
296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272;
300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275;
304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277;
307 100 276 278 109; 308 276 277 279 278; 309 277 99 108 279;

```
310 109 278 280 118; 311 278 279 281 280; 312 279 108 117 281;
313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283;
316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285;
319 135 284 286 144; 320 284 285 287 286; 321 285 134 143 287;
322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289;
325 159 288 290 161; 326 288 289 291 290; 327 289 158 160 291;
328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295;
331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260;
334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297;
337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299;
340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303;
343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
214 TO 345
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
_SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 345
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
```

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*****************************************
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
joint load
5   FX  -1.3    FY  -79.5   FZ  4.7
6   FX  -2.1    FY  -183.3  FZ  4.9

Load 2   LIVE LOAD
joint load
5   FX  -1.0    FY  -60.0   FZ  0.0
6   FX  -0.9    FY  -59.9   FZ  0.0

Load 3  MATERIAL LOAD
joint load
5   FX  -0.6    FY  -41.0   FZ  0.0
6   FX  -0.6    FY  -40.9   FZ  0.0

Load 4  WIND LOAD +Z
joint load
5   FX  1.2 FY  106.7   FZ  32.0
6   FX  -1.2    FY  -106.7  FZ  32.0

Load 5  WIND LOAD -Z
joint load
5   FX  -1.24   FY  -106.63 FZ  -31.94
6   FX  1.24    FY  106.63  FZ  -31.94

Load 6  WIND LOAD (+X)
joint load
5   FX  0.47    FY  0   FZ  0
6   FX  0.47    FY  0   FZ  0

Load 7  WIND LOAD (-X)
joint load
5   FX  -0.47   FY  0   FZ  0
6   FX  -0.47   FY  0   FZ  0

Load 8  TEMP LOAD 1
joint load
5   FX  -0.01   FY  -0.03   FZ  -0.01
6   FX  0.01    FY  0.03    FZ  0.02

Load 9  TEMP LOAD 2
joint load
5   FX  0.01    FY  0.03    FZ  0.01
```

```
6   FX  -0.01   FY  -0.03   FZ  -0.02

Load 10 EARTHQUAKE LOAD +X
joint load
5   FX  0.04    FY  -0.12   FZ  -0.03
6   FX  0.05    FY  0.02    FZ  -0.02

Load 11 EARTHQUAKE LOAD +Z
joint load
5   FX  -0.02   FY  7.38    FZ  3.58
6   FX  0.02    FY  -7.38   FZ  3.58

****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
```

```
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
```

```
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 25, 2020                *
*           Time=    20:36:19                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Conveyor CNV-3130-002\Civil\Staad\3130-CNV-002 -TR1-Fdn.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 25-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 -3 0; 2 0 -3 2; 3 0 1.49012E-008 0; 4 0 1.49012E-008 2; 5 0 0.35 0
     9. 6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7
    10. 11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75
    11. 15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7
    12. 19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7
    13. 24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75
    14. 28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5
    15. 33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25
    16. 38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25
    17. 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25
    18. 46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2
    19. 51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75
    20. 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75
    21. 60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5
    22. 64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5
    23. 69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25
    24. 73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25
    25. 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25
    26. 81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1
    27. 86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75
    28. 91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75
    29. 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75
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    32. 108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25
    33. 112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25
    34. 116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0
    35. 121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25
    36. 126 -1.6 -3 -0.25; 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25
    37. 130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25
    38. 134 -0.75 -3 -0.5; 135 -1.6 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5
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40. 142 1.6 -3 -0.5; 143 -0.75 -3 -0.75; 144 -1.6 -3 -0.75; 145 -0.5 -3 -0.75
41. 146 -0.25 -3 -0.75; 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75
42. 150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7
43. 154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7
44. 158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25
45. 162 -0.5 -3 -1; 163 -0.5 -3 -1.25; 164 -0.25 -3 -1; 165 -0.25 -3 -1.25
46. 166 0 -3 -1; 167 0 -3 -1.25; 168 0.25 -3 -1; 169 0.25 -3 -1.25; 170 0.5 -3 -1.
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52. 192 1.6 -3 3.25; 193 1.6 -3 3; 194 -0.75 -3 -1.475; 195 -1.6 -3 -1.475
53. 196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475
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55. 204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475
56. 208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475
57. 212 1.03333 -3 2.75; 213 1.03333 -3 2.5; 214 1.31667 -3 2.75
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68. 251 1.31667 -3 -1.475; 252 1.03333 -3 -1.7; 253 1.31667 -3 -1.7
69. 254 1.03333 -3 3.7; 255 1.03333 -3 3.475; 256 1.31667 -3 3.7
70. 257 1.31667 -3 3.475; 258 -1.31667 -3 2.75; 259 -1.31667 -3 2.5
71. 260 -1.03333 -3 2.75; 261 -1.03333 -3 2.5; 262 -1.31667 -3 2.25
72. 263 -1.03333 -3 2.25; 264 -1.31667 -3 2; 265 -1.03333 -3 2
73. 266 -1.31667 -3 1.75; 267 -1.03333 -3 1.75; 268 -1.31667 -3 1.5
74. 269 -1.03333 -3 1.5; 270 -1.31667 -3 1.25; 271 -1.03333 -3 1.25
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76. 275 -1.03333 -3 0.75; 276 -1.31667 -3 0.5; 277 -1.03333 -3 0.5
77. 278 -1.31667 -3 0.25; 279 -1.03333 -3 0.25; 280 -1.31667 -3 0
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85. 302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475
86. MEMBER INCIDENCES
87. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
88. ELEMENT INCIDENCES SHELL
89. 15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33
90. 20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41
91. 27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49
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93. 41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63
94. 48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69
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95. 55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77
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97. 67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93
98. 74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99
99. 80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104
100. 84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111
101. 90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117
102. 96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121
103. 100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127
104. 105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131
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129. 238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231
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131. 244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235
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137. 262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249
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139. 268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251
140. 271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253
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142. 277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248
143. 280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38
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146. 292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73
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152. 313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283
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157. 328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295
158. 331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260
159. 334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297
160. 337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299
161. 340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303
162. 343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294
163. START GROUP DEFINITION
164. ELEMENT
165. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
166. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
167. 108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
168. 165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
169. 214 TO 345
170. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
171. 111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
172. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
173. 214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
174. _SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
175. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
176. 214 TO 345
177. END GROUP DEFINITION
178. ELEMENT PROPERTY
179. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
180. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
181. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
182. 178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
183. DEFINE MATERIAL START
184. ISOTROPIC CONCRETE
185. E 2.17185E+007
186. POISSON 0.17
187. DENSITY 23.5616
188. ALPHA 1E-005
189. DAMP 0.05
190. TYPE CONCRETE
191. STRENGTH FCU 27579
192. END DEFINE MATERIAL
193. MEMBER PROPERTY AMERICAN
194. 1 TO 4 PRIS YD 0.7 ZD 0.5
195. 5 PRIS YD 0.5 ZD 0.3
196. CONSTANTS
197. MATERIAL CONCRETE ALL
198. ****************************************
199. SUPPORTS
200. 1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
```

JOINT	INFLUENCE AREA
1	0.062500
2	0.062500

7	0.028125
8	0.028125
9	0.015937
10	0.015937
11	0.015937
12	0.015937
13	0.030000
14	0.066666
15	0.035416
16	0.028125
17	0.062500
18	0.028125
19	0.062500
20	0.062500
21	0.028125
22	0.062500
23	0.028125
24	0.062500
25	0.030000
26	0.066666
27	0.035416
28	0.066666
29	0.035416
30	0.062500
31	0.062500
32	0.062500
33	0.062500
34	0.062500
35	0.066666
36	0.035416
37	0.066666
38	0.035416
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.066666
45	0.035416
46	0.066666
47	0.035416
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53	0.035416
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62	0.035416

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79	0.066666
80	0.035416
81	0.066666
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83	0.062500
84	0.062500
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86	0.062500
87	0.062500
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89	0.035416
90	0.066666
91	0.035416
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94	0.062500
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98	0.035416
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100	0.035416
101	0.062500
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105	0.062500
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110	0.062500
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112	0.062500
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114	0.062500
115	0.066666
116	0.035416
117	0.066666
118	0.035416

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131	0.062500
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134	0.066666
135	0.035416
136	0.062500
137	0.062500
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139	0.062500
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141	0.066666
142	0.035416
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145	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.066666
151	0.035416
152	0.030000
153	0.028125
154	0.028125
155	0.028125
156	0.028125
157	0.030000
158	0.066666
159	0.035416
160	0.063333
161	0.033645
162	0.062500
163	0.059375
164	0.062500
165	0.059375
166	0.062500
167	0.059375
168	0.062500
169	0.059375
170	0.062500
171	0.059375
172	0.066666
173	0.063333
174	0.035416

175	0.033645
176	0.063333
177	0.033645
178	0.066666
179	0.035416
180	0.059375
181	0.062500
182	0.059375
183	0.062500
184	0.059375
185	0.062500
186	0.059375
187	0.062500
188	0.059375
189	0.062500
190	0.063333
191	0.066666
192	0.033645
193	0.035416
194	0.060000
195	0.031875
196	0.056250
197	0.056250
198	0.056250
199	0.056250
200	0.056250
201	0.060000
202	0.031875
203	0.060000
204	0.031875
205	0.056250
206	0.056250
207	0.056250
208	0.056250
209	0.056250
210	0.060000
211	0.031875
212	0.070834
213	0.070834
214	0.070834
215	0.070834
216	0.070834
217	0.070834
218	0.070834
219	0.070834
220	0.070834
221	0.070834
222	0.070834
223	0.070834
224	0.070834
225	0.070834
226	0.070834
227	0.070834
228	0.070834
229	0.070834
230	0.070834

231	0.070834
232	0.070834
233	0.070834
234	0.070834
235	0.070834
236	0.070834
237	0.070834
238	0.070834
239	0.070834
240	0.070834
241	0.070834
242	0.070834
243	0.070834
244	0.067292
245	0.067292
246	0.067292
247	0.070834
248	0.067292
249	0.070834
250	0.063750
251	0.063750
252	0.031875
253	0.031875
254	0.031875
255	0.063750
256	0.031875
257	0.063750
258	0.070834
259	0.070834
260	0.070834
261	0.070834
262	0.070834
263	0.070834
264	0.070834
265	0.070834
266	0.070834
267	0.070834
268	0.070834
269	0.070834
270	0.070834
271	0.070834
272	0.070834
273	0.070834
274	0.070834
275	0.070834
276	0.070834
277	0.070834
278	0.070834
279	0.070834
280	0.070834
281	0.070834
282	0.070834
283	0.070834
284	0.070834
285	0.070834
286	0.070834

287	0.070834
288	0.070834
289	0.070834
290	0.067292
291	0.067292
292	0.067292
293	0.070834
294	0.067292
295	0.070834
296	0.063750
297	0.063750
298	0.031875
299	0.031875
300	0.031875
301	0.063750
302	0.031875
303	0.063750
TOTAL	17.280014

```
201. ****************************************
202. LOAD 1 DEAD LOAD DL
203. SELFWEIGHT Y -1
204. ELEMENT LOAD
205. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
206. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
207. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
208. 178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
209. JOINT LOAD
210. 5 FX 0 FY -35.5 FZ 4.6
211. 6 FX 0 FY -75.2 FZ -0.6
212. LOAD 2 LIVE LOAD
213. JOINT LOAD
214. 5 FX 0 FY -24.1 FZ -0.9
215. 6 FX 0 FY -23.9 FZ 0.9
216. LOAD 3 MATERIAL LOAD
217. JOINT LOAD
218. 5 FX 0 FY -17.1 FZ -1.7
219. 6 FX 0 FY -16.8 FZ 1.7
220. LOAD 4 WIND LOAD +Z
221. JOINT LOAD
222. 5 FX 0 FY 33.7 FZ 12.5
223. 6 FX 0 FY -33.7 FZ 12
224. LOAD 5 WIND LOAD -Z
225. JOINT LOAD
226. 5 FX -0.04 FY -33.65 FZ -12.55
227. 6 FX 0.04 FY 33.65 FZ -11.95
228. LOAD 6 WIND LOAD (+X)
229. JOINT LOAD
230. 5 FX 0 FY -0.1 FZ 0
231. 6 FX 0 FY 0.1 FZ 0
232. LOAD 7 WIND LOAD (-X)
233. JOINT LOAD
234. 5 FX 0 FY 0.1 FZ 0
235. 6 FX 0 FY -0.1 FZ 0
236. LOAD 8 TEMP LOAD 1
237. JOINT LOAD
238. 5 FX -0.01 FY -0.22 FZ -6.33
```

```
239. 6 FX 0.01 FY 0.22 FZ 6.32
240. LOAD 9 TEMP LOAD 2
241. JOINT LOAD
242. 5 FX 0.01 FY 0.22 FZ 6.33
243. 6 FX -0.01 FY -0.22 FZ -6.32
244. LOAD 10 EARTHQUAKE LOAD +X
245. JOINT LOAD
246. 5 FX 0.92 FY 0.36 FZ 0.02
247. 6 FX 0.92 FY 0.35 FZ -0.01
248. LOAD 11 EARTHQUAKE LOAD +Z
249. JOINT LOAD
250. 5 FX 0 FY 1.03 FZ 0.39
251. 6 FX 0 FY -1.03 FZ 0.37
252. ****************************************
253. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
254. LOAD 301 1.0 DL
255. REPEAT LOAD
256. 1 1.0
257. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
258. REPEAT LOAD
259. 1 1.0 8 1.0 2 1.0 3 1.0
260. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
261. REPEAT LOAD
262. 1 1.0 9 1.0 2 1.0 3 1.0
263. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
264. REPEAT LOAD
265. 1 1.0 4 0.75 2 0.75 3 0.75
266. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
267. REPEAT LOAD
268. 1 1.0 5 0.75 2 0.75 3 0.75
269. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
270. REPEAT LOAD
271. 1 1.0 6 0.75 2 0.75 3 0.75
272. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
273. REPEAT LOAD
274. 1 1.0 7 0.75 2 0.75 3 0.75
275. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
276. REPEAT LOAD
277. 1 1.0 10 0.525 2 0.75 3 0.75
278. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
279. REPEAT LOAD
280. 1 1.0 10 -0.525 2 0.75 3 0.75
281. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
282. REPEAT LOAD
283. 1 1.0 11 0.525 2 0.75 3 0.75
284. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
285. REPEAT LOAD
286. 1 1.0 11 -0.525 2 0.75 3 0.75
287. LOAD 312 0.6 DL + 1.0 WL(+Z)
288. REPEAT LOAD
289. 1 0.6 4 1.0
290. LOAD 313 0.6 DL + 1.0 WL(-Z)
291. REPEAT LOAD
292. 1 0.6 5 1.0
293. LOAD 314 0.6 DL + 1.0 WL(+X)
294. REPEAT LOAD
```

```
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
****************************************
*****LOAD COMBINATION FOR FOUNDATION DESIGN
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
```

```
351. 1 0.9 5 1.6
352. LOAD 414 0.9 DL + 1.6 WL(+X)
353. REPEAT LOAD
354. 1 0.9 6 1.6
355. LOAD 415 0.9 DL + 1.6 WL(-X)
356. REPEAT LOAD
357. 1 0.9 7 1.6
358. LOAD 416 0.9 DL + 1.0 EL(+X)
359. REPEAT LOAD
360. 1 0.9 10 1.0
361. LOAD 417 0.9 DL + 1.0 EL(-X)
362. REPEAT LOAD
363. 1 0.9 10 -1.0
364. LOAD 418 0.9 DL + 1.0 EL(+Z)
365. REPEAT LOAD
366. 1 0.9 11 1.0
367. LOAD 419 0.9 DL + 1.0 EL(-Z)
368. REPEAT LOAD
369. 1 0.9 11 -1.0
370. ****************************************
371. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**303**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**264**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**299**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 921
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

```
372. ****************************************
373. LOAD LIST 401 TO 419
374. START CONCRETE DESIGN
```

```
375. CODE ACI 2008
376. FC 35000 MEMB 1 TO 5
377. FYMAIN 420000 MEMB 1 TO 5
378. TRACK 1 ALL
379. CLB 0.05 MEMB 1 TO 5
380. CLT 0.05 MEMB 1 TO 5
381. CLS 0.05 ALL
382. MAXMAIN 25 ALL
383. MINMAIN 16 ALL
384. MINSEC 8 ALL
385. REINF 0 MEMB 1 TO 4
386. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	84.92	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	84.92	0.27	0.00105

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37     76.46      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37     76.46      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 700.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3500.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 12 - 20 MM               1.077      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    181.17      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    181.17      7.99    0.01470
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	161.97	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	161.97	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

387. DESIGN BEAM 5

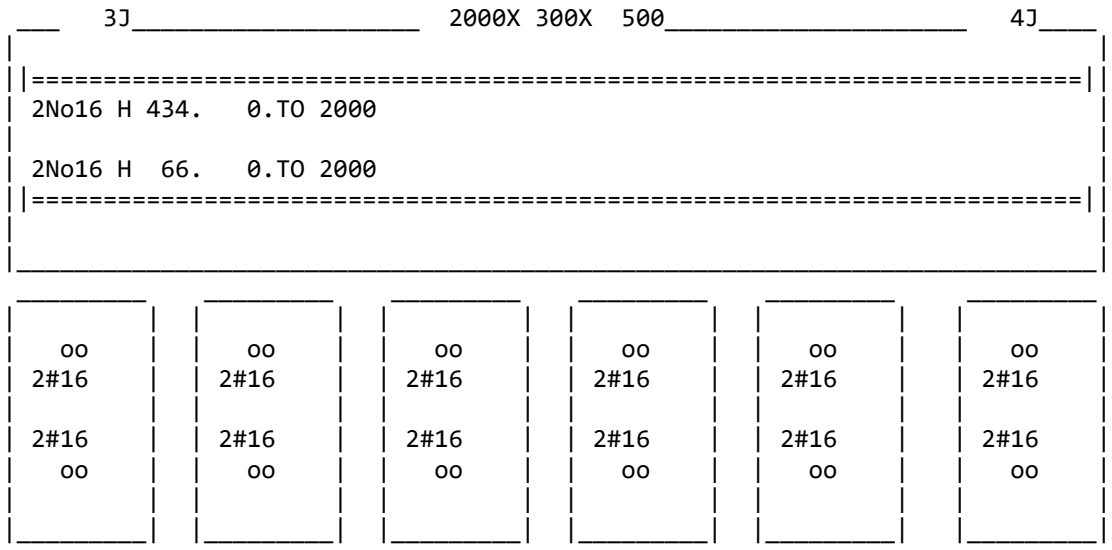
BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	2000.	YES	YES
	CRITICAL POS MOMENT= 29.76 KN-MET AT 0.MM, LOAD 404					
	REQD STEEL= 244.MM2, RHO=0.0019, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
2	434.	2 - 16MM	0.	2000.	YES	YES
	CRITICAL NEG MOMENT= 32.55 KN-MET AT 2000.MM, LOAD 404					
	REQD STEEL= 267.MM2, RHO=0.0020, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 0.69 KNS Vc= 122.74 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 6.29 KNS Vc= 143.40 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

388. END CONCRETE DESIGN
389. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 25,2020 TIME= 20:36:24 ****

```
     ************************************************************
     *   For technical assistance on STAAD.Pro, please visit    *
     *   http://selectservices.bentley.com/en-US/               *
     *                                                          *
     *   Details about additional assistance from               *
     *   Bentley and Partners can be found at program menu      *
     *   Help->Technical Support                                *
     *                                                          *
     *      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
     *                http://www.bentley.com                    *
     ************************************************************
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	26-03-2020	24-04-2020	01-11-2023				Date	26-03-2020	24-04-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 6 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-3**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-Mar-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 2; 3 0 1.49012e-008 0; 4 0 1.49012e-008 2; 5 0 0.35 0;
6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7;
11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75;
15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7;
19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7;
24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75;
28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5;
33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25;
38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25;
42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25;
46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2;
51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75;
56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75;
60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5;
64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5;
69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25;
73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25;
77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25;
81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1;
86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75;
91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75;
95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75;
99 -0.75 -3 0.5; 100 -1.6 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5;
103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 107 1.6 -3 0.5;
108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25;
112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25;
116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0;
121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25;
126 -1.6 -3 -0.25; 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25;
130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25;
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150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7;
154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7;
158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25;
162 -0.5 -3 -1; 163 -0.5 -3 -1.25; 164 -0.25 -3 -1; 165 -0.25 -3 -1.25;
166 0 -3 -1; 167 0 -3 -1.25; 168 0.25 -3 -1; 169 0.25 -3 -1.25; 170 0.5 -3 -1;
171 0.5 -3 -1.25; 172 0.75 -3 -1; 173 0.75 -3 -1.25; 174 1.6 -3 -1;
175 1.6 -3 -1.25; 176 -0.75 -3 3.25; 177 -1.6 -3 3.25; 178 -0.75 -3 3;
179 -1.6 -3 3; 180 -0.5 -3 3.25; 181 -0.5 -3 3; 182 -0.25 -3 3.25;
183 -0.25 -3 3; 184 0 -3 3.25; 185 0 -3 3; 186 0.25 -3 3.25; 187 0.25 -3 3;
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188 0.5 -3 3.25; 189 0.5 -3 3; 190 0.75 -3 3.25; 191 0.75 -3 3;
192 1.6 -3 3.25; 193 1.6 -3 3; 194 -0.75 -3 -1.475; 195 -1.6 -3 -1.475;
196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475;
200 0.5 -3 -1.475; 201 0.75 -3 -1.475; 202 1.6 -3 -1.475; 203 -0.75 -3 3.475;
204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475;
208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475;
212 1.03333 -3 2.75; 213 1.03333 -3 2.5; 214 1.31667 -3 2.75;
215 1.31667 -3 2.5; 216 1.03333 -3 2.25; 217 1.31667 -3 2.25; 218 1.03333 -3 2;
219 1.31667 -3 2; 220 1.03333 -3 1.75; 221 1.31667 -3 1.75; 222 1.03333 -3 1.5;
223 1.31667 -3 1.5; 224 1.03333 -3 1.25; 225 1.31667 -3 1.25; 226 1.03333 -3 1;
227 1.31667 -3 1; 228 1.03333 -3 0.75; 229 1.31667 -3 0.75; 230 1.03333 -3 0.5;
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244 1.03333 -3 -1.25; 245 1.31667 -3 -1.25; 246 1.03333 -3 3.25;
247 1.03333 -3 3; 248 1.31667 -3 3.25; 249 1.31667 -3 3; 250 1.03333 -3 -1.475;
251 1.31667 -3 -1.475; 252 1.03333 -3 -1.7; 253 1.31667 -3 -1.7;
254 1.03333 -3 3.7; 255 1.03333 -3 3.475; 256 1.31667 -3 3.7;
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263 -1.03333 -3 2.25; 264 -1.31667 -3 2; 265 -1.03333 -3 2;
266 -1.31667 -3 1.75; 267 -1.03333 -3 1.75; 268 -1.31667 -3 1.5;
269 -1.03333 -3 1.5; 270 -1.31667 -3 1.25; 271 -1.03333 -3 1.25;
272 -1.31667 -3 1; 273 -1.03333 -3 1; 274 -1.31667 -3 0.75;
275 -1.03333 -3 0.75; 276 -1.31667 -3 0.5; 277 -1.03333 -3 0.5;
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287 -1.03333 -3 -0.75; 288 -1.31667 -3 -1; 289 -1.03333 -3 -1;
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293 -1.31667 -3 3; 294 -1.03333 -3 3.25; 295 -1.03333 -3 3;
296 -1.31667 -3 -1.475; 297 -1.03333 -3 -1.475; 298 -1.31667 -3 -1.7;
299 -1.03333 -3 -1.7; 300 -1.31667 -3 3.7; 301 -1.31667 -3 3.475;
302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
ELEMENT INCIDENCES SHELL
15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33;
20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41;
27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49;
34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56;
41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63;
48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69;
55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77;
60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85;
67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93;
74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99;
80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104;
84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111;

90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117;
96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121;
100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127;
105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131;
111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137;
114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140;
119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146;
122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149;
137 143 145 162 158; 138 158 162 163 160; 140 145 146 164 162;
141 162 164 165 163; 143 146 147 166 164; 144 164 166 167 165;
146 147 148 168 166; 147 166 168 169 167; 149 148 149 170 168;
150 168 170 171 169; 152 149 150 172 170; 153 170 172 173 171;
162 176 180 181 178; 163 178 181 17 14; 165 180 182 183 181; 166 181 183 19 17;
168 182 184 185 183; 169 183 185 20 19; 171 184 186 187 185; 172 185 187 22 20;
174 186 188 189 187; 175 187 189 24 22; 177 188 190 191 189; 178 189 191 26 24;
184 160 163 196 194; 185 194 196 153 152; 186 163 165 197 196;
187 196 197 154 153; 188 165 167 198 197; 189 197 198 7 154;
190 167 169 199 198; 191 198 199 155 7; 192 169 171 200 199;
193 199 200 156 155; 194 171 173 201 200; 195 200 201 157 156;
200 13 16 205 203; 201 203 205 180 176; 202 16 18 206 205; 203 205 206 182 180;
204 18 8 207 206; 205 206 207 184 182; 206 8 21 208 207; 207 207 208 186 184;
208 21 23 209 208; 209 208 209 188 186; 210 23 25 210 209; 211 209 210 190 188;
214 26 212 213 35; 215 212 214 215 213; 216 214 27 36 215; 217 35 213 216 44;
218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218;
222 217 45 53 219; 223 52 218 220 61; 224 218 219 221 220; 225 219 53 62 221;
226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79;
230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226;
234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229;
238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231;
241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233;
244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235;
247 123 234 236 132; 248 234 235 237 236; 249 235 124 133 237;
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253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241;
256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243;
259 172 242 244 173; 260 242 243 245 244; 261 243 174 175 245;
262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249;
265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214;
268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251;
271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253;
274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257;
277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248;
280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38;
284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264;
288 263 37 46 265; 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267;
292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73;
296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272;
300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275;
304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277;
307 100 276 278 109; 308 276 277 279 278; 309 277 99 108 279;

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310 109 278 280 118; 311 278 279 281 280; 312 279 108 117 281;
313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283;
316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285;
319 135 284 286 144; 320 284 285 287 286; 321 285 134 143 287;
322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289;
325 159 288 290 161; 326 288 289 291 290; 327 289 158 160 291;
328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295;
331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260;
334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297;
337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299;
340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303;
343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
214 TO 345
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
_SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 345
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
```

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*****************************************
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
joint load
5   FX  -0.1    FY  -77.7   FZ  -2.3
6   FX  0.0 FY  -234.1  FZ  13.6

Load 2   LIVE LOAD
joint load
5   FX  0.0 FY  -71.9   FZ  -3.7
6   FX  0.0 FY  -72.1   FZ  3.7

Load 3  MATERIAL LOAD
joint load
5   FX  0.0 FY  -48.3   FZ  -2.5
6   FX  0.0 FY  -47.9   FZ  2.4

Load 4  WIND LOAD +Z
joint load
5   FX  0.0 FY  267.2   FZ  40.7
6   FX  0.0 FY  -267.2  FZ  40.7

Load 5  WIND LOAD -Z
joint load
5   FX  -0.03   FY  -269.88 FZ  -41.76
6   FX  0.03    FY  269.88  FZ  -41.71

Load 6  WIND LOAD (+X)
joint load
5   FX  1.52    FY  0   FZ  0
6   FX  1.52    FY  0   FZ  0

Load 7  WIND LOAD (-X)
joint load
5   FX  -1.52   FY  0   FZ  0
6   FX  -1.52   FY  0   FZ  0

Load 8  TEMP LOAD 1
joint load
5   FX  0   FY  -0.08   FZ  -17.66
6   FX  0   FY  0.08    FZ  17.65

Load 9  TEMP LOAD 2
joint load
5   FX  0   FY  0.08    FZ  17.66
```

```
6   FX  0   FY  -0.08   FZ  -17.65

Load 10 EARTHQUAKE LOAD +X
joint load
5   FX  0.02    FY  -0.06   FZ  0
6   FX  0.02    FY  0.05    FZ  0

Load 11 EARTHQUAKE LOAD +Z
joint load
5   FX  -0.01   FY  13.72   FZ  2.35
6   FX  0.01    FY  -13.72  FZ  2.35


****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
```

```
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
```

```
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
```

FINISH

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 25, 2020                *
*           Time=    20:22:15                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Conveyor CNV-3130-002\Civil\Staad\3130-CNV-002-TR3-Fdn.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 25-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 -3 0; 2 0 -3 2; 3 0 1.49012E-008 0; 4 0 1.49012E-008 2; 5 0 0.35 0
     9. 6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7
    10. 11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75
    11. 15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7
    12. 19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7
    13. 24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75
    14. 28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5
    15. 33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25
    16. 38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25
    17. 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25
    18. 46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2
    19. 51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75
    20. 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75
    21. 60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5
    22. 64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5
    23. 69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25
    24. 73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25
    25. 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25
    26. 81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1
    27. 86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75
    28. 91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75
    29. 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75
    30. 99 -0.75 -3 0.5; 100 -1.6 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5
    31. 103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 107 1.6 -3 0.5
    32. 108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25
    33. 112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25
    34. 116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0
    35. 121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25
    36. 126 -1.6 -3 -0.25; 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25
    37. 130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25
    38. 134 -0.75 -3 -0.5; 135 -1.6 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5
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```
39. 138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5
40. 142 1.6 -3 -0.5; 143 -0.75 -3 -0.75; 144 -1.6 -3 -0.75; 145 -0.5 -3 -0.75
41. 146 -0.25 -3 -0.75; 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75
42. 150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7
43. 154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7
44. 158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25
45. 162 -0.5 -3 -1; 163 -0.5 -3 -1.25; 164 -0.25 -3 -1; 165 -0.25 -3 -1.25
46. 166 0 -3 -1; 167 0 -3 -1.25; 168 0.25 -3 -1; 169 0.25 -3 -1.25; 170 0.5 -3 -1.
47. 171 0.5 -3 -1.25; 172 0.75 -3 -1; 173 0.75 -3 -1.25; 174 1.6 -3 -1.
48. 175 1.6 -3 -1.25; 176 -0.75 -3 3.25; 177 -1.6 -3 3.25; 178 -0.75 -3 3
49. 179 -1.6 -3 3; 180 -0.5 -3 3.25; 181 -0.5 -3 3; 182 -0.25 -3 3.25
50. 183 -0.25 -3 3; 184 0 -3 3.25; 185 0 -3 3; 186 0.25 -3 3.25; 187 0.25 -3 3
51. 188 0.5 -3 3.25; 189 0.5 -3 3; 190 0.75 -3 3.25; 191 0.75 -3 3
52. 192 1.6 -3 3.25; 193 1.6 -3 3; 194 -0.75 -3 -1.475; 195 -1.6 -3 -1.475
53. 196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475
54. 200 0.5 -3 -1.475; 201 0.75 -3 -1.475; 202 1.6 -3 -1.475; 203 -0.75 -3 3.475
55. 204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475
56. 208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475
57. 212 1.03333 -3 2.75; 213 1.03333 -3 2.5; 214 1.31667 -3 2.75
58. 215 1.31667 -3 2.5; 216 1.03333 -3 2.25; 217 1.31667 -3 2.25; 218 1.03333 -3 2
59. 219 1.31667 -3 2; 220 1.03333 -3 1.75; 221 1.31667 -3 1.75; 222 1.03333 -3 1.5
60. 223 1.31667 -3 1.5; 224 1.03333 -3 1.25; 225 1.31667 -3 1.25; 226 1.03333 -3 1
61. 227 1.31667 -3 1; 228 1.03333 -3 0.75; 229 1.31667 -3 0.75; 230 1.03333 -3 0.5
62. 231 1.31667 -3 0.5; 232 1.03333 -3 0.25; 233 1.31667 -3 0.25; 234 1.03333 -3 0
63. 235 1.31667 -3 0; 236 1.03333 -3 -0.25; 237 1.31667 -3 -0.25
64. 238 1.03333 -3 -0.5; 239 1.31667 -3 -0.5; 240 1.03333 -3 -0.75
65. 241 1.31667 -3 -0.75; 242 1.03333 -3 -1; 243 1.31667 -3 -1.
66. 244 1.03333 -3 -1.25; 245 1.31667 -3 -1.25; 246 1.03333 -3 3.25
67. 247 1.03333 -3 3; 248 1.31667 -3 3.25; 249 1.31667 -3 3; 250 1.03333 -3 -1.475
68. 251 1.31667 -3 -1.475; 252 1.03333 -3 -1.7; 253 1.31667 -3 -1.7
69. 254 1.03333 -3 3.7; 255 1.03333 -3 3.475; 256 1.31667 -3 3.7
70. 257 1.31667 -3 3.475; 258 -1.31667 -3 2.75; 259 -1.31667 -3 2.5
71. 260 -1.03333 -3 2.75; 261 -1.03333 -3 2.5; 262 -1.31667 -3 2.25
72. 263 -1.03333 -3 2.25; 264 -1.31667 -3 2; 265 -1.03333 -3 2
73. 266 -1.31667 -3 1.75; 267 -1.03333 -3 1.75; 268 -1.31667 -3 1.5
74. 269 -1.03333 -3 1.5; 270 -1.31667 -3 1.25; 271 -1.03333 -3 1.25
75. 272 -1.31667 -3 1; 273 -1.03333 -3 1; 274 -1.31667 -3 0.75
76. 275 -1.03333 -3 0.75; 276 -1.31667 -3 0.5; 277 -1.03333 -3 0.5
77. 278 -1.31667 -3 0.25; 279 -1.03333 -3 0.25; 280 -1.31667 -3 0
78. 281 -1.03333 -3 0; 282 -1.31667 -3 -0.25; 283 -1.03333 -3 -0.25
79. 284 -1.31667 -3 -0.5; 285 -1.03333 -3 -0.5; 286 -1.31667 -3 -0.75
80. 287 -1.03333 -3 -0.75; 288 -1.31667 -3 -1; 289 -1.03333 -3 -1.
81. 290 -1.31667 -3 -1.25; 291 -1.03333 -3 -1.25; 292 -1.31667 -3 3.25
82. 293 -1.31667 -3 3; 294 -1.03333 -3 3.25; 295 -1.03333 -3 3
83. 296 -1.31667 -3 -1.475; 297 -1.03333 -3 -1.475; 298 -1.31667 -3 -1.7
84. 299 -1.03333 -3 -1.7; 300 -1.31667 -3 3.7; 301 -1.31667 -3 3.475
85. 302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475
86. MEMBER INCIDENCES
87. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
88. ELEMENT INCIDENCES SHELL
89. 15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33
90. 20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41
91. 27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49
92. 34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56
93. 41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63
94. 48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69
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95. 55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77
96. 60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85
97. 67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93
98. 74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99
99. 80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104
100. 84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111
101. 90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117
102. 96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121
103. 100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127
104. 105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131
105. 111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137
106. 114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140
107. 119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146
108. 122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149
109. 137 143 145 162 158; 138 158 162 163 160; 140 145 146 164 162
110. 141 162 164 165 163; 143 146 147 166 164; 144 164 166 167 165
111. 146 147 148 168 166; 147 166 168 169 167; 149 148 149 170 168
112. 150 168 170 171 169; 152 149 150 172 170; 153 170 172 173 171
113. 162 176 180 181 178; 163 178 181 17 14; 165 180 182 183 181; 166 181 183 19 17
114. 168 182 184 185 183; 169 183 185 20 19; 171 184 186 187 185; 172 185 187 22 20
115. 174 186 188 189 187; 175 187 189 24 22; 177 188 190 191 189; 178 189 191 26 24
116. 184 160 163 196 194; 185 194 196 153 152; 186 163 165 197 196
117. 187 196 197 154 153; 188 165 167 198 197; 189 197 198 7 154
118. 190 167 169 199 198; 191 198 199 155 7; 192 169 171 200 199
119. 193 199 200 156 155; 194 171 173 201 200; 195 200 201 157 156
120. 200 13 16 205 203; 201 203 205 180 176; 202 16 18 206 205; 203 205 206 182 180
121. 204 18 8 207 206; 205 206 207 184 182; 206 8 21 208 207; 207 207 208 186 184
122. 208 21 23 209 208; 209 208 209 188 186; 210 23 25 210 209; 211 209 210 190 188
123. 214 26 212 213 35; 215 212 214 215 213; 216 214 27 36 215; 217 35 213 216 44
124. 218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218
125. 222 217 45 53 219; 223 52 218 220 61; 224 218 219 221 220; 225 219 53 62 221
126. 226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79
127. 230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226
128. 234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229
129. 238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231
130. 241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233
131. 244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235
132. 247 123 234 236 132; 248 234 235 237 236; 249 235 124 133 237
133. 250 132 236 238 141; 251 236 237 239 238; 252 237 133 142 239
134. 253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241
135. 256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243
136. 259 172 242 244 173; 260 242 243 245 244; 261 243 174 175 245
137. 262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249
138. 265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214
139. 268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251
140. 271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253
141. 274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257
142. 277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248
143. 280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38
144. 284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264
145. 288 263 37 46 265; 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267
146. 292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73
147. 296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272
148. 300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275
149. 304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277
150. 307 100 276 278 109; 308 276 277 279 278; 309 277 99 108 279

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151. 310 109 278 280 118; 311 278 279 281 280; 312 279 108 117 281
152. 313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283
153. 316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285
154. 319 135 284 286 144; 320 284 285 287 286; 321 285 134 143 287
155. 322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289
156. 325 159 288 290 161; 326 288 289 291 290; 327 289 158 160 291
157. 328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295
158. 331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260
159. 334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297
160. 337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299
161. 340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303
162. 343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294
163. START GROUP DEFINITION
164. ELEMENT
165. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
166. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
167. 108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
168. 165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
169. 214 TO 345
170. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
171. 111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
172. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
173. 214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
174. _SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
175. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
176. 214 TO 345
177. END GROUP DEFINITION
178. ELEMENT PROPERTY
179. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
180. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
181. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
182. 178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
183. DEFINE MATERIAL START
184. ISOTROPIC CONCRETE
185. E 2.17185E+007
186. POISSON 0.17
187. DENSITY 23.5616
188. ALPHA 1E-005
189. DAMP 0.05
190. TYPE CONCRETE
191. STRENGTH FCU 27579
192. END DEFINE MATERIAL
193. MEMBER PROPERTY AMERICAN
194. 1 TO 4 PRIS YD 0.7 ZD 0.5
195. 5 PRIS YD 0.5 ZD 0.3
196. CONSTANTS
197. MATERIAL CONCRETE ALL
198. ****************************************
199. SUPPORTS
200. 1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
```

JOINT	INFLUENCE AREA
1	0.062500
2	0.062500

7	0.028125
8	0.028125
9	0.015937
10	0.015937
11	0.015937
12	0.015937
13	0.030000
14	0.066666
15	0.035416
16	0.028125
17	0.062500
18	0.028125
19	0.062500
20	0.062500
21	0.028125
22	0.062500
23	0.028125
24	0.062500
25	0.030000
26	0.066666
27	0.035416
28	0.066666
29	0.035416
30	0.062500
31	0.062500
32	0.062500
33	0.062500
34	0.062500
35	0.066666
36	0.035416
37	0.066666
38	0.035416
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.066666
45	0.035416
46	0.066666
47	0.035416
48	0.062500
49	0.062500
50	0.062500
51	0.062500
52	0.066666
53	0.035416
54	0.066666
55	0.035416
56	0.062500
57	0.062500
58	0.062500
59	0.062500
60	0.062500
61	0.066666
62	0.035416

63	0.066666
64	0.035416
65	0.062500
66	0.062500
67	0.062500
68	0.062500
69	0.062500
70	0.066666
71	0.035416
72	0.066666
73	0.035416
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.066666
80	0.035416
81	0.066666
82	0.035416
83	0.062500
84	0.062500
85	0.062500
86	0.062500
87	0.062500
88	0.066666
89	0.035416
90	0.066666
91	0.035416
92	0.062500
93	0.062500
94	0.062500
95	0.062500
96	0.062500
97	0.066666
98	0.035416
99	0.066666
100	0.035416
101	0.062500
102	0.062500
103	0.062500
104	0.062500
105	0.062500
106	0.066666
107	0.035416
108	0.066666
109	0.035416
110	0.062500
111	0.062500
112	0.062500
113	0.062500
114	0.062500
115	0.066666
116	0.035416
117	0.066666
118	0.035416

119	0.062500
120	0.062500
121	0.062500
122	0.062500
123	0.066666
124	0.035416
125	0.066666
126	0.035416
127	0.062500
128	0.062500
129	0.062500
130	0.062500
131	0.062500
132	0.066666
133	0.035416
134	0.066666
135	0.035416
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.066666
142	0.035416
143	0.066666
144	0.035416
145	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.066666
151	0.035416
152	0.030000
153	0.028125
154	0.028125
155	0.028125
156	0.028125
157	0.030000
158	0.066666
159	0.035416
160	0.063333
161	0.033645
162	0.062500
163	0.059375
164	0.062500
165	0.059375
166	0.062500
167	0.059375
168	0.062500
169	0.059375
170	0.062500
171	0.059375
172	0.066666
173	0.063333
174	0.035416

175	0.033645
176	0.063333
177	0.033645
178	0.066666
179	0.035416
180	0.059375
181	0.062500
182	0.059375
183	0.062500
184	0.059375
185	0.062500
186	0.059375
187	0.062500
188	0.059375
189	0.062500
190	0.063333
191	0.066666
192	0.033645
193	0.035416
194	0.060000
195	0.031875
196	0.056250
197	0.056250
198	0.056250
199	0.056250
200	0.056250
201	0.060000
202	0.031875
203	0.060000
204	0.031875
205	0.056250
206	0.056250
207	0.056250
208	0.056250
209	0.056250
210	0.060000
211	0.031875
212	0.070834
213	0.070834
214	0.070834
215	0.070834
216	0.070834
217	0.070834
218	0.070834
219	0.070834
220	0.070834
221	0.070834
222	0.070834
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224	0.070834
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226	0.070834
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238	0.070834
239	0.070834
240	0.070834
241	0.070834
242	0.070834
243	0.070834
244	0.067292
245	0.067292
246	0.067292
247	0.070834
248	0.067292
249	0.070834
250	0.063750
251	0.063750
252	0.031875
253	0.031875
254	0.031875
255	0.063750
256	0.031875
257	0.063750
258	0.070834
259	0.070834
260	0.070834
261	0.070834
262	0.070834
263	0.070834
264	0.070834
265	0.070834
266	0.070834
267	0.070834
268	0.070834
269	0.070834
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273	0.070834
274	0.070834
275	0.070834
276	0.070834
277	0.070834
278	0.070834
279	0.070834
280	0.070834
281	0.070834
282	0.070834
283	0.070834
284	0.070834
285	0.070834
286	0.070834

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       287          0.070834
       288          0.070834
       289          0.070834
       290          0.067292
       291          0.067292
       292          0.067292
       293          0.070834
       294          0.067292
       295          0.070834
       296          0.063750
       297          0.063750
       298          0.031875
       299          0.031875
       300          0.031875
       301          0.063750
       302          0.031875
       303          0.063750
    TOTAL          17.280014
201. ****************************************
202. LOAD 1 DEAD LOAD
203. SELFWEIGHT Y -1
204. ELEMENT LOAD
205. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
206. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
207. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
208. 178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
209. JOINT LOAD
210. 5 FX -0.1 FY -77.7 FZ -2.3
211. 6 FX 0.0 FY -234.1 FZ 13.6
213. LOAD 2  LIVE LOAD
214. JOINT LOAD
215. 5 FX 0.0 FY -71.9 FZ -3.7
216. 6 FX 0.0 FY -72.1 FZ 3.7
218. LOAD 3 MATERIAL LOAD
219. JOINT LOAD
220. 5 FX 0.0 FY -48.3 FZ -2.5
221. 6 FX 0.0 FY -47.9 FZ 2.4
223. LOAD 4 WIND LOAD +Z
224. JOINT LOAD
225. 5 FX 0.0 FY 267.2 FZ 40.7
226. 6 FX 0.0 FY -267.2 FZ 40.7
228. LOAD 5 WIND LOAD -Z
229. JOINT LOAD
230. 5 FX -0.03 FY -269.88 FZ -41.76
231. 6 FX 0.03 FY 269.88 FZ -41.71
233. LOAD 6 WIND LOAD (+X)
234. JOINT LOAD
235. 5 FX 1.52 FY 0 FZ 0
236. 6 FX 1.52 FY 0 FZ 0
238. LOAD 7 WIND LOAD (-X)
239. JOINT LOAD
240. 5 FX -1.52 FY 0 FZ 0
241. 6 FX -1.52 FY 0 FZ 0
243. LOAD 8 TEMP LOAD 1
244. JOINT LOAD
245. 5 FX 0 FY -0.08 FZ -17.66
```

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246. 6 FX 0 FY 0.08 FZ 17.65
248. LOAD 9 TEMP LOAD 2
249. JOINT LOAD
250. 5 FX 0 FY 0.08 FZ 17.66
251. 6 FX 0 FY -0.08 FZ -17.65
253. LOAD 10 EARTHQUAKE LOAD +X
254. JOINT LOAD
255. 5 FX 0.02 FY -0.06 FZ 0
256. 6 FX 0.02 FY 0.05 FZ 0
258. LOAD 11 EARTHQUAKE LOAD +Z
259. JOINT LOAD
260. 5 FX -0.01 FY 13.72 FZ 2.35
261. 6 FX 0.01 FY -13.72 FZ 2.35
264. ****************************************
265. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
266. LOAD 301 1.0 DL
267. REPEAT LOAD
268. 1 1.0
269. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
270. REPEAT LOAD
271. 1 1.0 8 1.0 2 1.0 3 1.0
272. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
273. REPEAT LOAD
274. 1 1.0 9 1.0 2 1.0 3 1.0
275. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
276. REPEAT LOAD
277. 1 1.0 4 0.75 2 0.75 3 0.75
278. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
279. REPEAT LOAD
280. 1 1.0 5 0.75 2 0.75 3 0.75
281. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
282. REPEAT LOAD
283. 1 1.0 6 0.75 2 0.75 3 0.75
284. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
285. REPEAT LOAD
286. 1 1.0 7 0.75 2 0.75 3 0.75
287. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
288. REPEAT LOAD
289. 1 1.0 10 0.525 2 0.75 3 0.75
290. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
291. REPEAT LOAD
292. 1 1.0 10 -0.525 2 0.75 3 0.75
293. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
294. REPEAT LOAD
295. 1 1.0 11 0.525 2 0.75 3 0.75
296. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
297. REPEAT LOAD
298. 1 1.0 11 -0.525 2 0.75 3 0.75
299. LOAD 312 0.6 DL + 1.0 WL(+Z)
300. REPEAT LOAD
301. 1 0.6 4 1.0
302. LOAD 313 0.6 DL + 1.0 WL(-Z)
303. REPEAT LOAD
304. 1 0.6 5 1.0
305. LOAD 314 0.6 DL + 1.0 WL(+X)
306. REPEAT LOAD
```

```
307. 1 0.6 6 1.0
308. LOAD 315 0.6 DL + 1.0 WL(-X)
309. REPEAT LOAD
310. 1 0.6 7 1.0
311. LOAD 316 0.6 DL + 0.7 EL(+X)
312. REPEAT LOAD
313. 1 0.6 10 0.7
314. LOAD 317 0.6 DL + 0.7 EL(-X)
315. REPEAT LOAD
316. 1 0.6 10 -0.7
317. LOAD 318 0.6 DL + 0.7 EL(+Z)
318. REPEAT LOAD
319. 1 0.6 11 0.7
320. LOAD 319 0.6 DL + 0.7 EL(-Z)
321. REPEAT LOAD
322. 1 0.6 11 -0.7
323. ****************************************
324. *****LOAD COMBINATION FOR FOUNDATION DESIGN
325. LOAD 401 1.4 DL
326. REPEAT LOAD
327. 1 1.4
328. LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
329. REPEAT LOAD
330. 1 1.4 8 1.4 2 1.7 3 1.7
331. LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
332. REPEAT LOAD
333. 1 1.4 9 1.4 2 1.7 3 1.7
334. LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
335. REPEAT LOAD
336. 1 1.2 4 1.6 2 0.5 3 0.5
337. LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
338. REPEAT LOAD
339. 1 1.2 5 1.6 2 0.5 3 0.5
340. LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
341. REPEAT LOAD
342. 1 1.2 6 1.6 2 0.5 3 0.5
343. LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
344. REPEAT LOAD
345. 1 1.2 7 1.6 2 0.5 3 0.5
346. LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
347. REPEAT LOAD
348. 1 1.2 10 1.0 2 0.5 3 0.5
349. LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
350. REPEAT LOAD
351. 1 1.2 10 -1.0 2 0.5 3 0.5
352. LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
353. REPEAT LOAD
354. 1 1.2 11 1.0 2 0.5 3 0.5
355. LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
356. REPEAT LOAD
357. 1 1.2 11 -1.0 2 0.5 3 0.5
358. LOAD 412 0.9 DL + 1.6 WL(+Z)
359. REPEAT LOAD
360. 1 0.9 4 1.6
361. LOAD 413 0.9 DL + 1.6 WL(-Z)
362. REPEAT LOAD
```

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363. 1 0.9 5 1.6
364. LOAD 414 0.9 DL + 1.6 WL(+X)
365. REPEAT LOAD
366. 1 0.9 6 1.6
367. LOAD 415 0.9 DL + 1.6 WL(-X)
368. REPEAT LOAD
369. 1 0.9 7 1.6
370. LOAD 416 0.9 DL + 1.0 EL(+X)
371. REPEAT LOAD
372. 1 0.9 10 1.0
373. LOAD 417 0.9 DL + 1.0 EL(-X)
374. REPEAT LOAD
375. 1 0.9 10 -1.0
376. LOAD 418 0.9 DL + 1.0 EL(+Z)
377. REPEAT LOAD
378. 1 0.9 11 1.0
379. LOAD 419 0.9 DL + 1.0 EL(-Z)
380. REPEAT LOAD
381. 1 0.9 11 -1.0
382. ****************************************
383. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 303 NUMBER OF MEMBERS 5
NUMBER OF PLATES 264 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 299

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 921
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

```
384. ****************************************
385. LOAD LIST 401 TO 419
386. START CONCRETE DESIGN
```

```
387. CODE ACI 2008
388. FC 35000 MEMB 1 TO 5
389. FYMAIN 420000 MEMB 1 TO 5
390. TRACK 1 ALL
391. CLB 0.05 MEMB 1 TO 5
392. CLT 0.05 MEMB 1 TO 5
393. CLS 0.05 ALL
394. MAXMAIN 25 ALL
395. MINMAIN 16 ALL
396. MINSEC 8 ALL
397. REINF 0 MEMB 1 TO 4
398. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	167.28	0.05	0.00010

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	167.28	10.14	0.02020

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    167.35      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    167.35      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 700.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3500.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
---------------------------------------------------------

 12 - 20 MM               1.077      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    531.95      0.03    0.00002
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    531.95      9.16    0.00574
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	504.22	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	504.22	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

399. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

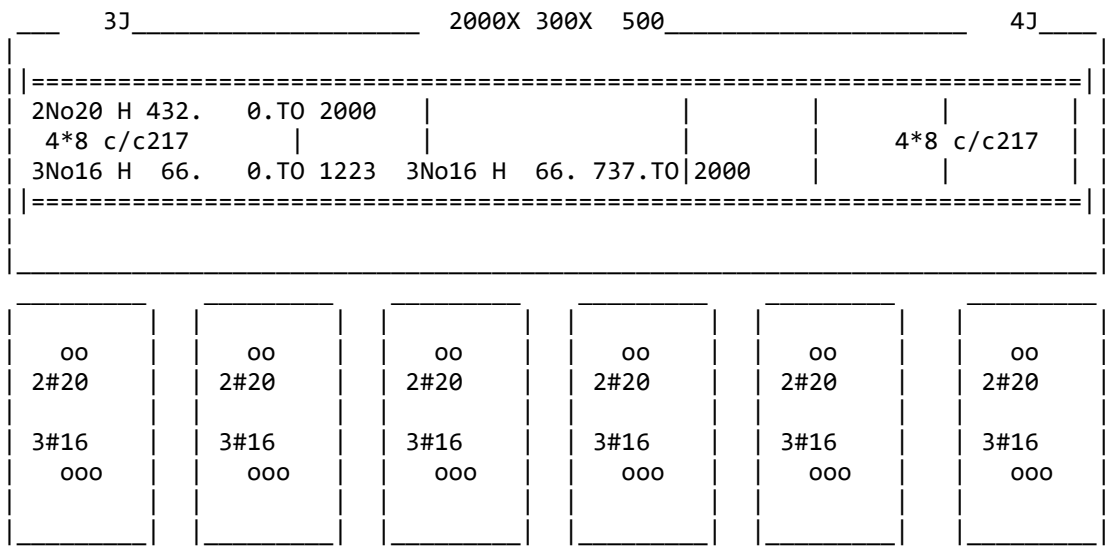
LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	1223.	YES	NO
	CRITICAL POS MOMENT= 91.03 KN-MET AT 0.MM, LOAD 404					
	REQD STEEL= 573.MM2, RHO=0.0044, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
2	66.	3 - 16MM	737.	2000.	NO	YES
	CRITICAL POS MOMENT= 81.25 KN-MET AT 2000.MM, LOAD 413					
	REQD STEEL= 509.MM2, RHO=0.0039, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
3	432.	2 - 20MM	0.	2000.	YES	YES
	CRITICAL NEG MOMENT= 98.35 KN-MET AT 2000.MM, LOAD 404					
	REQD STEEL= 620.MM2, RHO=0.0048, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 45./ 164. MMS					
	REQD. DEVELOPMENT LENGTH = 513. MMS					
	Cracked Moment of Inertia Iz at above location = 60046.8 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 92.29 KNS Vc= 125.48 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM

AT END SUPPORT - Vu= 97.09 KNS Vc= 125.48 KNS Vs= 3.98 KNS
Tu= 0.01 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT IS REQUIRED FOR SHEAR.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM



**********END OF BEAM DESIGN**********

400. END CONCRETE DESIGN
401. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 25,2020 TIME= 20:22:23 ****

```
     ************************************************************
     *   For technical assistance on STAAD.Pro, please visit    *
     *   http://selectservices.bentley.com/en-US/               *
     *                                                          *
     *   Details about additional assistance from               *
     *   Bentley and Partners can be found at program menu      *
     *   Help->Technical Support                                *
     *                                                          *
     *      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
     *                http://www.bentley.com                    *
     ************************************************************
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	26-03-2020	24-04-2020	01-11-2023				Date	26-03-2020	24-04-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 7 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR TR-4**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-Mar-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 6; 3 0 1.49012e-008 0; 4 0 1.49012e-008 6; 5 0 0.35 0;
6 0 0.35 6;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.6 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 2 FIXED
*****************************************
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
joint load
5   FX  0.0 FY  -132.9  FZ  -26.8
6   FX  0.0 FY  -228.3  FZ  38.1

Load 2   LIVE LOAD
joint load
5   FX  0.0 FY  -72.2   FZ  -13.1
6   FX  0.0 FY  -71.8   FZ  13.0

Load 3  MATERIAL LOAD
joint load
5   FX  0.0 FY  -55.1   FZ  -10.0
6   FX  0.0 FY  -55.1   FZ  10.0

Load 4  WIND LOAD +Z
joint load
5   FX  0.0 FY  182.1   FZ  24.4
6   FX  0.0 FY  -182.1  FZ  24.3
```

```
Load 5  WIND LOAD -Z
joint load
5   FX  -0.02   FY  -244.81 FZ  -48.99
6   FX  0.02    FY  244.81  FZ  -48.98

Load 6  WIND LOAD (+X)
joint load
5   FX  4.61    FY  0   FZ  0
6   FX  4.61    FY  0   FZ  0

Load 7  WIND LOAD (-X)
joint load
5   FX  -4.61   FY  0   FZ  0
6   FX  -4.61   FY  0   FZ  0

Load 8  TEMP LOAD 1
joint load
5   FX  0   FY  0.02    FZ  -57.72
6   FX  0   FY  -0.02   FZ  57.73

Load 9  TEMP LOAD 2
joint load
5   FX  0   FY  -0.02   FZ  57.72
6   FX  0   FY  0.02    FZ  -57.73

Load 10 EARTHQUAKE LOAD +X
joint load
5   FX  0.03    FY  -0.05   FZ  0
6   FX  0.03    FY  0.05    FZ  -0.01

Load 11 EARTHQUAKE LOAD +Z
joint load
5   FX  -0.01   FY  11.74   FZ  2.18
6   FX  0.01    FY  -11.74  FZ  2.18


****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
```

```
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL(+X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL(-X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL(+X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL(-X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
1 0.6 11 -0.7
****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
```

```
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL(+X)
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL(-X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL(+X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL(-X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 1 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 25, 2020                *
*           Time=    20:28:11                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Conveyor CNV-3130-002\Civil\Staad\3130-CNV-002-TR4-Fdn.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 25-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 -3 0; 2 0 -3 6; 3 0 1.49012E-008 0; 4 0 1.49012E-008 6; 5 0 0.35 0
     9. 6 0 0.35 6
    10. MEMBER INCIDENCES
    11. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
    12. DEFINE MATERIAL START
    13. ISOTROPIC CONCRETE
    14. E 2.17185E+007
    15. POISSON 0.17
    16. DENSITY 23.5616
    17. ALPHA 1E-005
    18. DAMP 0.05
    19. TYPE CONCRETE
    20. STRENGTH FCU 27579
    21. END DEFINE MATERIAL
    22. MEMBER PROPERTY AMERICAN
    23. 1 TO 4 PRIS YD 0.7 ZD 0.5
    24. 5 PRIS YD 0.6 ZD 0.3
    25. CONSTANTS
    26. MATERIAL CONCRETE ALL
    27. SUPPORTS
    28. 1 2 FIXED
    29. ****************************************
    30. LOAD 1 DEAD LOAD DL
    31. SELFWEIGHT Y -1
    32. JOINT LOAD
    33. 5 FX 0.0 FY -132.9 FZ -26.8
    34. 6 FX 0.0 FY -228.3 FZ 38.1
    36. LOAD 2  LIVE LOAD
    37. JOINT LOAD
    38. 5 FX 0.0 FY -72.2 FZ -13.1
    39. 6 FX 0.0 FY -71.8 FZ 13.0
```

```
 41. LOAD 3 MATERIAL LOAD
 42. JOINT LOAD
 43. 5 FX 0.0 FY -55.1 FZ -10.0
 44. 6 FX 0.0 FY -55.1 FZ 10.0
 46. LOAD 4 WIND LOAD +Z
 47. JOINT LOAD
 48. 5 FX 0.0 FY 182.1 FZ 24.4
 49. 6 FX 0.0 FY -182.1 FZ 24.3
 51. LOAD 5 WIND LOAD -Z
 52. JOINT LOAD
 53. 5 FX -0.02 FY -244.81 FZ -48.99
 54. 6 FX 0.02 FY 244.81 FZ -48.98
 56. LOAD 6 WIND LOAD (+X)
 57. JOINT LOAD
 58. 5 FX 4.61 FY 0 FZ 0
 59. 6 FX 4.61 FY 0 FZ 0
 61. LOAD 7 WIND LOAD (-X)
 62. JOINT LOAD
 63. 5 FX -4.61 FY 0 FZ 0
 64. 6 FX -4.61 FY 0 FZ 0
 66. LOAD 8 TEMP LOAD 1
 67. JOINT LOAD
 68. 5 FX 0 FY 0.02 FZ -57.72
 69. 6 FX 0 FY -0.02 FZ 57.73
 71. LOAD 9 TEMP LOAD 2
 72. JOINT LOAD
 73. 5 FX 0 FY -0.02 FZ 57.72
 74. 6 FX 0 FY 0.02 FZ -57.73
 76. LOAD 10 EARTHQUAKE LOAD +X
 77. JOINT LOAD
 78. 5 FX 0.03 FY -0.05 FZ 0
 79. 6 FX 0.03 FY 0.05 FZ -0.01
 81. LOAD 11 EARTHQUAKE LOAD +Z
 82. JOINT LOAD
 83. 5 FX -0.01 FY 11.74 FZ 2.18
 84. 6 FX 0.01 FY -11.74 FZ 2.18
 87. ****************************************
 88. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
 89. LOAD COMB 301 1.0 DL
 90. 1 1.0
 91. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
 92. 1 1.0 8 1.0 2 1.0 3 1.0
 93. LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
 94. 1 1.0 9 1.0 2 1.0 3 1.0
 95. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
 96. 1 1.0 4 0.75 2 0.75 3 0.75
 97. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
 98. 1 1.0 5 0.75 2 0.75 3 0.75
 99. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
100. 1 1.0 6 0.75 2 0.75 3 0.75
101. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
102. 1 1.0 7 0.75 2 0.75 3 0.75
103. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
104. 1 1.0 10 0.525 2 0.75 3 0.75
105. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
106. 1 1.0 10 -0.525 2 0.75 3 0.75
```

```
107. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
108. 1 1.0 11 0.525 2 0.75 3 0.75
109. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
110. 1 1.0 11 -0.525 2 0.75 3 0.75
111. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
112. 1 0.6 4 1.0
113. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
114. 1 0.6 5 1.0
115. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
116. 1 0.6 6 1.0
117. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
118. 1 0.6 7 1.0
119. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
120. 1 0.6 10 0.7
121. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
122. 1 0.6 10 -0.7
123. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
124. 1 0.6 11 0.7
125. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
126. 1 0.6 11 -0.7
127. ****************************************
128. *****LOAD COMBINATION FOR FOUNDATION DESIGN
129. LOAD COMB 401 1.4 DL
130. 1 1.4
131. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
132. 1 1.4 8 1.4 2 1.7 3 1.7
133. LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
134. 1 1.4 9 1.4 2 1.7 3 1.7
135. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
136. 1 1.2 4 1.6 2 0.5 3 0.5
137. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
138. 1 1.2 5 1.6 2 0.5 3 0.5
139. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
140. 1 1.2 6 1.6 2 0.5 3 0.5
141. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
142. 1 1.2 7 1.6 2 0.5 3 0.5
143. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
144. 1 1.2 10 1.0 2 0.5 3 0.5
145. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
146. 1 1.2 10 -1.0 2 0.5 3 0.5
147. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
148. 1 1.2 11 1.0 2 0.5 3 0.5
149. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
150. 1 1.2 11 -1.0 2 0.5 3 0.5
151. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
152. 1 0.9 4 1.6
153. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
154. 1 0.9 5 1.6
155. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
156. 1 0.9 6 1.6
157. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
158. 1 0.9 7 1.6
159. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
160. 1 0.9 10 1.0
161. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
162. 1 0.9 10 -1.0
```

```
163. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
164. 1 0.9 11 1.0
165. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
166. 1 0.9 11 -1.0
167. ****************************************
168. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**6**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 38 SO FAR.

```
169. LOAD LIST 301 TO 319
170. PRINT SUPPORT REACTION ALL
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	0.00	170.95	-4.49	-10.05	0.00	0.00
	302	0.00	298.24	-14.29	-16.99	0.00	0.00
	303	0.00	298.29	-0.20	-6.92	0.00	0.00
	304	0.00	122.39	-24.82	-50.24	0.00	0.00
	305	0.01	465.22	30.18	66.07	-0.00	-0.04
	306	-3.46	266.44	-6.56	-11.48	0.00	11.58
	307	3.46	266.44	-6.56	-11.48	0.00	-11.58
	308	-0.02	266.46	-6.55	-11.47	0.00	0.05
	309	0.02	266.41	-6.56	-11.48	0.00	-0.05
	310	0.00	259.80	-7.70	-13.90	-0.00	-0.02
	311	-0.00	273.07	-5.41	-9.06	0.00	0.02
	312	0.00	-89.50	-27.04	-57.71	0.00	0.00
	313	0.02	367.61	46.29	97.36	-0.00	-0.06
	314	-4.61	102.57	-2.69	-6.03	0.00	15.44
	315	4.61	102.57	-2.69	-6.03	0.00	-15.44
	316	-0.02	102.60	-2.69	-6.02	0.00	0.07
	317	0.02	102.53	-2.70	-6.04	0.00	-0.07
	318	0.01	93.72	-4.22	-9.26	-0.00	-0.02
	319	-0.01	111.42	-1.17	-2.81	0.00	0.02
2	301	0.00	270.95	-6.81	-13.98	0.00	0.00
	302	0.00	397.85	3.08	-6.86	0.00	0.00
	303	0.00	397.81	-10.99	-16.88	0.00	0.00
	304	0.00	510.16	-22.93	-51.16	0.00	0.00
	305	-0.01	167.33	32.07	65.15	-0.00	0.04
	306	-3.46	366.11	-4.67	-12.40	0.00	11.58
	307	3.46	366.11	-4.67	-12.40	0.00	-11.58
	308	-0.02	366.08	-4.67	-12.39	0.00	0.05
	309	0.02	366.14	-4.67	-12.40	0.00	-0.05
	310	-0.00	372.75	-5.81	-14.82	-0.00	0.02
	311	0.00	359.48	-3.52	-9.98	0.00	-0.02
	312	0.00	354.63	-28.44	-60.08	0.00	0.00
	313	-0.02	-102.47	44.90	95.01	-0.00	0.06
	314	-4.61	162.57	-4.09	-8.39	0.00	15.44
	315	4.61	162.57	-4.09	-8.39	0.00	-15.44
	316	-0.02	162.54	-4.09	-8.38	0.00	0.07
	317	0.02	162.61	-4.09	-8.40	0.00	-0.07
	318	-0.01	171.42	-5.61	-11.61	-0.00	0.02
	319	0.01	153.72	-2.56	-5.17	0.00	-0.02

********** END OF LATEST ANALYSIS RESULT **********

```
171. ****************************************
172. LOAD LIST 401 TO 419
173. START CONCRETE DESIGN
174. CODE ACI 2008
175. FC 35000 ALL
176. FYMAIN 420000 ALL
177. TRACK 1 ALL
178. CLB 0.05 ALL
179. CLT 0.05 ALL
180. CLS 0.05 ALL
181. MAXMAIN 25 ALL
182. MINMAIN 16 ALL
183. MINSEC 8 ALL
184. REINF 0 MEMB 1 TO 4
185. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	314.91	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	314.91	7.33	0.00776

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    286.25      0.00    0.00000
 ------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    286.25      0.00    0.00000
 ------------------------------------------------------


======================================================================

     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 700.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3500.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
------------------------------------------------------

 12 - 20 MM               1.077      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    530.30      0.00    0.00000
 ------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    530.30     13.91    0.00875
 ------------------------------------------------------


======================================================================

     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	491.72	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	491.72	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

186. DESIGN BEAM 5

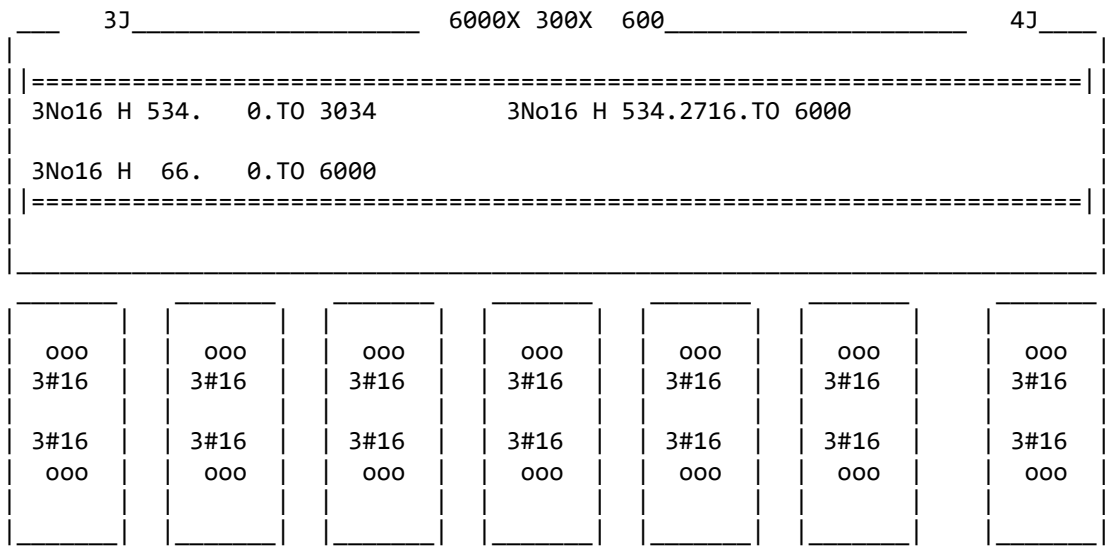
BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	6000.	YES	YES
	CRITICAL POS MOMENT= 79.31 KN-MET AT 6000.MM, LOAD 413					
	REQD STEEL= 532.MM2, RHO=0.0033, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
2	534.	3 - 16MM	0.	3034.	YES	NO
	CRITICAL NEG MOMENT= 105.15 KN-MET AT 0.MM, LOAD 405					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
3	534.	3 - 16MM	2716.	6000.	NO	YES
	CRITICAL NEG MOMENT= 72.36 KN-MET AT 6000.MM, LOAD 404					
	REQD STEEL= 485.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.42 KNS Vc= 146.28 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 6.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 17.87 KNS Vc= 146.28 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 6.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

187. END CONCRETE DESIGN
188. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 25,2020 TIME= 20:28:12 ****

```
     ************************************************************
     *   For technical assistance on STAAD.Pro, please visit    *
     *   http://selectservices.bentley.com/en-US/               *
     *                                                          *
     *   Details about additional assistance from               *
     *   Bentley and Partners can be found at program menu      *
     *   Help->Technical Support                                *
     *                                                          *
     *      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
     *                http://www.bentley.com                    *
     ************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	26-03-2020	24-04-2020	01-11-2023				Date	26-03-2020	24-04-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 8 **STAAD INPUT & OUTPUT FILE OF FOUNDATION FOR VGTU**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 25-Mar-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 4.5; 3 4 -3 0; 4 4 -3 4.5; 5 8 -3 0; 6 8 -3 4.5; 7 0 0.35 0;
8 0 0.35 4.5; 9 4 0.35 0; 10 4 0.35 4.5; 11 8 0.35 0; 12 8 0.35 4.5; 13 0 1 0;
14 0 1 4.5; 15 4 1 0; 16 4 1 4.5; 17 8 1 0; 18 8 1 4.5;
MEMBER INCIDENCES
1 1 7; 2 7 13; 3 2 8; 4 8 14; 5 3 9; 6 9 15; 7 4 10; 8 10 16; 9 5 11; 10 11 17;
11 6 12; 12 12 18; 13 7 8; 14 9 10; 15 11 12; 16 7 9; 17 9 11; 18 8 10;
19 10 12;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
9 TO 12 PRIS YD 0.6 ZD 0.6
13 TO 15 PRIS YD 0.5 ZD 0.3
MEMBER PROPERTY AMERICAN
16 TO 19 PRIS YD 0.6 ZD 0.3
MEMBER PROPERTY AMERICAN
1 TO 8 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 TO 6 FIXED
*****************************************
LOAD 1 DEAD LOA
SELFWEIGHT Y -1
JOINT LOAD
13  FX  -23.1   FY  -256.5  FZ  -11.3
14  FX  -22.9   FY  -306.2  FZ  16.5

15  FX  -21.1   FY  -60.3   FZ  1.5
16  FX  -20.9   FY  -117.7  FZ  4.2

17  FX  0.0 FY  -15.7   FZ  -0.1
18  FX  0.0 FY  -17.8   FZ  0.6

*****************************************
LOAD 2 LIVE LOAD
```

```
JOINT LOAD
13  FX  0.1 FY  -138.6  FZ  -7.3
14  FX  -1.0    FY  -126.7  FZ  7.3

15  FX  1.0 FY  -138.5  FZ  -0.6
16  FX  -0.1    FY  -165.7  FZ  1.4


17  FX  0.0 FY  -51.3   FZ  -0.5
18  FX  0.0 FY  -48.5   FZ  -0.5

*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
13  FX  0.1 FY  -23.9   FZ  -1.4
14  FX  -0.1    FY  -23.4   FZ  1.2

15  FX  0.1 FY  -27.6   FZ  0.1
16  FX  0.0 FY  -28.3   FZ  0.2

17  FX  0.0 FY  -0.4    FZ  -0.1
18  FX  0.0 FY  0.4 FZ  -0.1

*****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
13  FX  -1.3    FY  122.8   FZ  30.5
14  FX  1.3 FY  -122.8  FZ  30.7

15  FX  -1.8    FY  122.9   FZ  25.6
16  FX  1.8 FY  -122.9  FZ  25.6

17  FX  0.0 FY  29.2    FZ  11.5
18  FX  0.0 FY  -29.2   FZ  11.5

*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
13  FX  1.26    FY  -122.84 FZ  -30.5
14  FX  -1.26   FY  122.84  FZ  -30.65

15  FX  1.78    FY  -122.87 FZ  -25.62
16  FX  -1.77   FY  122.87  FZ  -25.61

17  FX  0   FY  -29.18  FZ  -11.5
18  FX  -0.01   FY  29.18   FZ  -11.5

*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
```

```
13   FX   49.09    FY   206.65   FZ   7.94
14   FX   48.86    FY   207.97   FZ   -8.33

15   FX   49.07    FY   -205.41  FZ   0.19
16   FX   50.03    FY   -209.21  FZ   0.82

17   FX   1.02     FY   -1.24    FZ   -0.31
18   FX   -0.11    FY   1.24     FZ   -0.31

*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
13   FX   -49.09   FY   -206.65  FZ   -7.94
14   FX   -48.86   FY   -207.97  FZ   8.33

15   FX   -49.07   FY   205.41   FZ   -0.19
16   FX   -50.03   FY   209.21   FZ   -0.82

17   FX   -1.02    FY   1.24     FZ   0.31
18   FX   0.11     FY   -1.24    FZ   0.31

*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
13   FX   -1.74    FY   -4.12    FZ   -38.21
14   FX   -1.25    FY   4.12     FZ   37.18

15   FX   1.17     FY   5.37     FZ   0.22
16   FX   1.21     FY   -5.37    FZ   1.45

17   FX   0.1 FY   -1.22    FZ   -0.8
18   FX   0.51     FY   1.22     FZ   0.19

*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
13   FX   1.74     FY   4.12     FZ   38.21
14   FX   1.25     FY   -4.12    FZ   -37.18

15   FX   -1.17    FY   -5.37    FZ   -0.22
16   FX   -1.21    FY   5.37     FZ   -1.45

17   FX   -0.1     FY   1.22     FZ   0.8
18   FX   -0.51    FY   -1.22    FZ   -0.19

*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
13   FX   1.45     FY   7.15     FZ   0.32
14   FX   1.55     FY   7.42     FZ   -0.27
```

```
15  FX  1.45    FY  -7.13   FZ  -0.02
16  FX  1.55    FY  -7.42   FZ  0

17  FX  0   FY  -0.05   FZ  -0.01
18  FX  -0.01   FY  0.05    FZ  -0.01

*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
13  FX  -0.12   FY  6.67    FZ  1.56
14  FX  0.12    FY  -6.67   FZ  1.57

15  FX  -0.15   FY  6.66    FZ  1.2
16  FX  0.15    FY  -6.66   FZ  1.2

17  FX  0   FY  0.94    FZ  0.24
18  FX  0   FY  -0.94   FZ  0.24

*****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL(+X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL(-X)
```

```
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL(+X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL(-X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
1 0.6 11 -0.7
****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL(+X)
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL(-X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL(+X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL(-X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
1 0.9 11 -1.0
****************************************
```

```
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
*******************************************************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 1 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 12
DESIGN COLUMN 1 TO 12
DESIGN BEAM 13 TO 19
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro CONNECT Edition            *
*           Version  21.00.01.12                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAR 25, 2020                *
*           Time=    20:32:46                    *
*                                                *
*  Licensed to: MMH KOLKATTA                     *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\Conveyor CNV-3130-002\Civil\Staad\3130-CNV-002-VGTU-Fdn.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 25-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 -3 0; 2 0 -3 4.5; 3 4 -3 0; 4 4 -3 4.5; 5 8 -3 0; 6 8 -3 4.5; 7 0 0.35 0
     9. 8 0 0.35 4.5; 9 4 0.35 0; 10 4 0.35 4.5; 11 8 0.35 0; 12 8 0.35 4.5; 13 0 1 0
    10. 14 0 1 4.5; 15 4 1 0; 16 4 1 4.5; 17 8 1 0; 18 8 1 4.5
    11. MEMBER INCIDENCES
    12. 1 1 7; 2 7 13; 3 2 8; 4 8 14; 5 3 9; 6 9 15; 7 4 10; 8 10 16; 9 5 11; 10 11 17
    13. 11 6 12; 12 12 18; 13 7 8; 14 9 10; 15 11 12; 16 7 9; 17 9 11; 18 8 10
    14. 19 10 12
    15. DEFINE MATERIAL START
    16. ISOTROPIC CONCRETE
    17. E 2.17185E+007
    18. POISSON 0.17
    19. DENSITY 23.5616
    20. ALPHA 1E-005
    21. DAMP 0.05
    22. TYPE CONCRETE
    23. STRENGTH FCU 27579
    24. END DEFINE MATERIAL
    25. MEMBER PROPERTY AMERICAN
    26. 9 TO 12 PRIS YD 0.6 ZD 0.6
    27. 13 TO 15 PRIS YD 0.5 ZD 0.3
    28. MEMBER PROPERTY AMERICAN
    29. 16 TO 19 PRIS YD 0.6 ZD 0.3
    30. MEMBER PROPERTY AMERICAN
    31. 1 TO 8 PRIS YD 0.6 ZD 0.6
    32. CONSTANTS
    33. MATERIAL CONCRETE ALL
    34. SUPPORTS
    35. 1 TO 6 FIXED
    36. ****************************************
    37. LOAD 1 DEAD LOA
    38. SELFWEIGHT Y -1
```

```
 39. JOINT LOAD
 40. 13 FX -23.1 FY -256.5 FZ -11.3
 41. 14 FX -22.9 FY -306.2 FZ 16.5
 43. 15 FX -21.1 FY -60.3 FZ 1.5
 44. 16 FX -20.9 FY -117.7 FZ 4.2
 46. 17 FX 0.0 FY -15.7 FZ -0.1
 47. 18 FX 0.0 FY -17.8 FZ 0.6
 49. ****************************************
 50. LOAD 2 LIVE LOAD
 51. JOINT LOAD
 52. 13 FX 0.1 FY -138.6 FZ -7.3
 53. 14 FX -1.0 FY -126.7 FZ 7.3
 55. 15 FX 1.0 FY -138.5 FZ -0.6
 56. 16 FX -0.1 FY -165.7 FZ 1.4
 59. 17 FX 0.0 FY -51.3 FZ -0.5
 60. 18 FX 0.0 FY -48.5 FZ -0.5
 62. ****************************************
 63. LOAD 3 MATERIAL LOAD LL
 64. JOINT LOAD
 65. 13 FX 0.1 FY -23.9 FZ -1.4
 66. 14 FX -0.1 FY -23.4 FZ 1.2
 68. 15 FX 0.1 FY -27.6 FZ 0.1
 69. 16 FX 0.0 FY -28.3 FZ 0.2
 71. 17 FX 0.0 FY -0.4 FZ -0.1
 72. 18 FX 0.0 FY 0.4 FZ -0.1
 74. ****************************************
 75. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
 76. JOINT LOAD
 77. 13 FX -1.3 FY 122.8 FZ 30.5
 78. 14 FX 1.3 FY -122.8 FZ 30.7
 80. 15 FX -1.8 FY 122.9 FZ 25.6
 81. 16 FX 1.8 FY -122.9 FZ 25.6
 83. 17 FX 0.0 FY 29.2 FZ 11.5
 84. 18 FX 0.0 FY -29.2 FZ 11.5
 86. ****************************************
 87. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
 88. JOINT LOAD
 89. 13 FX 1.26 FY -122.84 FZ -30.5
 90. 14 FX -1.26 FY 122.84 FZ -30.65
 92. 15 FX 1.78 FY -122.87 FZ -25.62
 93. 16 FX -1.77 FY 122.87 FZ -25.61
 95. 17 FX 0 FY -29.18 FZ -11.5
 96. 18 FX -0.01 FY 29.18 FZ -11.5
 98. ****************************************
 99. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
100. JOINT LOAD
101. 13 FX 49.09 FY 206.65 FZ 7.94
102. 14 FX 48.86 FY 207.97 FZ -8.33
104. 15 FX 49.07 FY -205.41 FZ 0.19
105. 16 FX 50.03 FY -209.21 FZ 0.82
107. 17 FX 1.02 FY -1.24 FZ -0.31
108. 18 FX -0.11 FY 1.24 FZ -0.31
110. ****************************************
111. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
112. JOINT LOAD
113. 13 FX -49.09 FY -206.65 FZ -7.94
```

```
114. 14 FX -48.86 FY -207.97 FZ 8.33
116. 15 FX -49.07 FY 205.41 FZ -0.19
117. 16 FX -50.03 FY 209.21 FZ -0.82
119. 17 FX -1.02 FY 1.24 FZ 0.31
120. 18 FX 0.11 FY -1.24 FZ 0.31
122. ****************************************
123. LOAD 8 TEMP LOAD   TL1
124. JOINT LOAD
125. 13 FX -1.74 FY -4.12 FZ -38.21
126. 14 FX -1.25 FY 4.12 FZ 37.18
128. 15 FX 1.17 FY 5.37 FZ 0.22
129. 16 FX 1.21 FY -5.37 FZ 1.45
131. 17 FX 0.1 FY -1.22 FZ -0.8
132. 18 FX 0.51 FY 1.22 FZ 0.19
134. ****************************************
135. LOAD 9 TEMP LOAD   TL2
136. JOINT LOAD
137. 13 FX 1.74 FY 4.12 FZ 38.21
138. 14 FX 1.25 FY -4.12 FZ -37.18
140. 15 FX -1.17 FY -5.37 FZ -0.22
141. 16 FX -1.21 FY 5.37 FZ -1.45
143. 17 FX -0.1 FY 1.22 FZ 0.8
144. 18 FX -0.51 FY -1.22 FZ -0.19
146. ****************************************
147. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
148. JOINT LOAD
149. 13 FX 1.45 FY 7.15 FZ 0.32
150. 14 FX 1.55 FY 7.42 FZ -0.27
152. 15 FX 1.45 FY -7.13 FZ -0.02
153. 16 FX 1.55 FY -7.42 FZ 0
155. 17 FX 0 FY -0.05 FZ -0.01
156. 18 FX -0.01 FY 0.05 FZ -0.01
158. ****************************************
159. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
160. JOINT LOAD
161. 13 FX -0.12 FY 6.67 FZ 1.56
162. 14 FX 0.12 FY -6.67 FZ 1.57
164. 15 FX -0.15 FY 6.66 FZ 1.2
165. 16 FX 0.15 FY -6.66 FZ 1.2
167. 17 FX 0 FY 0.94 FZ 0.24
168. 18 FX 0 FY -0.94 FZ 0.24
170. ****************************************
171. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
172. LOAD COMB 301 1.0 DL
173. 1 1.0
174. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
175. 1 1.0 8 1.0 2 1.0 3 1.0
176. LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
177. 1 1.0 9 1.0 2 1.0 3 1.0
178. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
179. 1 1.0 4 0.75 2 0.75 3 0.75
180. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
181. 1 1.0 5 0.75 2 0.75 3 0.75
182. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
183. 1 1.0 6 0.75 2 0.75 3 0.75
184. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
```

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185. 1 1.0 7 0.75 2 0.75 3 0.75
186. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
187. 1 1.0 10 0.525 2 0.75 3 0.75
188. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
189. 1 1.0 10 -0.525 2 0.75 3 0.75
190. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
191. 1 1.0 11 0.525 2 0.75 3 0.75
192. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
193. 1 1.0 11 -0.525 2 0.75 3 0.75
194. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
195. 1 0.6 4 1.0
196. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
197. 1 0.6 5 1.0
198. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
199. 1 0.6 6 1.0
200. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
201. 1 0.6 7 1.0
202. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
203. 1 0.6 10 0.7
204. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
205. 1 0.6 10 -0.7
206. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
207. 1 0.6 11 0.7
208. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
209. 1 0.6 11 -0.7
210. ****************************************
211. *****LOAD COMBINATION FOR FOUNDATION DESIGN
212. LOAD COMB 401 1.4 DL
213. 1 1.4
214. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
215. 1 1.4 8 1.4 2 1.7 3 1.7
216. LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
217. 1 1.4 9 1.4 2 1.7 3 1.7
218. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
219. 1 1.2 4 1.6 2 0.5 3 0.5
220. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
221. 1 1.2 5 1.6 2 0.5 3 0.5
222. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
223. 1 1.2 6 1.6 2 0.5 3 0.5
224. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
225. 1 1.2 7 1.6 2 0.5 3 0.5
226. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
227. 1 1.2 10 1.0 2 0.5 3 0.5
228. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
229. 1 1.2 10 -1.0 2 0.5 3 0.5
230. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
231. 1 1.2 11 1.0 2 0.5 3 0.5
232. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
233. 1 1.2 11 -1.0 2 0.5 3 0.5
234. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
235. 1 0.9 4 1.6
236. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
237. 1 0.9 5 1.6
238. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
239. 1 0.9 6 1.6
240. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
```

```
241. 1 0.9 7 1.6
242. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
243. 1 0.9 10 1.0
244. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
245. 1 0.9 10 -1.0
246. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
247. 1 0.9 11 1.0
248. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
249. 1 0.9 11 -1.0
250. ****************************************
251. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**18**	**NUMBER OF MEMBERS**	**19**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**6**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 72
TOTAL LOAD COMBINATION CASES = 38 SO FAR.

```
252. LOAD LIST 301 TO 319
253. PRINT SUPPORT REACTION ALL
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	12.87	314.72	-3.51	-7.83	0.06	-32.52
	302	12.62	481.63	-13.66	-16.33	0.57	-32.37
	303	13.03	472.84	2.79	-2.64	-0.07	-32.30
	304	13.51	332.43	-27.59	-70.98	-0.51	-34.07
	305	12.17	540.82	17.66	52.78	0.91	-30.72
	306	-5.58	265.66	-3.48	-7.57	0.25	18.14
	307	31.26	607.56	-6.43	-10.58	0.16	-82.91
	308	12.46	432.53	-4.93	-9.07	0.20	-31.35
	309	13.21	440.69	-4.98	-9.07	0.21	-33.42
	310	12.88	432.69	-5.76	-11.27	0.17	-32.48
	311	12.80	440.53	-4.15	-6.87	0.24	-32.28
	312	8.62	49.93	-32.28	-87.23	-0.91	-21.77
	313	6.84	327.78	28.05	77.78	0.98	-17.30
	314	-16.84	-39.09	-0.14	-2.69	0.10	47.85
	315	32.28	416.76	-4.07	-6.70	-0.02	-86.88
	316	7.22	183.39	-2.07	-4.70	0.03	-18.14
	317	8.22	194.28	-2.13	-4.70	0.05	-20.89
	318	7.78	183.61	-3.18	-7.63	-0.01	-19.65
	319	7.67	194.06	-1.03	-1.76	0.09	-19.38
2	301	12.85	367.29	-1.77	-6.83	-0.05	-32.36
	302	13.22	513.92	9.15	3.89	0.05	-33.39
	303	13.85	522.45	-8.70	-13.71	0.09	-34.24
	304	12.69	584.64	-22.87	-67.26	-0.67	-31.76
	305	14.03	376.25	22.31	56.45	0.75	-35.11
	306	-4.99	308.48	-1.57	-6.41	0.14	16.89
	307	31.71	652.44	1.02	-4.36	-0.06	-83.79
	308	12.97	476.23	-0.32	-5.45	0.04	-32.36
	309	13.76	484.69	-0.23	-5.32	0.04	-34.55
	310	13.32	484.38	-1.08	-7.59	0.00	-33.35
	311	13.40	476.55	0.53	-3.19	0.08	-33.55
	312	6.81	359.28	-31.20	-86.59	-0.98	-17.16
	313	8.59	81.42	29.05	78.35	0.91	-21.63
	314	-16.76	-8.94	-2.79	-5.46	0.11	47.70
	315	32.18	449.68	0.66	-2.73	-0.17	-86.53
	316	7.18	214.73	-1.13	-4.18	-0.04	-17.96
	317	8.23	226.01	-1.00	-4.01	-0.02	-20.88
	318	7.65	225.59	-2.14	-7.03	-0.08	-19.28
	319	7.76	215.15	0.01	-1.16	0.02	-19.55
3	301	17.57	118.02	-0.62	-5.00	-0.26	-37.11
	302	17.60	277.52	-2.22	-8.20	-0.11	-37.18
	303	16.75	288.91	-0.27	-4.33	-0.38	-35.92
	304	18.48	139.95	-19.49	-56.49	-2.12	-38.93
	305	16.09	343.86	17.32	44.61	1.62	-34.48
	306	-12.04	395.81	-1.34	-6.63	-0.27	24.67
	307	46.59	88.02	-0.83	-5.27	-0.22	-98.05
	308	16.68	245.65	-1.08	-5.94	-0.25	-35.43

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	309	17.88	238.18	-1.09	-5.96	-0.24	-37.95
	310	17.34	238.11	-1.68	-7.58	-0.34	-36.82
	311	17.21	245.72	-0.49	-4.32	-0.16	-36.56
	312	12.15	-65.14	-24.90	-70.39	-2.65	-25.25
	313	8.96	206.74	24.17	64.41	2.33	-19.32
	314	-28.54	276.01	-0.71	-3.91	-0.19	59.55
	315	49.63	-134.38	-0.03	-2.10	-0.12	-104.08
	316	9.74	75.79	-0.36	-2.99	-0.16	-20.58
	317	11.34	65.83	-0.37	-3.01	-0.15	-23.94
	318	10.63	65.74	-1.16	-5.18	-0.28	-22.44
	319	10.45	75.89	0.42	-0.82	-0.03	-22.09
4	301	17.27	177.67	-4.41	-9.03	-0.35	-36.65
	302	17.76	375.83	-4.57	-10.82	-0.51	-37.71
	303	17.67	365.04	-4.33	-8.65	-0.28	-37.94
	304	16.40	424.21	-22.84	-60.10	-2.25	-35.29
	305	18.79	220.30	13.97	41.00	1.48	-39.74
	306	-11.45	479.25	-4.68	-10.24	-0.37	23.45
	307	46.66	165.24	-4.20	-8.88	-0.40	-98.51
	308	16.97	326.13	-4.44	-9.55	-0.39	-36.20
	309	18.24	318.36	-4.44	-9.57	-0.38	-38.87
	310	17.53	326.05	-5.03	-11.19	-0.47	-37.40
	311	17.67	318.44	-3.85	-7.92	-0.29	-37.66
	312	8.75	242.55	-27.18	-72.81	-2.70	-19.00
	313	11.95	-29.32	21.89	62.00	2.28	-24.94
	314	-28.38	315.94	-2.97	-6.32	-0.19	59.31
	315	49.10	-102.74	-2.32	-4.51	-0.23	-103.29
	316	9.51	111.78	-2.64	-5.41	-0.21	-20.21
	317	11.21	101.42	-2.65	-5.43	-0.20	-23.77
	318	10.27	111.68	-3.44	-7.59	-0.33	-21.82
	319	10.45	101.52	-1.86	-3.24	-0.09	-22.16
5	301	13.92	56.24	1.67	0.79	-0.37	-32.61
	302	13.41	110.21	2.16	2.37	-0.65	-32.14
	303	13.06	107.67	2.05	1.62	-0.28	-31.43
	304	14.60	67.36	-7.77	-25.13	-2.04	-34.18
	305	12.22	124.15	11.76	28.52	1.16	-29.83
	306	-13.22	112.82	2.15	2.13	-0.51	25.40
	307	40.03	78.72	1.84	1.25	-0.38	-89.38
	308	12.85	96.13	2.00	1.71	-0.44	-30.80
	309	13.96	95.40	1.99	1.68	-0.44	-33.18
	310	13.47	95.12	1.82	1.19	-0.52	-32.12
	311	13.34	96.41	2.17	2.19	-0.36	-31.86
	312	9.94	-4.13	-12.01	-35.29	-2.36	-22.49
	313	6.77	71.59	14.02	36.24	1.91	-16.68
	314	-27.15	56.48	1.21	1.06	-0.31	56.95
	315	43.85	11.01	0.79	-0.11	-0.14	-96.08
	316	7.61	34.23	1.01	0.49	-0.23	-17.98

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	317	9.09	33.26	0.99	0.45	-0.22	-21.15
	318	8.44	32.89	0.76	-0.19	-0.33	-19.73
	319	8.26	34.61	1.24	1.14	-0.12	-19.40
6	301	13.53	59.40	-2.76	-3.83	-0.37	-32.08
	302	13.29	106.73	-1.99	-1.99	-0.62	-32.35
	303	13.53	108.92	-2.61	-3.21	-0.31	-32.96
	304	12.25	124.12	-12.18	-29.73	-2.04	-30.31
	305	14.62	67.33	7.35	23.92	1.16	-34.67
	306	-13.27	110.70	-2.26	-2.47	-0.51	24.82
	307	40.15	80.74	-2.58	-3.35	-0.37	-89.84
	308	12.85	96.04	-2.41	-2.89	-0.44	-31.25
	309	14.03	95.40	-2.42	-2.93	-0.44	-33.77
	310	13.38	96.36	-2.60	-3.41	-0.52	-32.38
	311	13.51	95.07	-2.24	-2.41	-0.36	-32.63
	312	6.52	73.51	-14.67	-38.06	-2.36	-16.32
	313	9.69	-2.21	11.36	33.47	1.92	-22.13
	314	-27.49	55.61	-1.44	-1.71	-0.32	57.19
	315	43.73	15.67	-1.87	-2.88	-0.12	-95.69
	316	7.33	36.07	-1.65	-2.28	-0.22	-17.57
	317	8.90	35.21	-1.66	-2.32	-0.22	-20.92
	318	8.03	36.50	-1.89	-2.96	-0.32	-19.08
	319	8.21	34.78	-1.42	-1.63	-0.12	-19.41

********** END OF LATEST ANALYSIS RESULT **********

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254. ********************************************************************************
255. LOAD LIST 401 TO 419
256. START CONCRETE DESIGN
257. CODE ACI 2008
258. FC 35000 ALL
259. FYMAIN 420000 ALL
260. TRACK 1 ALL
261. CLB 0.05 ALL
262. CLT 0.05 ALL
263. CLS 0.05 ALL
264. MAXMAIN 25 ALL
265. MINMAIN 16 ALL
266. MINSEC 8 ALL
267. REINF 0 MEMB 1 TO 12
268. DESIGN COLUMN 1 TO 12
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COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	616.66	22.81	0.01104

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	616.66	8.44	0.00408

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COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

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 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    552.46      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    552.46      0.00    0.00000
 ---------------------------------------------------------


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     COLUMN  NO.     3  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    729.87     22.98    0.00940
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    729.87      1.90    0.00078
 ---------------------------------------------------------


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     COLUMN  NO.     4  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	659.51	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	659.51	0.00	0.00000

==========

COLUMN NO. 5 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	193.00	46.87	0.07250

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    193.00      6.33    0.00980
 ---------------------------------------------------------
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COLUMN NO. 6 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    129.88      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    129.88      0.00    0.00000
 ---------------------------------------------------------
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COLUMN NO. 7 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM
```

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	321.47	45.65	0.04239

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	321.47	12.41	0.01152

==========

COLUMN NO. 8 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	253.51	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	253.51	0.00	0.00000

==========

COLUMN NO. 9 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	59.94	30.16	0.15023

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	59.94	10.34	0.05151

==========

COLUMN NO. 10 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37     33.82      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37     33.82      0.00    0.00000
 ---------------------------------------------------------


=====================================================================

     COLUMN  NO.    11  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
---------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37     66.73     28.53    0.12760
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37     66.73     11.65    0.05212
 ---------------------------------------------------------


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     COLUMN  NO.    12  DESIGN PER ACI 318-08 - AXIAL + BENDING
```

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	38.34	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	38.34	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

269. DESIGN BEAM 13 TO 19

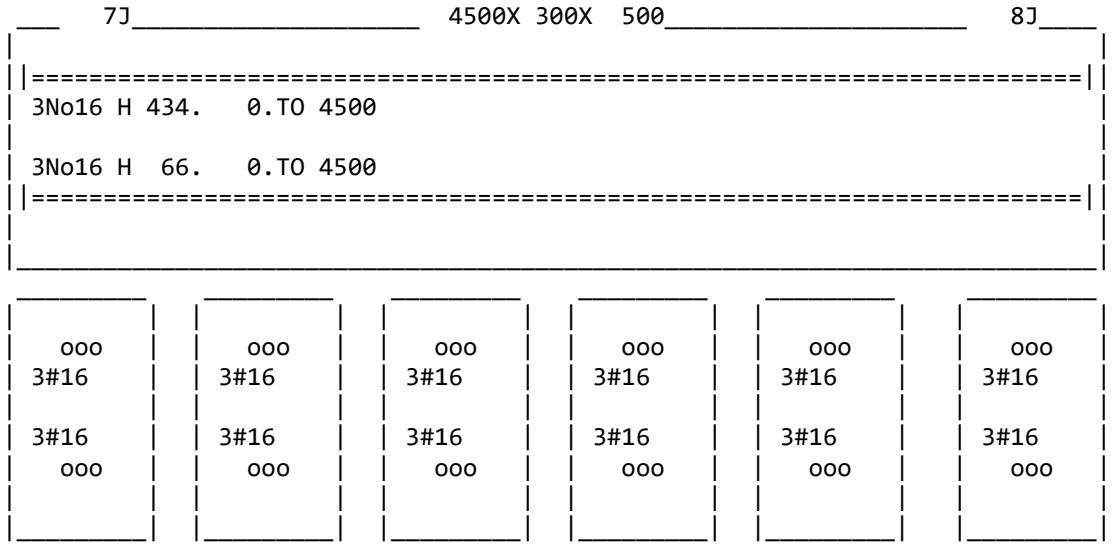
BEAM NO. 13 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4500.	YES	YES
	CRITICAL POS MOMENT= 57.43 KN-MET AT 0.MM, LOAD 412					
	REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
2	434.	3 - 16MM	0.	4500.	YES	YES
	CRITICAL NEG MOMENT= 72.13 KN-MET AT 4500.MM, LOAD 404					
	REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					

B E A M N O. 13 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.00 KNS Vc= 122.82 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.97 KNS Vc= 122.82 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



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BEAM NO. 14 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

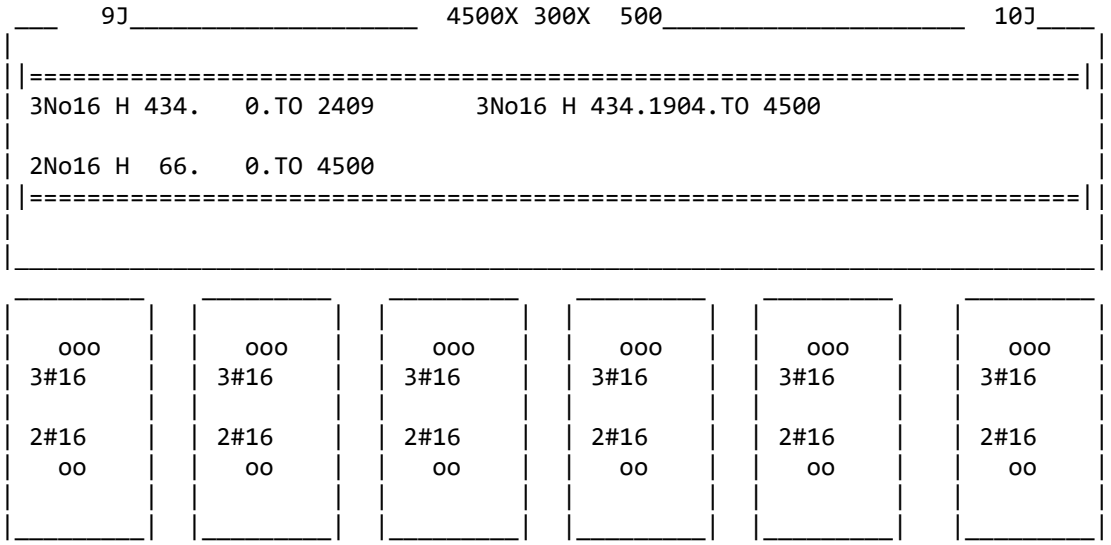
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	4500.	YES	YES
	CRITICAL POS MOMENT= 46.77 KN-MET AT 0.MM, LOAD 412					
	REQD STEEL= 385.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
2	434.	3 - 16MM	0.	2409.	YES	NO
	CRITICAL NEG MOMENT= 51.84 KN-MET AT 0.MM, LOAD 405					
	REQD STEEL= 428.MM2, RHO=0.0033, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
3	434.	3 - 16MM	1904.	4500.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     59.34 KN-MET  AT  4500.MM, LOAD  404|
|   REQD STEEL=   457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 58050.0 cm^4

B E A M N O. 14 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 7.12 KNS Vc= 224.01 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.86 KNS Vc= 126.28 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



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BEAM NO. 15 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 16MM	0.	4500.	YES	YES

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=     22.75 KN-MET  AT     0.MM, LOAD  412|
|   REQD STEEL=   186.MM2, RHO=0.0014, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4

  2        434.     2 - 16MM         0.         2409.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     32.75 KN-MET  AT     0.MM, LOAD  405|
|   REQD STEEL=   268.MM2, RHO=0.0021, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|

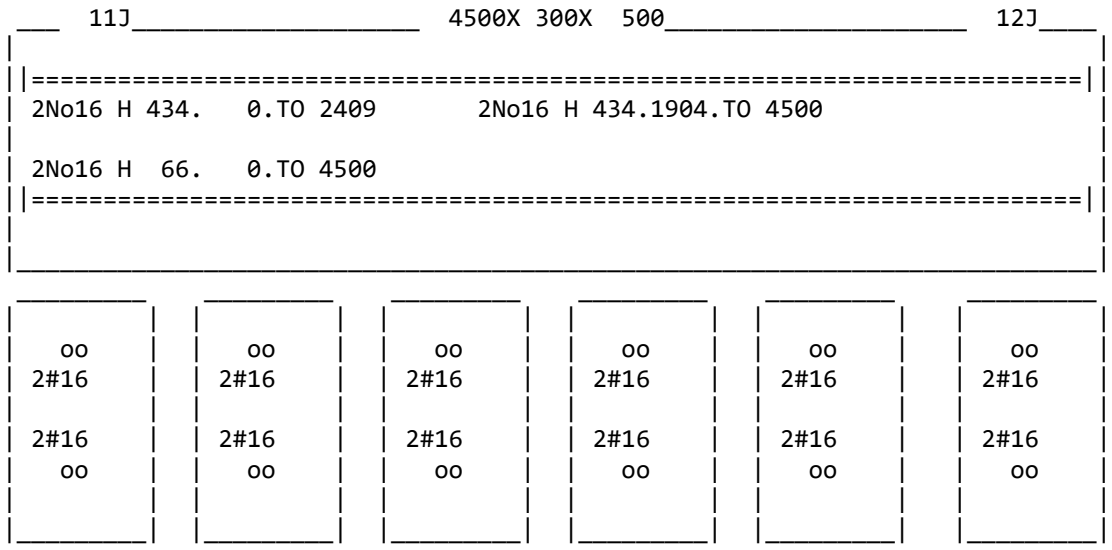
  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4

  3        434.     2 - 16MM      1904.         4500.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     33.96 KN-MET  AT  4500.MM, LOAD  404|
|   REQD STEEL=   278.MM2, RHO=0.0021, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41100.8 cm^4
```

B E A M N O. 15 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    8.53 KNS  Vc=  127.96 KNS  Vs=    0.00 KNS
Tu=    0.01 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=    9.44 KNS  Vc=  125.27 KNS  Vs=    0.00 KNS
Tu=    0.01 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 16 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES

CRITICAL POS MOMENT= 75.57 KN-MET AT 4000.MM, LOAD 407
REQD STEEL= 506.MM2, RHO=0.0032, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

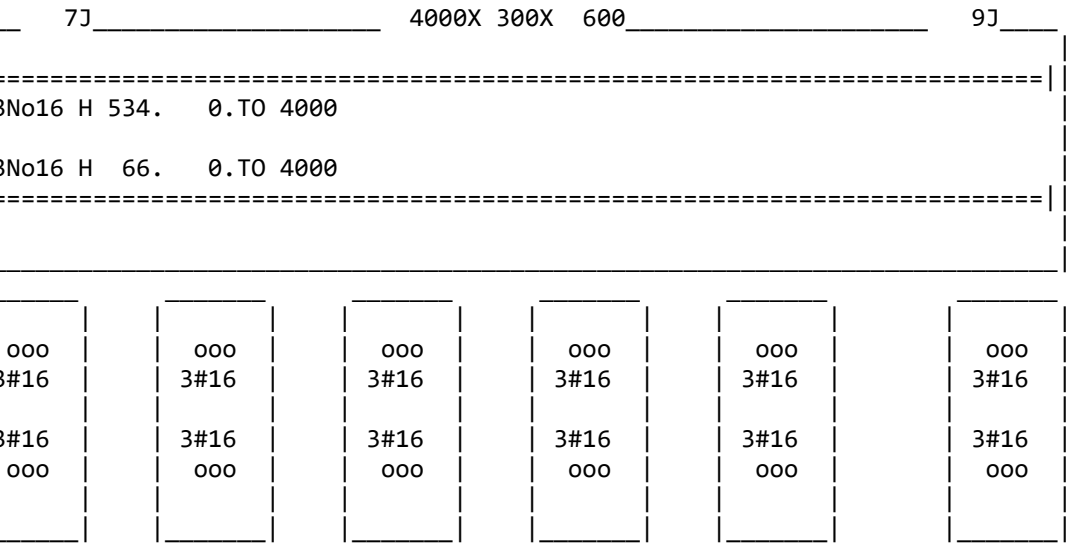
2	534.	3 - 16MM	0.	4000.	YES	YES

CRITICAL NEG MOMENT= 105.59 KN-MET AT 0.MM, LOAD 407
REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

B E A M N O. 16 D E S I G N R E S U L T S - SHEAR

```
  AT START SUPPORT - Vu=   21.70 KNS  Vc=  153.76 KNS  Vs=    0.00 KNS
  Tu=    0.35 KN-MET  Tc=    6.5 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                   STIRRUPS ARE NOT REQUIRED.
  AT END   SUPPORT - Vu=    4.29 KNS  Vc=  153.76 KNS  Vs=    0.00 KNS
  Tu=    0.35 KN-MET  Tc=    6.5 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                   STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 17 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL POS MOMENT= 84.99 KN-MET AT 4000.MM, LOAD 407					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
2	534.	3 - 16MM	0.	2200.	YES	NO

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     95.89 KN-MET  AT     0.MM, LOAD  407|
|   REQD STEEL=   562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./   84. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

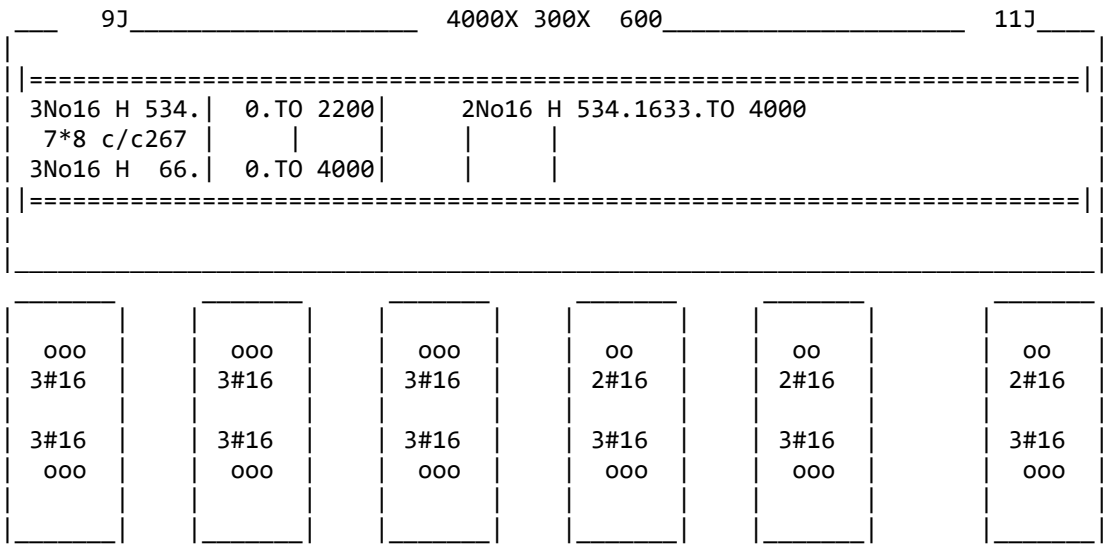
```
  3        534.     2 - 16MM      1633.         4000.        NO   YES
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     56.27 KN-MET  AT  4000.MM, LOAD  414|
|   REQD STEEL=   375.MM2, RHO=0.0023, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

B E A M N O. 17 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 52.68 KNS Vc= 139.28 KNS Vs= 0.00 KNS
Tu= 0.22 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 267. MM C/C FOR 1466. MM

AT END SUPPORT - Vu= 4.59 KNS Vc= 152.49 KNS Vs= 0.00 KNS
Tu= 0.34 KN-MET Tc= 6.4 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



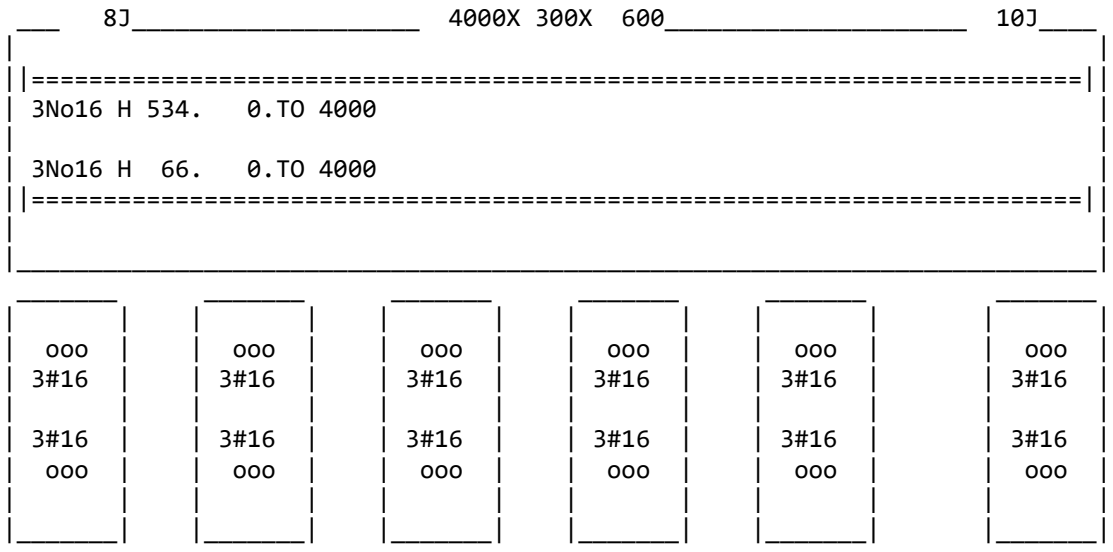
BEAM NO. 18 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL POS MOMENT= 76.38 KN-MET AT 4000.MM, LOAD 407					
	REQD STEEL= 512.MM2, RHO=0.0032, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
2	534.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL NEG MOMENT= 106.04 KN-MET AT 0.MM, LOAD 407					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					

B E A M N O. 18 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 21.73 KNS Vc= 153.82 KNS Vs= 0.00 KNS
Tu= 0.38 KN-MET Tc= 6.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 4.32 KNS Vc= 153.82 KNS Vs= 0.00 KNS
Tu= 0.38 KN-MET Tc= 6.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 19 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL POS MOMENT= 83.87 KN-MET AT 4000.MM, LOAD 407					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
2	534.	3 - 16MM	0.	2200.	YES	NO
	CRITICAL NEG MOMENT= 95.10 KN-MET AT 0.MM, LOAD 407					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
3	534.	2 - 16MM	1633.	4000.	NO	YES

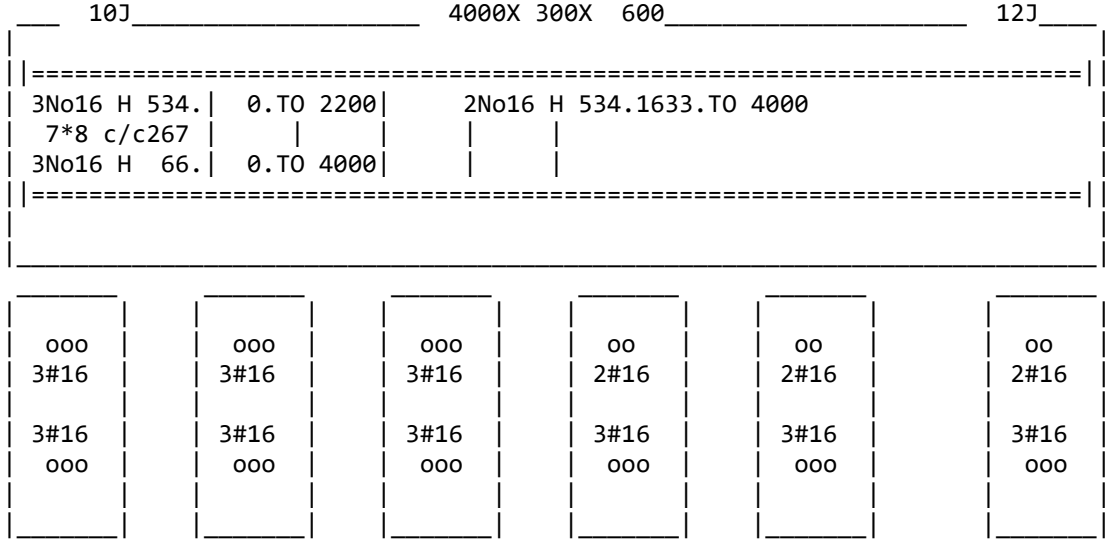
```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     56.58 KN-MET  AT  4000.MM, LOAD  414|
|   REQD STEEL=   378.MM2, RHO=0.0024, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   41./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 63898.6 cm^4

B E A M N O. 19 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 52.20 KNS Vc= 138.77 KNS Vs= 0.00 KNS
Tu= 0.31 KN-MET Tc= 5.9 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 267. MM C/C FOR 1466. MM

AT END SUPPORT - Vu= 4.02 KNS Vc= 152.56 KNS Vs= 0.00 KNS
Tu= 0.42 KN-MET Tc= 6.5 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

270. END CONCRETE DESIGN
271. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAR 25,2020 TIME= 20:32:48 ****

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* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
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